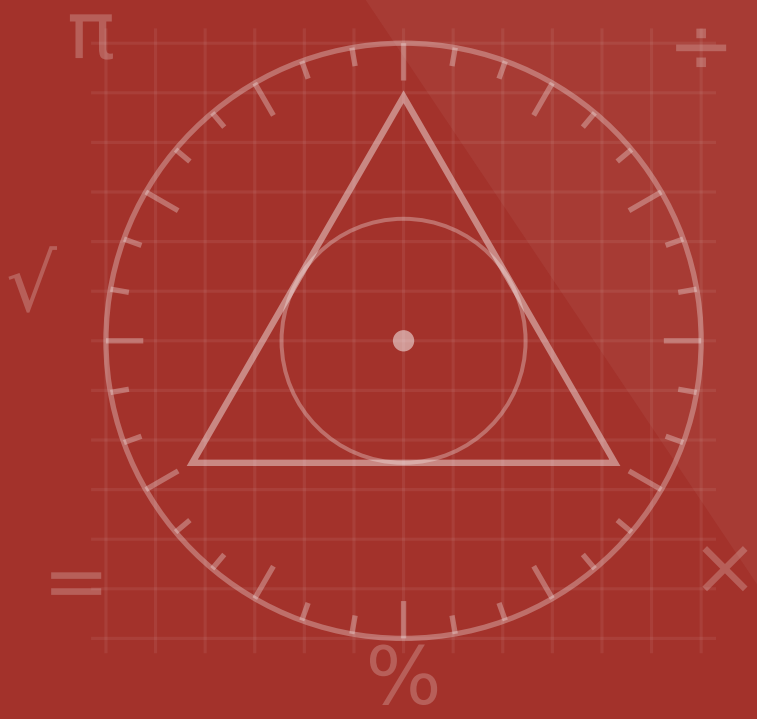
VENVLEARNING

STUDENT TEXTBOOK



Mathematics

GRADE 7

Lessons, examples, and practice for the whole year

2026–2027 School Year · 9 units

VENVLEARNING

Grade 7
Mathematics

STUDENT TEXTBOOK

2026–2027 School Year

Unit 1 Proportional Relationships
Unit 2 Percents
Unit 3 Adding and Subtracting Integers
Unit 4 Adding and Subtracting Rational Numbers
Unit 5 Multiplying and Dividing Rational Numbers
Unit 6 Expressions, Equations, and Inequalities
Unit 7 Statistics and Probability
Unit 8 Scale Drawings and Geometric Figures
Unit 9 Angles, Circles, Area, and Volume

This book belongs to _____

A VenvLearning family homeschool curriculum

How to use this book

Each lesson starts with an I can goal and a Let'S Explore question. Read the lesson with your teacher. Study the Worked Examples step by step, and try the Try It problems. Your practice pages are in your Student Workbook, labeled with the same lesson number.

- Key Idea boxes hold the most important rule or fact. Learn these.
- Watch Out boxes warn you about common mistakes.
- Every Friday there is a short Weekly Check in your workbook. The last Friday of a unit is the Unit Test.

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UNIT 1

Proportional Relationships

Ratios, unit rates, the constant of proportionality, and proportional tables, graphs, and equations.

Many real-world quantities grow together at a constant rate: pay and hours, miles and gallons, cost and pounds. In this unit you will find unit rates, recognize proportional relationships, and represent them with tables, graphs, and equations.

In this unit you will…

- Compute unit rates, including with fractions.
- Decide whether two quantities are proportional.
- Find the constant of proportionality and write $y = kx$.
- Graph and interpret proportional relationships.
- Solve multistep ratio and rate problems.

MATH IN REAL LIFE

Unit prices at the store, gas mileage, hourly pay, recipe scaling, and map distances all use proportional reasoning.

UNIT VOCABULARY

ratio	A comparison of two quantities.
unit rate	Amount per one unit.
proportional relationship	$y \div x$ is always the same.
constant of proportionality	$k = y \div x$.

Lessons in this unit

Lesson	Title
1.1	Ratios and Rates

Lesson	Title
1.2	Unit Rates
1.3	Unit Rates with Fractions
1.4	Comparing Unit Rates
1.5	What Is a Proportional Relationship?
1.6	The Constant of Proportionality
1.7	Equations for Proportional Relationships
1.8	Is It Proportional? Tables and Equations
1.9	Graphing Proportional Relationships
1.10	Recognizing Proportional Graphs
1.11	Interpreting Points on Proportional Graphs
1.12	Comparing Proportional Relationships
1.13	Solving Proportions
1.14	Proportional Reasoning in the Real World
1.15	Multi-Step Ratio Problems
1.16	Proportional Relationships Problem Solving

UNIT 1 · LESSON 1.1 · 7.RP.1 (review of 6.RP.1–3)

Lesson 1.1: Ratios and Rates

I can write ratios and rates and explain what they mean.

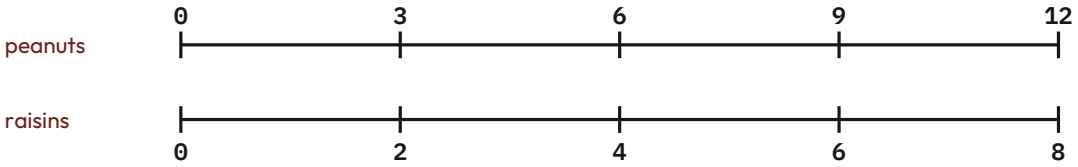
WORDS TO KNOW

ratio	A comparison of two quantities, written 3 to 2, 3:2, or $^{3}/_{2}$.
rate	A ratio that compares two quantities with different units, like 120 miles per 2 hours.
equivalent ratios	Ratios that make the same comparison, like 3:2 and 6:4.

LET'S EXPLORE

A trail mix recipe uses 3 cups of peanuts for every 2 cups of raisins. If you want a bigger batch, how much of each could you use?

A ratio compares two amounts. The recipe's ratio of peanuts to raisins is 3:2. To make more, multiply both quantities by the same number. That keeps the taste the same.



A rate compares quantities with different units, such as miles and hours or dollars and pounds.

WORKED EXAMPLE · EQUIVALENT RATIOS

Is 9 cups of peanuts to 6 cups of raisins the same mix as 3:2?

1 $3 \times 3 = 9$ and $2 \times 3 = 6$.

2 Both parts were multiplied by 3.

Answer: Yes, 9:6 is equivalent to 3:2.

WORKED EXAMPLE · WRITE A RATE

A car travels 150 miles in 3 hours. Write the rate.

1 Compare miles to hours: 150 miles : 3 hours.

Answer: 150 miles per 3 hours (50 miles per hour)

KEY IDEA

Equivalent ratios come from multiplying or dividing both quantities by the same nonzero number.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Write the ratio of 4 dogs to 7 cats three ways.
2. Is 10:15 equivalent to 2:3?
3. A recipe uses 2 eggs per 3 cups of flour. How many eggs for 12 cups of flour?

More practice: Workbook, Lesson 1.1.

LESSON SUMMARY

Ratios compare quantities; rates compare different units. Multiply or divide both parts by the same number to get equivalent ratios.

UNIT 1 · LESSON 1.2 · 7.RP.1

Lesson 1.2: Unit Rates

I can find a unit rate and use it to solve problems.

WORDS TO KNOW

unit rate	A rate for 1 unit of the second quantity, like 55 miles per 1 hour or $3 per pound.

LET'S EXPLORE

A 12-pack of water costs $4.20. How much does one bottle cost?

A unit rate tells how much for one. Divide the first quantity by the second.

WORKED EXAMPLE · PRICE PER ITEM

$4.20 for 12 bottles

1 $4.20 ÷ 12 = 0.35.

Answer: $0.35 per bottle

WORKED EXAMPLE · SPEED

A cyclist rides 54 miles in 4.5 hours. Find the speed.

1 $54 ÷ 4.5 = 12$.

Answer: 12 miles per hour

WORKED EXAMPLE · USE A UNIT RATE

At $0.35 per bottle, how much do 30 bottles cost?

1 $30 × 0.35 = 10.50$.

Answer: $10.50

KEY IDEA

Unit rate = first quantity ÷ second quantity. Then multiply the unit rate to scale up.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $18 for 6 notebooks: unit price?
2. 390 miles on 15 gallons: miles per gallon?
3. Type 210 words in 5 minutes: words per minute?

More practice: Workbook, Lesson 1.2.

LESSON SUMMARY

A unit rate is the amount per one unit. Divide to find it; multiply to use it.

UNIT 1 · LESSON 1.3 · 7.RP.1

Lesson 1.3: Unit Rates with Fractions

I can find unit rates when the quantities are fractions, like $\frac{1}{2}$ mile in $\frac{1}{4}$ hour.

WORDS TO KNOW

complex fraction	A fraction with a fraction in its numerator, denominator, or both.

LET'S EXPLORE

Ellie walks $\frac{1}{2}$ mile in $\frac{1}{6}$ hour. How fast does she walk in miles per hour?

Find the unit rate the same way: divide. $\frac{1}{2} \div \frac{1}{6}$ asks "how many miles in 1 whole hour?"

WORKED EXAMPLE · DIVIDE FRACTIONS

$\frac{1}{2}$ mile in $\frac{1}{6}$ hour

1 $\frac{1}{2} \div \frac{1}{6} = \frac{1}{2} \times \frac{6}{1} = \frac{6}{2} = 3$.

Answer: 3 miles per hour

WORKED EXAMPLE · THINK WITH A TABLE

A painter covers $\frac{2}{3}$ of a wall in $\frac{1}{3}$ hour. How much per hour?

1 In $\frac{1}{3}$ hour: $\frac{2}{3}$ wall. In 1 hour (3 times as long): $3 \times \frac{2}{3} = 2$.

Answer: 2 walls per hour

KEY IDEA

To divide by a fraction, multiply by its reciprocal: $a \div {}^{b}/_{c} = a \times {}^{c}/_{b}$.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. ${}^{3}/_{4}$ cup of sugar per ${}^{1}/_{2}$ batch. Cups per batch?
2. ${}^{1}/_{3}$ mile in ${}^{1}/_{12}$ hour. mph?
3. $2.50 for ${}^{1}/_{2}$ pound. Price per pound?

More practice: Workbook, Lesson 1.3.

LESSON SUMMARY

Divide the quantities even when they are fractions; multiply by the reciprocal.

UNIT 1 · LESSON 1.4 · 7.RP.1, 7.RP.3

Lesson 1.4: Comparing Unit Rates

I can use unit rates to decide which deal or option is better.

WORDS TO KNOW

unit price	The cost for one unit, such as dollars per ounce.

LET'S EXPLORE

A 16-ounce box of cereal costs $3.84. A 24-ounce box costs $5.28. Which is the better buy?

WORKED EXAMPLE · BETTER BUY

16 oz for $3.84 vs 24 oz for $5.28

1 $3.84 \div 16 = 0.24$ per ounce.

2 $5.28 \div 24 = 0.22$ per ounce.

Answer: The 24-ounce box ($0.22/oz)

WORKED EXAMPLE · FASTER RUNNER

Ellie runs 2 miles in 18 minutes. Lillie runs 3 miles in 26 minutes. Who is faster?

1 Ellie: $18 \div 2 = 9$ minutes per mile.

2 Lillie: $26 \div 3 \approx 8.67$ minutes per mile.

Answer: Lillie (fewer minutes per mile)

KEY IDEA

Compare like rates: dollars per ounce with dollars per ounce. For time per mile, smaller is faster.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 12 eggs for $3.00 or 18 eggs for $4.14?
2. Car A: 300 mi on 12 gal. Car B: 270 mi on 10 gal. Better mileage?
3. Who types faster: 150 words in 3 min or 200 words in 5 min?

More practice: Workbook, Lesson 1.4.

LESSON SUMMARY

Find unit rates, then compare them. Think about whether bigger or smaller is better.

UNIT 1 · LESSON 1.5 · 7.RP.2a

Lesson 1.5: What Is a Proportional Relationship?

I can tell when two quantities are in a proportional relationship.

WORDS TO KNOW

proportional relationship	A relationship where the ratio of two quantities is always the same: one is always the same multiple of the other.

LET'S EXPLORE

Tickets cost $8 each. Is the total cost proportional to the number of tickets? What if there is also a $5 parking fee?

Two quantities are proportional when one is always the same number times the other. Every ratio in the table is equivalent.

Tickets	Cost
1	$8
2	$16
3	$24
5	$40

Tickets	Cost + parking
1	$13
2	$21
3	$29
5	$45

In the first table, cost ÷ tickets is always 8: proportional. In the second, 13 ÷ 1 = 13 but 21 ÷ 2 = 10.5: not proportional. The $5 fee breaks the pattern.

KEY IDEA

Proportional: y ÷ x is the same for every pair, and 0 of one quantity means 0 of the other.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Is distance proportional to time at a steady 60 mph?
2. Is a $20 gym fee plus $5 per visit proportional?
3. x: 2, 4, 6 → y: 7, 14, 21. Proportional?

More practice: Workbook, Lesson 1.5.

LESSON SUMMARY

If y ÷ x is the same for every pair, the relationship is proportional.

UNIT 1 · LESSON 1.6 · 7.RP.2b

Lesson 1.6: The Constant of Proportionality

I can find the constant of proportionality k from a table.

WORDS TO KNOW

constant of proportionality	The number k that y is always multiplied from x by: k = y ÷ x. It is the unit rate.

LET'S EXPLORE

A pancake recipe uses 2.5 cups of milk for every 2 cups of flour. How much milk per 1 cup of flour?

In a proportional relationship, y ÷ x is always the same number. That number is the constant of proportionality, k. It is the same as the unit rate.

Flour (cups)	Milk (cups)	Milk ÷ Flour
2	2.5	1.25
4	5	1.25
6	7.5	1.25
1	1.25	1.25

WORKED EXAMPLE · FIND K

x: 3, 5, 8 → y: 12, 20, 32

1 12 ÷ 3 = 4, 20 ÷ 5 = 4, 32 ÷ 8 = 4.

Answer: k = 4

KEY IDEA

k = y ÷ x. It tells how much y for 1 of x.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. x: 2, 6, 10 → y: 9, 27, 45. Find k.

2. Hours 3 → $36 pay. k?

3. x: 4, 8 → y: 1, 2. k?

More practice: Workbook, Lesson 1.6.

LESSON SUMMARY

The constant of proportionality k = y ÷ x is the unit rate.

UNIT 1 · LESSON 1.7 · 7.RP.2c

Lesson 1.7: Equations for Proportional Relationships

I can write an equation y = kx for a proportional relationship.

WORDS TO KNOW

equation	A statement that two expressions are equal.

LET'S EXPLORE

Strawberries cost $3.50 per pound. Write a rule for the cost of any number of pounds.

If y is always k times x, the equation is y = kx. For strawberries: cost = 3.50 × pounds, or y = 3.5x.

WORKED EXAMPLE · FROM A TABLE

x: 2, 5, 9 → y: 16, 40, 72

1 k = 16 ÷ 2 = 8.

Answer: y = 8x

WORKED EXAMPLE · USE THE EQUATION

Use y = 3.5x to find the cost of 6 pounds.

1 y = 3.5 × 6 = 21.

Answer: $21

KEY IDEA

Proportional equation: y = kx, where k is the unit rate.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A plane flies 480 mph. Equation for distance d in t hours?

2. x: 3, 6 → y: 2, 4. Equation?

3. y = 12x. Find y when x = 7.5.

More practice: Workbook, Lesson 1.7.

LESSON SUMMARY

A proportional relationship can be written y = kx.

UNIT 1 · LESSON 1.8 · 7.RP.2a

Lesson 1.8: Is It Proportional? Tables and Equations

I can decide if a table or equation represents a proportional relationship.

LET'S EXPLORE

Which of these is proportional: y = 4x, y = x + 4, or y = 4?

Only equations of the form y = kx are proportional. Adding a number (like y = x + 4) breaks the pattern, because when x = 0, y would not be 0.

Equation	Proportional?	Why
y = 4x	Yes	form y = kx
y = x + 4	No	adds 4
y = 4	No	y never changes
y = 0.5x	Yes	k = 0.5

WORKED EXAMPLE · TABLE TEST

x: 0, 2, 4, 6 → y: 3, 7, 11, 15

1 When x = 0, y = 3 (not 0).

2 7 ÷ 2 = 3.5 but 11 ÷ 4 = 2.75.

Answer: Not proportional

KEY IDEA

Proportional: y = kx, ratios all equal, and (0, 0) fits.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. y = 7x

2. $y = 2x - 1$

3. x: 0, 3, 6 → y: 0, 2, 4

More practice: Workbook, Lesson 1.8.

LESSON SUMMARY

Proportional relationships have the form $y = kx$ and pass through (0, 0).

UNIT 1 · LESSON 1.9 · 7.RP.2a

Lesson 1.9: Graphing Proportional Relationships

I can graph a proportional relationship on a coordinate plane.

WORDS TO KNOW

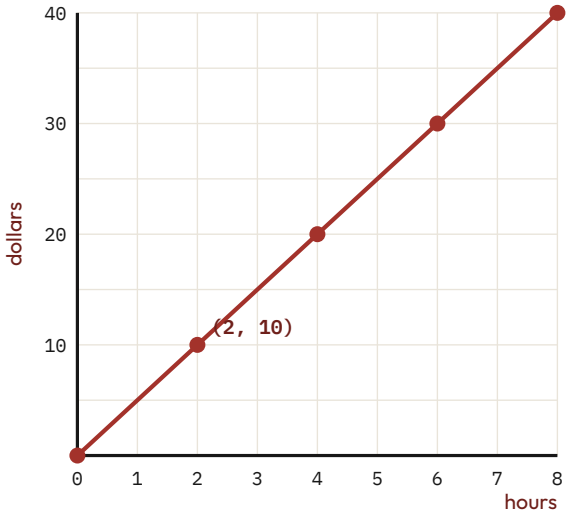
coordinate plane	A grid formed by a horizontal x-axis and a vertical y-axis.
origin	The point (0, 0) where the axes cross.

LET'S EXPLORE

Ellie earns $5 per hour. How could a graph show her earnings for 0 to 8 hours?

Make a table, write each pair as a point (x, y), and plot the points.

Hours (x)	Dollars (y)
0	0
2	10
4	20
6	30
8	40



The points lie on a straight line through the origin. That is what every proportional graph looks like.

KEY IDEA

A proportional relationship graphs as a straight line through the origin (0, 0).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Plot y = 3x for x = 0, 1, 2, 3. Name the points.
2. Does the graph of y = 3x pass through (0, 0)?
3. Is (5, 15) on y = 3x?

More practice: Workbook, Lesson 1.9.

LESSON SUMMARY

Proportional relationships graph as straight lines through the origin.

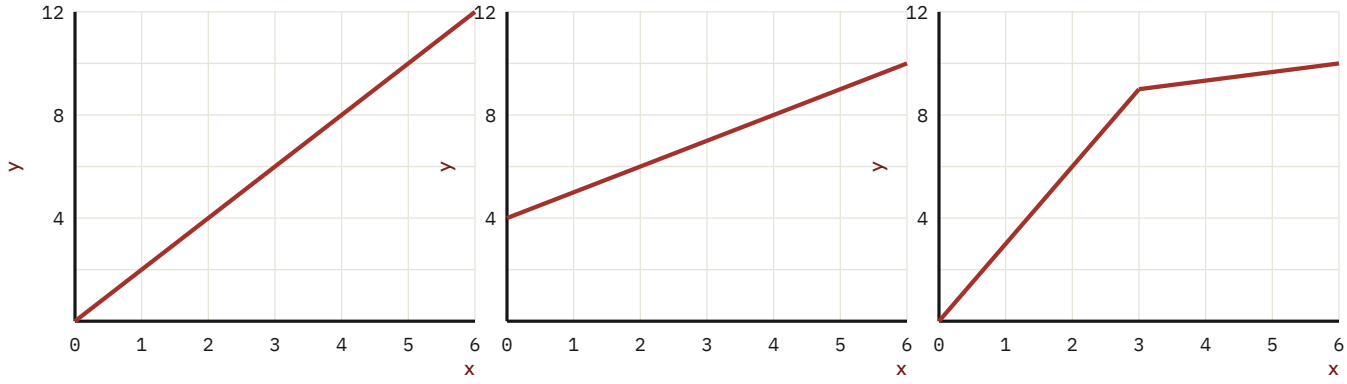
UNIT 1 · LESSON 1.10 · 7.RP.2a

Lesson 1.10: Recognizing Proportional Graphs

I can tell from a graph whether a relationship is proportional.

LET'S EXPLORE

Two graphs are straight lines. One starts at (0, 0) and one starts at (0, 4). Are both proportional?



proportional not: does not go through (0, 0) not: not a straight line

KEY IDEA

Proportional graph = straight line AND passes through the origin. Both are required.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A straight line through (0, 0) and (3, 12). Proportional? k?
2. A curved graph through (0, 0). Proportional?
3. A straight line through (0, 2) and (1, 5). Proportional?

More practice: Workbook, Lesson 1.10.

LESSON SUMMARY

A graph is proportional only if it is a straight line through the origin.

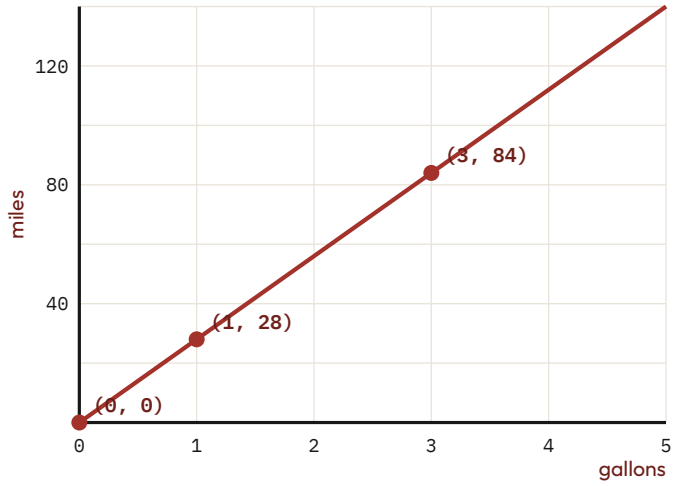
UNIT 1 · LESSON 1.11 · 7.RP.2d

Lesson 1.11: Interpreting Points on Proportional Graphs

I can explain what a point (x, y) means on a proportional graph, including (0, 0) and (1, k).

LET'S EXPLORE

On a graph of gallons and miles, what does the point (1, 28) tell you?



- (0, 0): 0 gallons, 0 miles.
- (1, 28): 1 gallon goes 28 miles. The unit rate k = 28 is the y-value when x = 1.
- (3, 84): 3 gallons go 84 miles.

KEY IDEA

On a proportional graph, the point (1, k) shows the unit rate.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. (1, 7.5) on a graph of hours and dollars means...
2. If (1, 12) is on a proportional graph, what is y when x = 5?
3. What does (0, 0) mean for pages read vs minutes?

More practice: Workbook, Lesson 1.11.

LESSON SUMMARY

(0, 0) means none of each; (1, k) shows the unit rate.

UNIT 1 · LESSON 1.12 · 7.RP.2

Lesson 1.12: Comparing Proportional Relationships

I can compare proportional relationships shown as tables, graphs, equations, and words.

LET'S EXPLORE

Ellie earns y = 9x. Lillie's table shows 3 hours → $30. Who earns more per hour?

Find k in each representation, then compare. Ellie: k = 9. Lillie: 30 ÷ 3 = 10. Lillie earns more per hour.

Representation	How to find k
Equation y = kx	the number times x
Table	y ÷ x
Graph	y-value at x = 1
Words	the unit rate (per 1)

KEY IDEA

Change every representation to its k (unit rate), then compare.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Car A: y = 32x. Car B: 150 miles on 5 gallons. Better mileage?
2. Graph through (1, 2.75) vs y = 2.5x. Greater k?
3. Store A: $4 per pound. Store B: (5, 18). Cheaper?

More practice: Workbook, Lesson 1.12.

LESSON SUMMARY

Find k in every representation, then compare.

UNIT 1 · LESSON 1.13 · 7.RP.3

Lesson 1.13: Solving Proportions

I can solve problems with proportions using unit rates, scale factors, or equations.

WORDS TO KNOW

proportion	An equation saying two ratios are equal, like $^{3}/_{4} = {}^{9}/_{12}$.

LET'S EXPLORE

If 5 notebooks cost $8.75, how much do 12 notebooks cost?

WORKED EXAMPLE · UNIT RATE METHOD

5 notebooks for $8.75; 12 notebooks?

1 8.75 ÷ 5 = 1.75 per notebook.

2 12 × 1.75 = 21.

Answer: $21

WORKED EXAMPLE · EQUATION METHOD

$^{5}/_{8.75} = {}^{12}/_{c}$

1 Equivalent ratios: c = 8.75 × 12 ÷ 5.

2 c = 105 ÷ 5 = 21.

Answer: $21

WORKED EXAMPLE · SCALE FACTOR

3 cups of flour make 24 cookies. How much flour for 96 cookies?

1 96 ÷ 24 = 4, so multiply by 4.

2 $3 \times 4 = 12$.

Answer: 12 cups

KEY IDEA

Use whichever method fits the numbers: unit rate, scale factor, or an equation.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4 tickets cost $34. Cost of 10 tickets?
2. Solve $\frac{6}{15} = \frac{x}{25}$
3. A map: 2 cm = 15 km. 7 cm = ?

More practice: Workbook, Lesson 1.13.

LESSON SUMMARY

Solve proportions with unit rates, scale factors, or equations; keep units matched.

UNIT 1 · LESSON 1.14 · 7.RP.3

Lesson 1.14: Proportional Reasoning in the Real World

I can solve real-world problems with proportional relationships, including recipes, maps, and speed.

LET'S EXPLORE

A recipe for 6 people uses $1\frac{1}{2}$ cups of rice. How much for 10 people?

WORKED EXAMPLE · RECIPE

$1\frac{1}{2}$ cups for 6 people → 10 people

1 Per person: $\frac{3}{2} \div 6 = \frac{1}{4}$ cup.

2 $10 \times \frac{1}{4} = 2\frac{1}{2}$.

Answer: $2\frac{1}{2}$ cups

WORKED EXAMPLE · MAP

On a map, 1 inch = 25 miles. Jackson to Hattiesburg is 3.6 inches.

1 $3.6 \times 25 = 90$.

Answer: about 90 miles

WORKED EXAMPLE · TIME AND DISTANCE

At 60 mph, how long to drive 210 miles?

1 210 ÷ 60 = 3.5.

Answer: 3.5 hours

KEY IDEA

Find the unit rate, then multiply or divide as needed. Check: does the answer make sense?

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A car uses 2 gallons per 54 miles. Gallons for 189 miles?

2. Map: 1 cm = 40 km. 6.5 cm = ?

3. $^{3}/_{4}$ cup sugar for 12 cookies. For 30 cookies?

More practice: Workbook, Lesson 1.14.

LESSON SUMMARY

Find the unit rate, scale up or down, and check that the answer makes sense.

UNIT 1 · LESSON 1.15 · 7.RP.3

Lesson 1.15: Multi-Step Ratio Problems

I can solve multi-step problems involving ratios and rates.

LET'S EXPLORE

Paint is mixed 3 parts blue to 2 parts white. You need 20 quarts of paint. How much of each color?

WORKED EXAMPLE · PART-TO-WHOLE

3 blue : 2 white, 20 quarts total

1 3 + 2 = 5 parts in the whole.

2 20 ÷ 5 = 4 quarts per part.

3 Blue: 3 × 4 = 12; white: 2 × 4 = 8.

Answer: 12 qt blue, 8 qt white

WORKED EXAMPLE · TWO RATES

Ellie drives 2 hours at 50 mph, then 3 hours at 65 mph. Average speed?

1 Distance: 100 + 195 = 295 miles.

2 Time: 5 hours.

3 295 ÷ 5 = 59.

Answer: 59 mph

KEY IDEA

Use the total number of parts for part-to-whole problems. Average speed = total distance ÷ total time.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Boys to girls 4:5; 36 students. How many girls?
2. Concrete mix 1:2:3 (cement:sand:gravel). 30 buckets total. Gravel?
3. Walk 1 hr at 3 mph, then jog 1 hr at 6 mph. Average speed?

More practice: Workbook, Lesson 1.15.

LESSON SUMMARY

Use parts of the whole for ratio totals; use totals for average rates.

UNIT 1 · LESSON 1.16 · 7.RP.1–3

Lesson 1.16: Proportional Relationships Problem Solving

I can choose the best representation to solve proportional relationship problems.

LET'S EXPLORE

Three phone plans: Plan A is y = 0.12x (dollars per text). Plan B: 50 texts for $5.50. Plan C: a graph through (1, 0.10). Which is cheapest per text?

WORKED EXAMPLE · COMPARE PLANS

Find k for each.

1 A: 0.12.

2 B: 5.50 ÷ 50 = 0.11.

3 C: 0.10.

Answer: Plan C

KEY IDEA

Tables, graphs, equations, and words all show the same idea: a constant rate k.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. y = 18x: what is k?

2. Is y = x + 18 proportional?

3. Table: 4 → 14, 10 → 35. Equation?

More practice: Workbook, Lesson 1.16.

LESSON SUMMARY

Find k in any representation and use it to compare and solve.

UNIT 1 REVIEW

Unit 1 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember...
1.1	Ratios compare quantities; rates compare different units. Multiply or divide both parts by the same number to get equivalent ratios.
1.2	A unit rate is the amount per one unit. Divide to find it; multiply to use it.
1.3	Divide the quantities even when they are fractions; multiply by the reciprocal.
1.4	Find unit rates, then compare them. Think about whether bigger or smaller is better.
1.5	If $y \div x$ is the same for every pair, the relationship is proportional.
1.6	The constant of proportionality $k = y \div x$ is the unit rate.
1.7	A proportional relationship can be written $y = kx$.
1.8	Proportional relationships have the form $y = kx$ and pass through (0, 0).
1.9	Proportional relationships graph as straight lines through the origin.
1.10	A graph is proportional only if it is a straight line through the origin.
1.11	(0, 0) means none of each; (1, k) shows the unit rate.
1.12	Find k in every representation, then compare.
1.13	Solve proportions with unit rates, scale factors, or equations; keep units matched.
1.14	Find the unit rate, scale up or down, and check that the answer makes sense.
1.15	Use parts of the whole for ratio totals; use totals for average rates.
1.16	Find k in any representation and use it to compare and solve.

UNIT 2

Percents

Percent as a rate per 100, percent equations, discounts, tax, tips, percent change, interest, and percent error.

Percents are rates per 100, and they are everywhere: sales, taxes, grades, interest, and statistics. You will use proportional reasoning and the percent equation to solve real percent problems.

In this unit you will…

- Understand percent as a rate per 100.
- Convert among percents, fractions, and decimals.
- Find a part, a percent, or a whole.
- Solve discount, markup, tax, tip, and commission problems.
- Find percent change, simple interest, and percent error.

MATH IN REAL LIFE

Percents decide sale prices, sales tax, tips, test grades, loan costs, and savings growth.

UNIT VOCABULARY

percent	A rate per 100.
discount	Amount off a price.
markup	Amount added to cost.
percent change	Change ÷ original.
simple interest	$I = P \times r \times t$.
percent error	|estimate − actual| ÷ actual.

Lessons in this unit

Lesson	Title
2.1	What Is a Percent?

Lesson	Title
2.2	Percents, Fractions, and Decimals
2.3	Finding a Percent of a Number
2.4	Finding What Percent
2.5	Finding the Whole
2.6	The Percent Equation
2.7	Percents Greater Than 100% and Less Than 1%
2.8	Estimating with Percents
2.9	Discounts and Sale Prices
2.10	Markups and Sales Tax
2.11	Percent Increase and Decrease
2.12	Tips, Commissions, and Fees
2.13	Simple Interest
2.14	Percent Error
2.15	Multi-Step Percent Problems
2.16	Percent Problem Solving

UNIT 2 · LESSON 2.1 · 7.RP.3

Lesson 2.1: What Is a Percent?

I can explain a percent as a rate per 100.

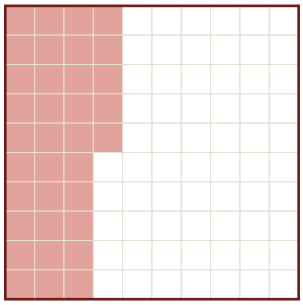
WORDS TO KNOW

percent	A rate per 100. 35% means 35 out of 100, or 35 per 100.

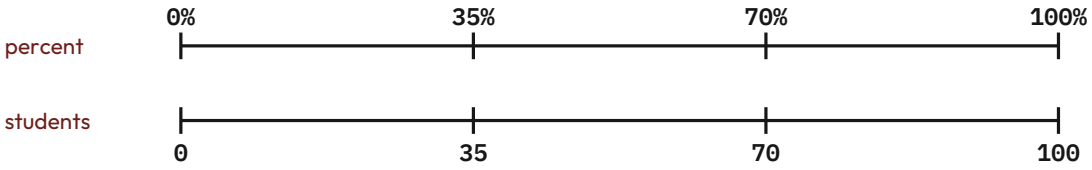
LET'S EXPLORE

A survey says 35% of students walk to school. What does that mean if a school has 100 students? 200 students?

Percent means "per hundred." 35% is 35 per 100. If there are 100 students, 35 walk. If there are 200, twice as many walk: 70.



35 out of 100 squares: 35%



WORKED EXAMPLE · PER 100

In a class, 18 out of 25 students got an A. What percent is that?

1 Scale to 100: 25 × 4 = 100.

2 18 × 4 = 72.

Answer: 72%

KEY IDEA

Percent is a rate per 100. Scale the ratio so the whole is 100.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 40 out of 100 = ___%
2. 7 out of 20 = ___%
3. What percent is 3 out of 4?

More practice: Workbook, Lesson 2.1.

LESSON SUMMARY

Percent means per 100. Scale a ratio to a whole of 100 to find the percent.

UNIT 2 · LESSON 2.2 · 7.RP.3, 7.NS.2d

Lesson 2.2: Percents, Fractions, and Decimals

I can write percents as fractions and decimals, and fractions and decimals as percents.

LET'S EXPLORE

Which is greater: $^{3}/_{8}$, 0.4, or 37%?

Percent	Fraction	Decimal
50%	$^{1}/_{2}$	0.5
25%	$^{1}/_{4}$	0.25
20%	$^{1}/_{5}$	0.2
10%	$^{1}/_{10}$	0.1
75%	$^{3}/_{4}$	0.75
12.5%	$^{1}/_{8}$	0.125

- Percent → decimal: divide by 100 (move the decimal point 2 places left). 37% = 0.37.
- Decimal → percent: multiply by 100. 0.4 = 40%.
- Fraction → percent: divide, then multiply by 100. $^{3}/_{8}$ = 3 ÷ 8 = 0.375 = 37.5%.

WORKED EXAMPLE · COMPARE

$^{3}/_{8}$, 0.4, 37%

1 $^{3}/_{8}$ = 37.5%; 0.4 = 40%; 37%.

Answer: 0.4 is greatest; 37% is least

KEY IDEA

Change everything to the same form to compare.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 0.06 as a percent
2. 145% as a decimal
3. $^{7}/_{20}$ as a percent

More practice: Workbook, Lesson 2.2.

LESSON SUMMARY

Percent ↔ decimal: ×100 or ÷100. Fraction → percent: divide, then ×100.

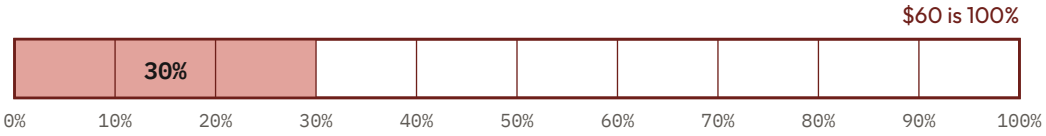
UNIT 2 · LESSON 2.3 · 7.RP.3

Lesson 2.3: Finding a Percent of a Number

I can find a percent of a number using models and multiplication.

LET'S EXPLORE

A $60 jacket is 30% off. How many dollars do you save?



10% of 60 is 6, so 30% is 3 × 6 = 18. Or multiply: 0.30 × 60 = 18.

WORKED EXAMPLE · USE 10%

30% of $60

1 10% of 60 = 6.

2 30% = 3 × 6 = 18.

Answer: $18

WORKED EXAMPLE · MULTIPLY BY THE DECIMAL

15% of 240

1 0.15 × 240 = 36.

Answer: 36

KEY IDEA

Percent of a number = percent (as a decimal) × number. 10% and 1% are quick building blocks.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 20% of 85
2. 7% of 300
3. 120% of 50

More practice: Workbook, Lesson 2.3.

LESSON SUMMARY

Multiply the number by the percent as a decimal, or build from 10% and 1%.

UNIT 2 · LESSON 2.4 · 7.RP.3

Lesson 2.4: Finding What Percent

I can find what percent one number is of another.

LET'S EXPLORE

Ellie scored 42 points out of 48 on a test. What percent is that?

Divide the part by the whole, then write the decimal as a percent.

WORKED EXAMPLE · PART ÷ WHOLE

42 out of 48

1 42 ÷ 48 = 0.875.

2 0.875 = 87.5%.

Answer: 87.5%

WORKED EXAMPLE · MORE THAN 100%

This year's garden produced 90 tomatoes. Last year it produced 60. This year is what percent of last year?

1 90 ÷ 60 = 1.5 = 150%.

Answer: 150%

KEY IDEA

Percent = part ÷ whole × 100%.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. What percent of 80 is 20?
2. What percent of 50 is 7?
3. What percent of 40 is 52?

More practice: Workbook, Lesson 2.4.

LESSON SUMMARY

Percent = part ÷ whole, then × 100%.

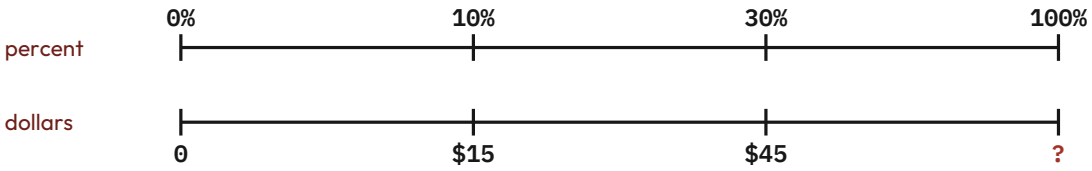
UNIT 2 · LESSON 2.5 · 7.RP.3

Lesson 2.5: Finding the Whole

I can find the whole when I know a part and its percent.

LET'S EXPLORE

Lillie has saved $45, which is 30% of the cost of a bike. How much does the bike cost?



WORKED EXAMPLE · USE 10%

30% is $45

1 10% is 45 ÷ 3 = 15.

2 100% is 15 × 10 = 150.

Answer: $150

WORKED EXAMPLE · DIVIDE BY THE PERCENT

30% of the whole is 45

1 $0.30 \times w = 45$.

2 $w = 45 \div 0.30 = 150$.

Answer: $150

KEY IDEA

Whole = part ÷ percent (as a decimal).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 20% of what number is 14?
2. 75% of what number is 60?
3. 8 is 40% of what?

More practice: Workbook, Lesson 2.5.

LESSON SUMMARY

Whole = part ÷ percent. Check: the whole should be larger than the part when the percent is under 100%.

UNIT 2 · LESSON 2.6 · 7.RP.3, 7.EE.4

Lesson 2.6: The Percent Equation

I can use part = percent × whole to solve any percent problem.

WORDS TO KNOW

percent equation part = percent × whole

LET'S EXPLORE

All percent problems have three numbers: a part, a percent, and a whole. If you know two, you can find the third.

Unknown	Equation	Example
part	p = 0.25 × 80	20
percent	20 = r × 80	r = 0.25 = 25%
whole	20 = 0.25 × w	w = 80

WORKED EXAMPLE · FIND THE WHOLE

36 is 45% of what number?

1 36 = 0.45 × w.

2 w = 36 ÷ 0.45 = 80.

Answer: 80

KEY IDEA

part = percent × whole. Substitute what you know, then solve.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. What is 64% of 75?

2. 27 is what percent of 90?

3. 33 is 55% of what?

More practice: Workbook, Lesson 2.6.

LESSON SUMMARY

Use part = percent × whole to find any missing value.

UNIT 2 · LESSON 2.7 · 7.RP.3

Lesson 2.7: Percents Greater Than 100% and Less Than 1%

I can work with percents greater than 100% and less than 1%.

LET'S EXPLORE

A town's population grew to 250% of what it was. A bank pays 0.5% interest. What do these mean?

250% means 2.5 times the whole. 0.5% means half of 1 percent: 0.005 times the whole.

WORKED EXAMPLE · GREATER THAN 100%

250% of 80

1 2.5 × 80 = 200.

Answer: 200

WORKED EXAMPLE · LESS THAN 1%

0.5% of $2,000

1 0.005 × 2,000 = 10.

Answer: $10

WATCH OUT

0.5% is not 0.5. As a decimal, 0.5% = 0.005.

KEY IDEA

Percent → decimal: divide by 100, even for 250% (2.5) or 0.5% (0.005).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 300% of 15
2. 0.2% of 5,000
3. 125% as a decimal

More practice: Workbook, Lesson 2.7.

LESSON SUMMARY

Percents over 100% mean more than the whole; under 1% means less than a hundredth.

UNIT 2 · LESSON 2.8 · 7.RP.3, 7.EE.3

Lesson 2.8: Estimating with Percents

I can estimate percents using benchmarks like 10%, 25%, and 50%.

LET'S EXPLORE

A $39.95 shirt is 25% off. About how much will it cost?

WORKED EXAMPLE · BENCHMARK

25% off $39.95

1 Round to $40.

2 25% of 40 = 10.

3 40 − 10 = 30.

Answer: about $30

WORKED EXAMPLE · TIP ESTIMATE

A 20% tip on $47.80

1 Round to $48. 10% = $4.80, so 20% ≈ $9.60.

Answer: about $10

KEY IDEA

10%: move the decimal point one place left. 5% is half of 10%. 25% is a quarter. 50% is half.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Estimate 10% of $83.20
2. Estimate 15% of $59
3. Estimate 48% of 202

More practice: Workbook, Lesson 2.8.

LESSON SUMMARY

Use 10%, 5%, 25%, and 50% benchmarks to estimate quickly.

UNIT 2 · LESSON 2.9 · 7.RP.3

Lesson 2.9: Discounts and Sale Prices

I can find the sale price after a percent discount.

WORDS TO KNOW

discount	An amount taken off the original price.
sale price	The price after the discount.

LET'S EXPLORE

A $80 pair of shoes is 25% off. What is the sale price?

WORKED EXAMPLE · SUBTRACT THE DISCOUNT

25% off $80

1 Discount: $0.25 \times 80 = 20$.

2 Sale price: $80 - 20 = 60$.

Answer: $60

WORKED EXAMPLE · MULTIPLY BY WHAT'S LEFT

25% off means you pay 75%

1 $0.75 \times 80 = 60$.

Answer: $60

KEY IDEA

Sale price = original × (100% − discount%).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 40% off $45
2. 15% off $120
3. A $90 coat is on sale for $63. What percent off?

More practice: Workbook, Lesson 2.9.

LESSON SUMMARY

Sale price = original × (100% − discount).

UNIT 2 · LESSON 2.10 · 7.RP.3

Lesson 2.10: Markups and Sales Tax

I can find prices after a markup or sales tax.

WORDS TO KNOW

markup	An amount added to the cost to set the selling price.
sales tax	A percent of the price added at the register.

LET'S EXPLORE

Mississippi's state sales tax is 7%. What is the total for a $40 purchase?

WORKED EXAMPLE · SALES TAX

7% tax on $40

1 Tax: 0.07 × 40 = 2.80.

2 Total: 40 + 2.80 = 42.80.

Answer: $42.80

WORKED EXAMPLE · MARKUP

A store buys a lamp for $25 and marks it up 60%.

1 Markup: 0.60 × 25 = 15.

2 Price: 25 + 15 = 40. Or 1.60 × 25 = 40.

Answer: $40

KEY IDEA

Total with tax or markup = original × (100% + percent).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 7% tax on $15
2. Markup 40% on $30
3. Total with 8% tax on $62.50

More practice: Workbook, Lesson 2.10.

LESSON SUMMARY

Add the percent: total = original × (100% + percent).

UNIT 2 · LESSON 2.11 · 7.RP.3

Lesson 2.11: Percent Increase and Decrease

I can find the percent of increase or decrease.

WORDS TO KNOW

percent change	The amount of change divided by the original amount, as a percent.

LET'S EXPLORE

A puppy weighed 8 pounds in March and 14 pounds in June. By what percent did its weight increase?

WORKED EXAMPLE · INCREASE

8 lb → 14 lb

1 Change: 14 − 8 = 6.

2 6 ÷ 8 = 0.75 = 75%.

Answer: 75% increase

WORKED EXAMPLE · DECREASE

A town had 2,500 people and now has 2,200.

1 Change: 300.

2 300 ÷ 2,500 = 0.12 = 12%.

Answer: 12% decrease

WATCH OUT

Always divide by the original amount, not the new amount.

KEY IDEA

Percent change = change ÷ original × 100%.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $40 → $50
2. 80 → 60
3. 150 → 165

More practice: Workbook, Lesson 2.11.

LESSON SUMMARY

Percent change = change ÷ original.

UNIT 2 · LESSON 2.12 · 7.RP.3

Lesson 2.12: Tips, Commissions, and Fees

I can solve problems with tips, commissions, and fees.

WORDS TO KNOW

tip (gratuity)	Money paid for a service, often a percent of the bill.
commission	Pay earned as a percent of sales.

LET'S EXPLORE

A salesperson earns 6% commission. She sells a $1,250 sofa. How much does she earn?

WORKED EXAMPLE · COMMISSION

6% of $1,250

1 0.06 × 1,250 = 75.

Answer: $75

WORKED EXAMPLE · TIP

18% tip on a $42 meal

1 0.18 × 42 = 7.56.

2 Total: 42 + 7.56 = 49.56.

Answer: $7.56 tip; $49.56 total

KEY IDEA

Tips, commissions, and fees are all “percent of” problems.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 15% tip on $36
2. 4% commission on $8,500
3. 3% card fee on $120

More practice: Workbook, Lesson 2.12.

LESSON SUMMARY

Tips, commissions, and fees are percents of an amount.

UNIT 2 · LESSON 2.13 · 7.RP.3, 7.EE.3

Lesson 2.13: Simple Interest

I can find simple interest and the total amount in an account or loan.

WORDS TO KNOW

interest	Money paid for using money: earned on savings or paid on loans.
principal	The original amount saved or borrowed.
simple interest	I = P × r × t: principal × yearly rate × years.

LET'S EXPLORE

Grandpa puts $500 in a savings account that pays 3% simple interest each year. How much interest after 4 years?

WORKED EXAMPLE · SIMPLE INTEREST

P = $500, r = 3%, t = 4 years

1 I = 500 × 0.03 × 4 = 60.

2 Total: 500 + 60 = 560.

Answer: $60 interest; $560 total

WORKED EXAMPLE · A LOAN

Borrow $1,200 at 6% simple interest for 2 years.

1 I = 1,200 × 0.06 × 2 = 144.

2 Total to repay: $1,344.

Answer: $1,344

KEY IDEA

$I = P \times r \times t$. Use r as a decimal and t in years.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $800 at 5% for 3 years
2. $2,000 at 4.5% for 2 years. Total?
3. $300 at 2% for 6 months (0.5 year)

More practice: Workbook, Lesson 2.13.

LESSON SUMMARY

Simple interest $I = P \times r \times t$.

UNIT 2 · LESSON 2.14 · 7.RP.3

Lesson 2.14: Percent Error

I can find the percent error of an estimate or measurement.

WORDS TO KNOW

percent error	How far off an estimate is, as a percent of the actual value: |estimate − actual| ÷ actual × 100%.

LET'S EXPLORE

Ellie guessed there were 120 jelly beans in a jar. There were actually 150. How far off was she, as a percent?

WORKED EXAMPLE · PERCENT ERROR

Guess 120, actual 150

1 Error: $|120 - 150| = 30$.

2 $30 \div 150 = 0.2 = 20\%$.

Answer: 20% error

KEY IDEA

Percent error = |estimate − actual| ÷ actual × 100%. Always divide by the actual value.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Estimate 45, actual 50
2. Measured 8.4 cm, actual 8 cm
3. Guessed 300, actual 240

More practice: Workbook, Lesson 2.14.

LESSON SUMMARY

Percent error = |estimate − actual| ÷ actual.

UNIT 2 · LESSON 2.15 · 7.RP.3

Lesson 2.15: Multi-Step Percent Problems

I can solve problems that combine discounts, tax, tips, and percent change.

LET'S EXPLORE

A $60 game is 25% off, and then 7% tax is added. What is the total?

WORKED EXAMPLE · DISCOUNT, THEN TAX

$60, 25% off, 7% tax

1 Sale: 0.75 × 60 = 45.

2 Total: 1.07 × 45 = 48.15.

Answer: $48.15

WORKED EXAMPLE · TWO CHANGES

A price rises 10%, then falls 10%. Start at $100.

1 100 × 1.10 = 110.

2 110 × 0.90 = 99.

Answer: $99, not $100!

KEY IDEA

Apply each percent in order to the current amount.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $40 shirt, 20% off, then 7% tax
2. $25 meal, 8% tax, then 20% tip on the pre-tax amount. Total?

3. $200 rises 25%, then falls 20%

More practice: Workbook, Lesson 2.15.

LESSON SUMMARY

Apply percents one step at a time to the current amount.

UNIT 2 · LESSON 2.16 · 7.RP.3

Lesson 2.16: Percent Problem Solving

I can choose and use the right percent strategy to solve real problems.

LET'S EXPLORE

Plan a back-to-school shopping trip: a $35 backpack (20% off), $48 shoes (25% off), 7% tax. You have $75. Is it enough?

WORKED EXAMPLE · PLAN

Backpack and shoes

1 Backpack: 0.8 × 35 = 28.

2 Shoes: 0.75 × 48 = 36.

3 Subtotal 64; tax 1.07 × 64 = 68.48.

Answer: Yes, $68.48 ≤ $75, with $6.52 left

KEY IDEA

Read carefully, decide what is the whole, and apply each percent in order.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. What percent of 64 is 16?

2. 12% of what is 9?

3. $75 → $90 percent change

More practice: Workbook, Lesson 2.16.

LESSON SUMMARY

Identify the part, percent, and whole; apply percents step by step.

UNIT 2 REVIEW

Unit 2 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
2.1	Percent means per 100. Scale a ratio to a whole of 100 to find the percent.
2.2	Percent ↔ decimal: ×100 or ÷100. Fraction → percent: divide, then ×100.
2.3	Multiply the number by the percent as a decimal, or build from 10% and 1%.
2.4	Percent = part ÷ whole, then × 100%.
2.5	Whole = part ÷ percent. Check: the whole should be larger than the part when the percent is under 100%.
2.6	Use part = percent × whole to find any missing value.
2.7	Percents over 100% mean more than the whole; under 1% means less than a hundredth.
2.8	Use 10%, 5%, 25%, and 50% benchmarks to estimate quickly.
2.9	Sale price = original × (100% − discount).
2.10	Add the percent: total = original × (100% + percent).
2.11	Percent change = change ÷ original.
2.12	Tips, commissions, and fees are percents of an amount.
2.13	Simple interest $I = P \times r \times t$.
2.14	Percent error = |estimate − actual| ÷ actual.
2.15	Apply percents one step at a time to the current amount.
2.16	Identify the part, percent, and whole; apply percents step by step.

UNIT 3

Adding and Subtracting Integers

Opposites, absolute value, adding and subtracting integers with chips and number lines, and real-world applications.

Negative numbers let us describe temperatures below zero, elevations below sea level, and money owed. In this unit you will add and subtract integers and explain why the rules work.

In this unit you will…

- Find opposites and absolute values.
- Add integers with the same and different signs.
- Subtract integers by adding the opposite.
- Find distances between numbers.
- Solve real-world integer problems.

MATH IN REAL LIFE

Temperatures, elevations, bank balances, football yardage, and golf scores all use integers.

UNIT VOCABULARY

integer	Whole numbers and their opposites.
opposites	Same distance from 0, opposite sides.
absolute value	Distance from 0.
additive inverse	A number's opposite; sums to 0.

Lessons in this unit

Lesson	Title
3.1	Integers, Opposites, and Absolute Value
3.2	Adding Integers with the Same Sign
3.3	Adding Integers with Different Signs

Lesson	Title
3.4	Additive Inverses
3.5	Adding Integers on a Number Line
3.6	Subtracting Integers
3.7	Subtraction as Distance
3.8	Distance Between Two Numbers
3.9	Integers: Temperature and Elevation
3.10	Integers and Money
3.11	Multi-Step Integer Problems
3.12	Properties and Mental Math with Integers
3.13	Integer Expressions
3.14	Explaining Integer Rules with Models
3.15	Finding and Fixing Integer Errors
3.16	Integer Problem Solving

UNIT 3 · LESSON 3.1 · 7.NS.1 (review of 6.NS.5–7)

Lesson 3.1: Integers, Opposites, and Absolute Value

I can use integers to describe real situations and find opposites and absolute values.

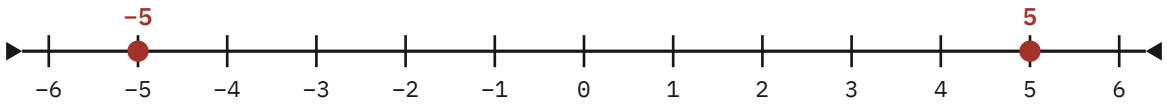
WORDS TO KNOW

integer	A whole number or its opposite: ..., −3, −2, −1, 0, 1, 2, 3, ...
opposites	Two numbers the same distance from 0 on opposite sides, like 4 and −4.
absolute value	A number's distance from 0, written |a|. |−7| = 7.

LET'S EXPLORE

The temperature in Jackson is 5°F. In Fairbanks, Alaska, it is 5 degrees below zero. How would you write both temperatures?

Integers include negative numbers. Negative numbers describe values below zero: temperatures, elevations below sea level, money owed.



5 and −5 are opposites: the same distance from 0 in opposite directions. Both have an absolute value of 5.

WORKED EXAMPLE · ABSOLUTE VALUE

Find \|−12\| and \|9\|.

1 −12 is 12 units from 0.

2 9 is 9 units from 0.

Answer: 12 and 9

KEY IDEA

Opposites have the same absolute value. Absolute value is a distance, so it is never negative.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Write an integer: a loss of 8 yards
2. Opposite of −15
3. |−23|

More practice: Workbook, Lesson 3.1.

LESSON SUMMARY

Integers include negatives. Opposites are the same distance from 0; absolute value is that distance.

UNIT 3 · LESSON 3.2 · 7.NS.1b

Lesson 3.2: Adding Integers with the Same Sign

I can add two integers that have the same sign.

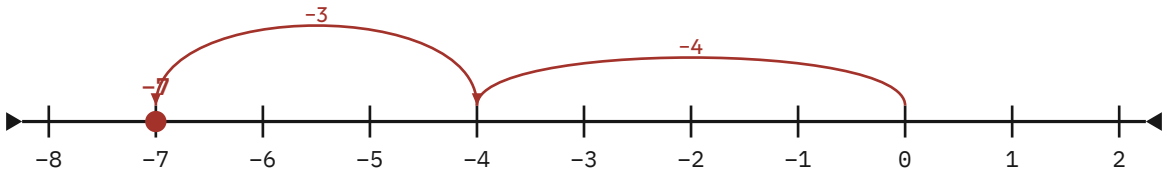
LET'S EXPLORE

You owe your sister $4. Then you borrow $3 more. How much do you owe in all?

Owing $4 is −4. Owing $3 more is −3. Together: −4 + (−3) = −7. You owe $7.



Seven negative chips: −7



WORKED EXAMPLE · SAME SIGN

−9 + (−6)

1 Both negative: add the absolute values, 9 + 6 = 15.

2 Keep the sign: negative.

Answer: −15

KEY IDEA

Same signs: add the absolute values and keep the common sign.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. −5 + (−8)
2. 12 + 9
3. −20 + (−15)

More practice: Workbook, Lesson 3.2.

LESSON SUMMARY

Same signs: add absolute values, keep the sign.

UNIT 3 · LESSON 3.3 · 7.NS.1b

Lesson 3.3: Adding Integers with Different Signs

I can add two integers with different signs.

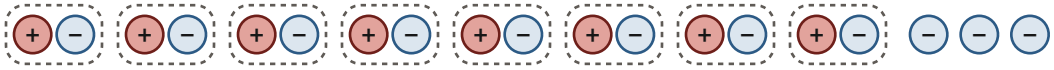
WORDS TO KNOW

zero pair	A positive and a negative chip together; they cancel to 0.

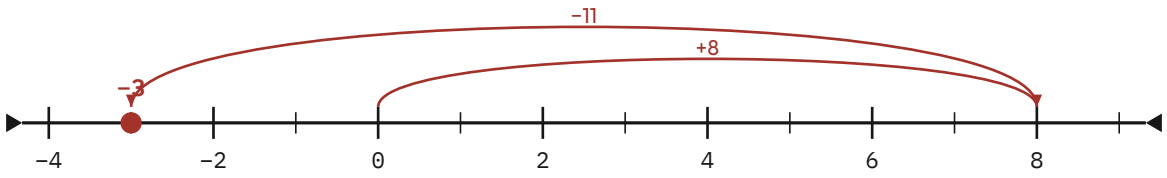
LET'S EXPLORE

You have $8. You spend $11. What is your balance?

Model 8 + (−11) with chips. Each positive cancels a negative to make a zero pair. After 8 zero pairs, 3 negative chips are left: −3.



8 zero pairs cancel; 3 negatives remain.



WORKED EXAMPLE · DIFFERENT SIGNS

−14 + 9

1 Subtract absolute values: 14 − 9 = 5.

2 Use the sign of the number with the larger absolute value: −14.

Answer: −5

KEY IDEA

Different signs: subtract the absolute values and use the sign of the number farther from 0.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 7 + (−10)

2. −6 + 15

3. −25 + 18

More practice: Workbook, Lesson 3.3.

LESSON SUMMARY

Different signs: subtract absolute values; the sign of the farther-from-zero number wins.

UNIT 3 · LESSON 3.4 · 7.NS.1a, 7.NS.1b

Lesson 3.4: Additive Inverses

I can explain why a number plus its opposite equals 0.

WORDS TO KNOW

additive inverse	A number's opposite. A number plus its additive inverse equals 0.

LET'S EXPLORE

A hydrogen atom has 1 proton (+1 charge) and 1 electron (−1 charge). What is its total charge?

+1 + (−1) = 0. Opposites always add to 0. The opposite of a number is called its additive inverse.

WORKED EXAMPLE · USE INVERSES

−13 + 8 + 13

1 −13 and 13 are opposites: they make 0.

2 0 + 8 = 8.

Answer: 8

WORKED EXAMPLE · FIND THE MISSING NUMBER

−35 + ___ = 0

1 The additive inverse of −35 is 35.

Answer: 35

KEY IDEA

a + (−a) = 0 for every number a. Look for opposites to make 0 and simplify.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. −42 + 42
2. 6 + (−9) + 9
3. ___ + 17 = 0

More practice: Workbook, Lesson 3.4.

LESSON SUMMARY

A number plus its opposite is 0.

UNIT 3 · LESSON 3.5 · 7.NS.1b

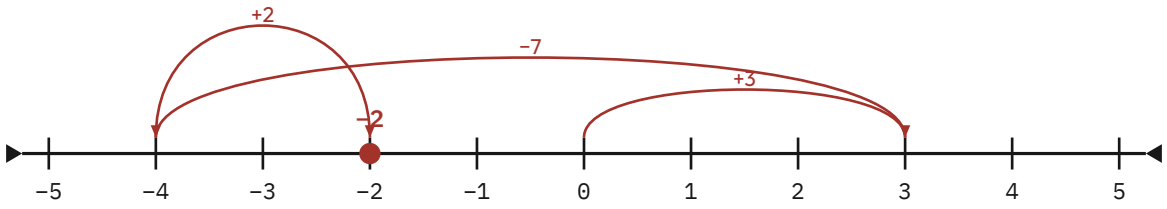
Lesson 3.5: Adding Integers on a Number Line

I can use a number line to add integers.

LET'S EXPLORE

A hiker starts at 3 m above a lake, climbs down 7 m, then up 2 m. Where does she end?

Start at the first number. Move right for a positive addend and left for a negative addend.



WORKED EXAMPLE · NUMBER LINE

3 + (−7) + 2

1 Start at 0, move right 3 to 3.

2 Move left 7 to −4.

3 Move right 2 to −2.

Answer: −2

KEY IDEA

Positive → right. Negative → left. The landing point is the sum.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. −4 + 6 (use a number line)

2. 2 + (−5) + (−1)

3. −3 + (−3) + 8

More practice: Workbook, Lesson 3.5.

LESSON SUMMARY

On a number line, positive moves right and negative moves left.

UNIT 3 · LESSON 3.6 · 7.NS.1c

Lesson 3.6: Subtracting Integers

I can subtract integers by adding the opposite.

LET'S EXPLORE

What is 3 − (−5)? How can subtracting make a number bigger?

Subtracting a number is the same as adding its opposite: a − b = a + (−b). So 3 − (−5) = 3 + 5 = 8.

With chips: start with 3 positives. To take away 5 negatives, first add 5 zero pairs (which doesn't change the value). Then remove the 5 negatives. 8 positives are left.



3 positives plus 5 zero pairs; remove the 5 negatives → 8.

WORKED EXAMPLE · ADD THE OPPOSITE

−6 − 4

1 −6 − 4 = −6 + (−4).

2 Same signs: −10.

Answer: −10

WORKED EXAMPLE · SUBTRACT A NEGATIVE

$-2 - (-9)$

1 $-2 - (-9) = -2 + 9 = 7$.

Answer: 7

KEY IDEA

$a - b = a + (-b)$. Keep the first number, change subtraction to addition, use the opposite of the second number.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $5 - 12$
2. $-8 - (-3)$
3. $-4 - 10$

More practice: Workbook, Lesson 3.6.

LESSON SUMMARY

Subtracting means adding the opposite: $a - b = a + (-b)$.

UNIT 3 · LESSON 3.7 · 7.NS.1c

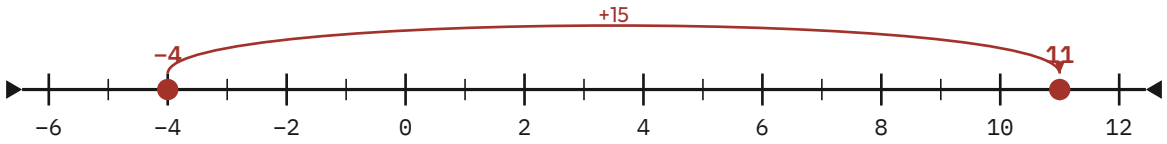
Lesson 3.7: Subtraction as Distance

I can use a number line to show subtraction as the change from one number to another.

LET'S EXPLORE

At 6 a.m. it was −4°F. At noon it was 11°F. How much did the temperature rise?

Change = end − start. 11 − (−4) = 11 + 4 = 15. On a number line, from −4 to 11 is 15 units to the right.



WORKED EXAMPLE · ELEVATION

A bird at 25 ft dives to a fish at −8 ft. Change in elevation?

1 −8 − 25 = −33.

Answer: −33 ft (33 ft down)

KEY IDEA

End − start tells the change. Positive means up or right; negative means down or left.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. From −3 to 9: change?
2. From 5 to −7: change?
3. From −10 to −2: change?

More practice: Workbook, Lesson 3.7.

LESSON SUMMARY

Change = end − start; on a number line it is the jump from start to end.

UNIT 3 · LESSON 3.8 · 7.NS.1c

Lesson 3.8: Distance Between Two Numbers

I can find the distance between two numbers using absolute value.

WORDS TO KNOW

distance	How far apart two numbers are; always positive: $\lvert a - b \rvert$.

LET'S EXPLORE

Death Valley is 282 ft below sea level. Mount Whitney is 14,505 ft above. How far apart are their elevations?

WORKED EXAMPLE · DISTANCE

$\lvert 14{,}505 - (-282) \rvert$

1 $14{,}505 + 282 = 14{,}787$.

Answer: 14,787 ft

WORKED EXAMPLE · ORDER DOESN'T MATTER

$\lvert -5 - 3 \rvert$ and $\lvert 3 - (-5) \rvert$

1 $\lvert -8 \rvert = 8$ and $\lvert 8 \rvert = 8$.

Answer: 8 both ways

KEY IDEA

Distance between a and b = $\lvert a - b \rvert$. It is never negative.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Distance between −7 and 2
2. Distance between −12 and −4
3. $\lvert -6 - 9 \rvert$

More practice: Workbook, Lesson 3.8.

LESSON SUMMARY

Distance between a and b is $\lvert a - b \rvert$.

UNIT 3 · LESSON 3.9 · 7.NS.1, 7.NS.3

Lesson 3.9: Integers: Temperature and Elevation

I can solve temperature and elevation problems with integers.

WORDS TO KNOW

sea level	Elevation 0; above is positive and below is negative.

LET'S EXPLORE

New Orleans sits about 6 feet below sea level in places. A levee rises 23 feet above that ground. How high above sea level is the top of the levee?

WORKED EXAMPLE · ELEVATION

Ground at −6 ft, levee 23 ft higher

1 −6 + 23 = 17.

Answer: 17 ft above sea level

WORKED EXAMPLE · TEMPERATURE

It is −3°F at 6 a.m. It rises 4° per hour for 3 hours, then drops 5°.

1 −3 + 12 = 9.

2 9 − 5 = 4.

Answer: 4°F

KEY IDEA

Choose a direction for positive (up, warmer) and use negatives for the opposite.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A diver at −30 ft rises 12 ft.
2. 8°F drops 15°.
3. A balloon at 400 ft descends 650 ft into a canyon. New elevation?

More practice: Workbook, Lesson 3.9.

LESSON SUMMARY

Use positive for up or warmer and negative for down or colder.

UNIT 3 · LESSON 3.10 · 7.NS.1, 7.NS.3

Lesson 3.10: Integers and Money

I can use integers to track money: deposits, withdrawals, profits, and debts.

WORDS TO KNOW

balance	The amount of money in an account; it can be negative if more is owed than saved.

LET'S EXPLORE

Lillie's account has $25. She spends $40, then deposits $30. What is her balance after each step?

Transaction	Integer	Balance
start		$25
spend $40	−40	−$15
deposit $30	+30	$15

WORKED EXAMPLE · PROFIT AND LOSS

A lemonade stand earns $18, $−7 (a loss), and $12 on three days.

1 $18 + (-7) + 12 = 23$.

Answer: $23 profit

KEY IDEA

Deposits and profits are positive; withdrawals, purchases, and losses are negative.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $10 balance, buy a $14 item
2. −$20 balance, deposit $35
3. Profits: $50, −$65, $20. Total?

More practice: Workbook, Lesson 3.10.

LESSON SUMMARY

Money in is positive; money out or owed is negative. Keep a running balance.

UNIT 3 · LESSON 3.11 · 7.NS.3, 7.EE.3

Lesson 3.11: Multi-Step Integer Problems

I can solve multi-step problems with integer addition and subtraction.

LET'S EXPLORE

A football team starts on its own 30-yard line. It loses 7 yards, gains 12, loses 3, and gains 25. Where is it now?

WORKED EXAMPLE · COMBINE

30 − 7 + 12 − 3 + 25

1 Gains: 12 + 25 = 37. Losses: −7 + (−3) = −10.

2 30 + 37 − 10 = 57.

Answer: the 57-yard line (in plain terms, 27 yards net gain)

KEY IDEA

Group the positives and the negatives, then combine.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. −8 + 15 − 6 + 4
2. A temperature of 3°F drops 10°, rises 4°, drops 2°.
3. Elevation −20 m: up 35, down 12, up 6

More practice: Workbook, Lesson 3.11.

LESSON SUMMARY

Group positives and negatives, then combine.

UNIT 3 · LESSON 3.12 · 7.NS.1d

Lesson 3.12: Properties and Mental Math with Integers

I can use properties of addition to add integers mentally.

WORDS TO KNOW

commutative property	Changing the order of addends doesn't change the sum: a + b = b + a.
associative property	Changing the grouping doesn't change the sum: (a + b) + c = a + (b + c).

LET'S EXPLORE

How would you quickly find −17 + 45 + 17 + (−5)?

WORKED EXAMPLE · REARRANGE

−17 + 45 + 17 + (−5)

1 Reorder: (−17 + 17) + (45 + (−5)).

2 0 + 40 = 40.

Answer: 40

KEY IDEA

Rewrite subtraction as addition, then reorder and regroup to make zeros or tens.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. −25 + 38 + 25

2. 14 + (−9) + (−14) + 9

3. −36 + 50 + (−14)

More practice: Workbook, Lesson 3.12.

LESSON SUMMARY

Reorder and regroup addends (with their signs) to simplify.

UNIT 3 · LESSON 3.13 · 7.NS.1, 7.EE.1

Lesson 3.13: Integer Expressions

I can evaluate expressions with integers and variables.

WORDS TO KNOW

evaluate	To find the value of an expression by substituting numbers.

LET'S EXPLORE

If $x = -4$, what is $x + 9 - x$?

WORKED EXAMPLE · SUBSTITUTE

$x + 9 - x$ when $x = -4$

1 $(-4) + 9 - (-4) = -4 + 9 + 4 = 9$.

Answer: 9

WORKED EXAMPLE · TWO VARIABLES

$a - b$ when $a = -6$, $b = -11$

1 $-6 - (-11) = -6 + 11 = 5$.

Answer: 5

KEY IDEA

Substitute with parentheses around negative values, then add the opposite to subtract.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $x - 7$ when $x = -3$
2. $-y + 5$ when $y = -2$
3. $a + b - c$ when $a = 4$, $b = -9$, $c = -2$

More practice: Workbook, Lesson 3.13.

LESSON SUMMARY

Substitute negatives in parentheses; subtract by adding the opposite.

UNIT 3 · LESSON 3.14 · 7.NS.1

Lesson 3.14: Explaining Integer Rules with Models

I can explain integer addition and subtraction rules using chips and number lines.

LET'S EXPLORE

Why does subtracting a negative make a number bigger? Can you prove it with a model?

Rules are easier to remember when you know why they work. Here are both models side by side.

Problem	Chip model	Number line
−3 + 5	3 zero pairs cancel; 2 positives left	start at −3, move right 5 → 2
4 − (−2)	add 2 zero pairs, remove 2 negatives → 6	the distance from −2 to 4 is 6
−2 − 5	add 5 zero pairs, remove 5 positives → −7	start at −2, move left 5 → −7

KEY IDEA

Every rule can be shown with a model. Use a model to check a rule you're unsure of.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Explain −6 + 6 = 0 with chips.
2. Explain 2 − 7 = −5 on a number line.
3. Explain −1 − (−4) = 3 with chips.

More practice: Workbook, Lesson 3.14.

LESSON SUMMARY

Chips and number lines show why integer rules work.

UNIT 3 · LESSON 3.15 · 7.NS.1

Lesson 3.15: Finding and Fixing Integer Errors

I can find and fix mistakes in integer calculations.

LET'S EXPLORE

Sam says $-8 - 3 = -5$. Is he right?

WORKED EXAMPLE · ERROR

$-8 - 3 = -5$?

1 Sam subtracted $8 - 3$ and kept a negative sign.

2 Correct: $-8 - 3 = -8 + (-3) = -11$.

Answer: -11

Common error	Fix
$-8 - 3 = -5$	Rewrite: $-8 + (-3) = -11$
$5 - (-2) = 3$	Subtracting a negative adds: $5 + 2 = 7$
$-4 + 9 = -5$	Different signs: $9 - 4 = 5$, sign of 9: $+5$
$\lvert -6 \rvert = -6$	Absolute value is a distance: 6

KEY IDEA

Rewrite subtraction as addition and check signs with a model.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Fix: $-7 + (-2) = -5$
2. Fix: $3 - 10 = 7$
3. Fix: $-6 - (-6) = -12$

More practice: Workbook, Lesson 3.15.

LESSON SUMMARY

Rewrite, check signs, and use a model to catch errors.

UNIT 3 · LESSON 3.16 · 7.NS.1, 7.NS.3

Lesson 3.16: Integer Problem Solving

I can solve problems with integer addition and subtraction from start to finish.

LET'S EXPLORE

A week of low temperatures: −4, 2, −7, −1, 3, −5, 0. What is the range? What is the sum?

WORKED EXAMPLE · RANGE AND SUM

−4, 2, −7, −1, 3, −5, 0

1 Range: 3 − (−7) = 10.

2 Sum: positives 5, negatives −17 → −12.

Answer: range 10°; sum −12

KEY IDEA

Organize the numbers, group signs, and check with a model.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Range of −15 and 22

2. −9 + 14 − 20

3. Distance from −6 to −18

More practice: Workbook, Lesson 3.16.

LESSON SUMMARY

Organize, group signs, and check with models.

UNIT 3 REVIEW

Unit 3 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
3.1	Integers include negatives. Opposites are the same distance from 0; absolute value is that distance.
3.2	Same signs: add absolute values, keep the sign.
3.3	Different signs: subtract absolute values; the sign of the farther-from-zero number wins.
3.4	A number plus its opposite is 0.
3.5	On a number line, positive moves right and negative moves left.
3.6	Subtracting means adding the opposite: $a - b = a + (-b)$.
3.7	Change = end − start; on a number line it is the jump from start to end.
3.8	Distance between a and b is $\lvert a - b \rvert$.
3.9	Use positive for up or warmer and negative for down or colder.
3.10	Money in is positive; money out or owed is negative. Keep a running balance.
3.11	Group positives and negatives, then combine.
3.12	Reorder and regroup addends (with their signs) to simplify.
3.13	Substitute negatives in parentheses; subtract by adding the opposite.
3.14	Chips and number lines show why integer rules work.
3.15	Rewrite, check signs, and use a model to catch errors.
3.16	Organize, group signs, and check with models.

UNIT 4

Adding and Subtracting Rational Numbers

Signed fractions and decimals, converting fractions to decimals, and real-world problems.

The integer rules from Unit 3 work for every rational number. In this unit you will add and subtract signed fractions, mixed numbers, and decimals, and use them to solve problems about money, temperature, and measurement.

In this unit you will…

- Place and compare signed fractions and decimals.
- Add and subtract signed fractions and mixed numbers.
- Convert fractions to terminating or repeating decimals.
- Add and subtract signed decimals.
- Solve multi-step rational-number problems.

MATH IN REAL LIFE

Bank balances, stock prices, tide levels, golf scores, and temperatures use signed fractions and decimals.

UNIT VOCABULARY

rational number	A number that can be written as a fraction of integers.
terminating decimal	A decimal that ends.
repeating decimal	A decimal with a repeating pattern.

Lessons in this unit

Lesson	Title
4.1	Rational Numbers on the Number Line
4.2	Adding Signed Fractions
4.3	Subtracting Signed Fractions

Lesson	Title
4.4	Adding and Subtracting Signed Mixed Numbers
4.5	Fractions as Decimals
4.6	Comparing and Ordering Rational Numbers
4.7	Adding Signed Decimals
4.8	Subtracting Signed Decimals
4.9	Rational Numbers and Money
4.10	Rational Numbers in Measurement
4.11	Properties for Rational Number Sums
4.12	Multi-Step Rational Number Problems
4.13	Distance Between Rational Numbers
4.14	Expressions with Rational Numbers
4.15	Error Analysis with Rational Numbers
4.16	Rational Number Problem Solving

UNIT 4 · LESSON 4.1 · 7.NS.1

Lesson 4.1: Rational Numbers on the Number Line

I can place positive and negative fractions and decimals on a number line.

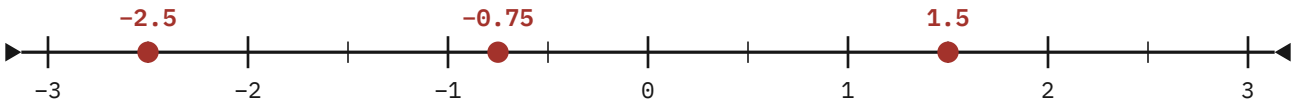
WORDS TO KNOW

rational number — A number that can be written as a fraction of integers, like $^{3}/_{4}$, −2.5, or 7.

LET'S EXPLORE

The temperature is −2.5°C. Is that warmer or colder than −2°C?

Rational numbers include integers, fractions, and decimals, positive and negative. −2.5 is halfway between −3 and −2, so it is farther from 0 and colder than −2.



WORKED EXAMPLE · COMPARE

$-^{3}/_{4}$ and $-^{2}/_{3}$

1 $^{3}/_{4} = ^{9}/_{12}$ and $^{2}/_{3} = ^{8}/_{12}$.

2 $-^{9}/_{12}$ is farther left.

Answer: $-^{3}/_{4} < -^{2}/_{3}$

KEY IDEA

For negative numbers, the one farther from 0 is less.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Which is less: −1.2 or −1.02?
2. Opposite of $-3\,^{1}/_{2}$
3. $|-4.75|$

More practice: Workbook, Lesson 4.1.

LESSON SUMMARY

Rational numbers include signed fractions and decimals. For negatives, farther from 0 is less.

UNIT 4 · LESSON 4.2 · 7.NS.1b, 7.NS.1d

Lesson 4.2: Adding Signed Fractions

I can add positive and negative fractions.

LET'S EXPLORE

A stock rose $\frac{3}{8}$ of a point on Monday and fell $\frac{5}{8}$ of a point on Tuesday. What was the total change?

The integer rules work for fractions too. Same signs: add absolute values, keep the sign. Different signs: subtract absolute values, use the sign of the one farther from 0.

WORKED EXAMPLE · DIFFERENT SIGNS

$\frac{3}{8} + (-\frac{5}{8})$

1 $|-\frac{5}{8}|$ is larger, so the answer is negative.

2 $\frac{5}{8} - \frac{3}{8} = \frac{2}{8} = \frac{1}{4}$.

Answer: $-\frac{1}{4}$

WORKED EXAMPLE · UNLIKE DENOMINATORS

$-\frac{1}{2} + (-\frac{1}{3})$

1 Common denominator 6: $-\frac{3}{6} + (-\frac{2}{6})$.

2 Same signs: $-\frac{5}{6}$.

Answer: $-\frac{5}{6}$

KEY IDEA

Use common denominators, then apply the integer sign rules to the numerators.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-\frac{2}{5} + \frac{4}{5}$
2. $\frac{3}{4} + (-\frac{5}{6})$
3. $-\frac{1}{6} + (-\frac{2}{3})$

More practice: Workbook, Lesson 4.2.

LESSON SUMMARY

Find a common denominator, then follow the integer sign rules.

UNIT 4 · LESSON 4.3 · 7.NS.1c

Lesson 4.3: Subtracting Signed Fractions

I can subtract positive and negative fractions by adding the opposite.

LET'S EXPLORE

At 6 a.m. a pond's ice was $-\frac{1}{4}$ inch below a mark. At noon it was $\frac{5}{8}$ inch above. How much did it rise?

WORKED EXAMPLE · ADD THE OPPOSITE

$\frac{5}{8} - (-\frac{1}{4})$

$= \frac{5}{8} + \frac{1}{4} = \frac{5}{8} + \frac{2}{8}$.

Answer: $\frac{7}{8}$ inch

WORKED EXAMPLE · NEGATIVE MINUS POSITIVE

$-\frac{2}{3} - \frac{1}{2}$

$= -\frac{2}{3} + (-\frac{1}{2}) = -\frac{4}{6} + (-\frac{3}{6})$.

Answer: $-\frac{7}{6} = -1\frac{1}{6}$

KEY IDEA

$a - b = a + (-b)$, for fractions too.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{1}{3} - \frac{2}{3}$
2. $-\frac{3}{4} - (-\frac{1}{8})$
3. $-\frac{1}{2} - \frac{3}{10}$

More practice: Workbook, Lesson 4.3.

LESSON SUMMARY

Subtract fractions by adding the opposite.

UNIT 4 · LESSON 4.4 · 7.NS.1

Lesson 4.4: Adding and Subtracting Signed Mixed Numbers

I can add and subtract signed mixed numbers like $-2\frac{1}{3} + 1\frac{1}{2}$.

LET'S EXPLORE

A hiker is at $-3\frac{1}{2}$ m (in a trench). She climbs $5\frac{3}{4}$ m. Where is she now?

Change mixed numbers to improper fractions, or work with whole and fraction parts carefully.

WORKED EXAMPLE · IMPROPER FRACTIONS

$-3\frac{1}{2} + 5\frac{3}{4}$

$= -\frac{14}{4} + \frac{23}{4} = \frac{9}{4}$.

Answer: $2\frac{1}{4}$ m

WORKED EXAMPLE · SUBTRACT

$-1\frac{2}{3} - 2\frac{1}{6}$

$= -\frac{10}{6} + \left(-\frac{13}{6}\right) = -\frac{23}{6}$.

Answer: $-3\frac{5}{6}$

KEY IDEA

Improper fractions make signed mixed-number problems safer.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-2\frac{1}{4} + 3\frac{1}{2}$
2. $1\frac{2}{3} - 3\frac{1}{4}$
3. $-2\frac{1}{5} + \left(-1\frac{1}{2}\right)$

More practice: Workbook, Lesson 4.4.

LESSON SUMMARY

Change signed mixed numbers to improper fractions, then use the sign rules. Remember $-3\frac{1}{2} = -3 - \frac{1}{2}$.

UNIT 4 · LESSON 4.5 · 7.NS.2d

Lesson 4.5: Fractions as Decimals

I can write fractions as terminating or repeating decimals using long division.

WORDS TO KNOW

terminating decimal	A decimal that ends, like 0.375.
repeating decimal	A decimal with a digit or digits that repeat forever, like 0.333... written $0.\overline{3}$

LET'S EXPLORE

Is $\frac{1}{3}$ closer to 0.33 or 0.34? Does it ever end?

Divide the numerator by the denominator. $\frac{3}{8} = 3 \div 8 = 0.375$: it terminates. $\frac{1}{3} = 0.3333\ldots$: it repeats forever. We write 0.3 with a bar over the 3.

Fraction	Decimal	Type
$\frac{1}{4}$	0.25	terminating
$\frac{3}{8}$	0.375	terminating
$\frac{2}{3}$	0.666...	repeating
$\frac{5}{11}$	0.4545...	repeating
$\frac{7}{20}$	0.35	terminating

WORKED EXAMPLE · LONG DIVISION

$\frac{5}{6}$

1 $5 \div 6$: $50 \div 6 = 8$ R2; $20 \div 6 = 3$ R2; $20 \div 6 = 3$ R2...

2 The 3 repeats.

Answer: 0.8333... (0.83 with a bar over the 3)

KEY IDEA

Every fraction is a terminating or repeating decimal. That's what makes it rational.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{5}{8}$ as a decimal
2. $\frac{4}{9}$ as a decimal

3. $-\frac{7}{4}$ as a decimal

More practice: Workbook, Lesson 4.5.

LESSON SUMMARY

Divide numerator by denominator; the decimal terminates or repeats.

UNIT 4 · LESSON 4.6 · 7.NS.1, 7.NS.2d

Lesson 4.6: Comparing and Ordering Rational Numbers

I can compare and order signed fractions and decimals.

LET'S EXPLORE

Order these golf scores (relative to par) from best (lowest) to worst: −2.5, $-2\frac{3}{4}$, 1.25, $-\frac{1}{2}$.

WORKED EXAMPLE · ORDER

$-2.5, -2\frac{3}{4}, 1.25, -\frac{1}{2}$

1 Decimals: −2.5, −2.75, 1.25, −0.5.

2 Least to greatest: −2.75, −2.5, −0.5, 1.25.

Answer: $-2\frac{3}{4}$, −2.5, $-\frac{1}{2}$, 1.25

KEY IDEA

Change all numbers to decimals, then compare on a number line.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-\frac{5}{8}$ ◯ −0.6

2. Order: −1.3, $-\frac{4}{3}$, −1.25

3. Greatest: −0.09, −0.9, −0.19

More practice: Workbook, Lesson 4.6.

LESSON SUMMARY

Convert to decimals and compare on a number line.

UNIT 4 · LESSON 4.7 · 7.NS.1

Lesson 4.7: Adding Signed Decimals

I can add positive and negative decimals.

LET'S EXPLORE

A bank account has −$12.50 (overdrawn). A deposit of $40.75 is made. What is the new balance?

WORKED EXAMPLE · DIFFERENT SIGNS

−12.50 + 40.75

1 Subtract absolute values: 40.75 − 12.50 = 28.25.

2 40.75 is farther from 0, so positive.

Answer: $28.25

WORKED EXAMPLE · SAME SIGNS

−3.6 + (−0.85)

1 Add: 3.6 + 0.85 = 4.45. Keep negative.

Answer: −4.45

KEY IDEA

Line up decimal points and use the integer sign rules.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. −7.2 + 3.5
2. −0.45 + (−1.8)
3. 12.06 + (−12.6)

More practice: Workbook, Lesson 4.7.

LESSON SUMMARY

Line up decimals and apply the sign rules.

UNIT 4 · LESSON 4.8 · 7.NS.1c

Lesson 4.8: Subtracting Signed Decimals

I can subtract positive and negative decimals.

LET'S EXPLORE

The lowest point in Louisiana is about −2.4 m (in New Orleans). Driskill Mountain is about 163.1 m. What is the difference in elevation?

WORKED EXAMPLE · SUBTRACT A NEGATIVE

163.1 − (−2.4)

1 = 163.1 + 2.4.

Answer: 165.5 m

WORKED EXAMPLE · NEGATIVE MINUS POSITIVE

−0.75 − 1.6

1 = −0.75 + (−1.6) = −2.35.

Answer: −2.35

KEY IDEA

Subtract decimals by adding the opposite.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4.2 − 6.5
2. −3.8 − (−1.25)
3. −0.9 − 0.45

More practice: Workbook, Lesson 4.8.

LESSON SUMMARY

Subtract decimals by adding the opposite.

UNIT 4 · LESSON 4.9 · 7.NS.3, 7.EE.3

Lesson 4.9: Rational Numbers and Money

I can solve money problems with positive and negative decimals.

LET'S EXPLORE

Lillie's account starts at $24.60. She buys a book for $18.95 and a snack for $7.40. Then she deposits $15. What is her balance?

WORKED EXAMPLE · RUNNING BALANCE

24.60 − 18.95 − 7.40 + 15

1 24.60 − 18.95 = 5.65.

2 5.65 − 7.40 = −1.75.

3 −1.75 + 15 = 13.25.

Answer: $13.25

KEY IDEA

Write each transaction as a signed number and keep a running balance.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $5.20 − $9.85
2. −$32.10 + $50
3. Profits: $12.40, −$20.15, $6.80

More practice: Workbook, Lesson 4.9.

LESSON SUMMARY

Use signed decimals for deposits and purchases and keep a running balance.

UNIT 4 · LESSON 4.10 · 7.NS.3

Lesson 4.10: Rational Numbers in Measurement

I can solve elevation, temperature, and measurement problems with signed fractions and decimals.

LET'S EXPLORE

A carpenter needs a board $4\frac{3}{8}$ ft long. Hers is $5\frac{1}{4}$ ft. How much must she cut off?

WORKED EXAMPLE · FRACTIONS

$5\frac{1}{4} - 4\frac{3}{8}$

1 $= \frac{42}{8} - \frac{35}{8} = \frac{7}{8}$.

Answer: $\frac{7}{8}$ ft

WORKED EXAMPLE · TEMPERATURE

−4.5°C at night, 7.2°C at noon

1 $7.2 - (-4.5) = 11.7$.

Answer: 11.7°C warmer

KEY IDEA

Choose fractions or decimals, whichever makes the computation easier.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A plant $3\frac{1}{2}$ in. tall shrinks $\frac{3}{4}$ in. New height?
2. −1.8°F to 6.5°F: change
3. Water level $-\frac{2}{3}$ ft then rises $1\frac{1}{6}$ ft

More practice: Workbook, Lesson 4.10.

LESSON SUMMARY

Use signed fractions and decimals for real measurements.

UNIT 4 · LESSON 4.11 · 7.NS.1d

Lesson 4.11: Properties for Rational Number Sums

I can use properties to add and subtract rational numbers efficiently.

LET'S EXPLORE

Find $-\frac{3}{4} + 2.6 + \frac{3}{4} + (-0.6)$ in your head.

WORKED EXAMPLE · REARRANGE

$-\frac{3}{4} + 2.6 + \frac{3}{4} + (-0.6)$

1 $(-\frac{3}{4} + \frac{3}{4}) + (2.6 + (-0.6))$.

2 $0 + 2 = 2$.

Answer: 2

KEY IDEA

Group opposites and friendly decimals; rewrite subtraction as addition first.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-3.7 + 8.25 + 3.7$
2. $\frac{2}{5} - 1.5 - \frac{2}{5} + 0.5$
3. $-0.25 + 6 + (-0.75)$

More practice: Workbook, Lesson 4.11.

LESSON SUMMARY

Regroup opposites and friendly numbers to compute mentally.

UNIT 4 · LESSON 4.12 · 7.EE.3

Lesson 4.12: Multi-Step Rational Number Problems

I can solve multi-step problems with signed fractions and decimals.

LET'S EXPLORE

A tide chart shows changes of $-1\frac{1}{4}$ ft, $+2\frac{1}{2}$ ft, $-\frac{3}{4}$ ft from a starting level of $\frac{1}{2}$ ft. Final level?

WORKED EXAMPLE · TIDE

$\frac{1}{2} - 1\frac{1}{4} + 2\frac{1}{2} - \frac{3}{4}$

1 In fourths: $2 - 5 + 10 - 3 = 4$ fourths.

Answer: 1 ft

KEY IDEA

Convert to one form (all fourths, or all decimals) and combine in order.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-2.4 + 5.1 - 3.25$
2. $\frac{2}{3} - \frac{5}{6} + (-\frac{1}{2})$
3. A hiker at −15.5 m climbs 40.25 m then descends 8.75 m.

More practice: Workbook, Lesson 4.12.

LESSON SUMMARY

Convert to one form and combine step by step.

UNIT 4 · LESSON 4.13 · 7.NS.1c

Lesson 4.13: Distance Between Rational Numbers

I can find the distance between two rational numbers using absolute value.

LET'S EXPLORE

How far apart are $-2\frac{1}{2}$ and $1\frac{3}{4}$ on a number line?

WORKED EXAMPLE · DISTANCE

$|1\frac{3}{4} - (-2\frac{1}{2})|$

$= |\frac{7}{4} + \frac{10}{4}| = \frac{17}{4}$.

Answer: $4\frac{1}{4}$

KEY IDEA

Distance = $|a - b|$, for fractions and decimals too.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Distance between -3.2 and 1.5
2. Distance between $-\frac{5}{6}$ and $-\frac{1}{3}$
3. $|-0.8 - 2.45|$

More practice: Workbook, Lesson 4.13.

LESSON SUMMARY

Distance between rational numbers is $|a - b|$.

UNIT 4 · LESSON 4.14 · 7.NS.1, 7.EE.1

Lesson 4.14: Expressions with Rational Numbers

I can evaluate expressions with signed fractions and decimals.

LET'S EXPLORE

Evaluate $x - y + 2.5$ when $x = -1.75$ and $y = -3$.

WORKED EXAMPLE · SUBSTITUTE

$x - y + 2.5$

1 $(-1.75) - (-3) + 2.5 = -1.75 + 3 + 2.5$.

Answer: 3.75

KEY IDEA

Substitute with parentheses, rewrite subtraction, then combine.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $a + b$ when $a = -0.6$, $b = 2.35$
2. $x - {}^{1}\!/_{2}$ when $x = -{}^{3}\!/_{4}$
3. $-m + n$ when $m = -4.2$, $n = -1.8$

More practice: Workbook, Lesson 4.14.

LESSON SUMMARY

Substitute in parentheses and follow the sign rules.

UNIT 4 · LESSON 4.15 · 7.NS.1

Lesson 4.15: Error Analysis with Rational Numbers

I can find and fix errors in rational-number computations.

LET'S EXPLORE

Sam says $-\frac{1}{2} - \frac{1}{3} = -\frac{1}{6}$. Is he right?

WORKED EXAMPLE · FIX IT

$-\frac{1}{2} - \frac{1}{3}$

1 Sam subtracted absolute values. It should be $-\frac{1}{2} + (-\frac{1}{3})$.

2 $= -\frac{3}{6} + (-\frac{2}{6}) = -\frac{5}{6}$.

Answer: $-\frac{5}{6}$

Error	Fix
−2.5 + 1.3 = −3.8	Different signs: −1.2
0.4 − (−0.6) = −0.2	Add the opposite: 1.0
$-1\frac{1}{2} = -1 + \frac{1}{2}$	$-1\frac{1}{2} = -1 - \frac{1}{2} = -1.5$

KEY IDEA

Estimate the sign and size first; then check the computed answer.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Fix: −3.2 − 1.5 = −1.7
2. Fix: $\frac{1}{4} - (-\frac{3}{4}) = -\frac{1}{2}$
3. Fix: −0.8 + 0.25 = −1.05

More practice: Workbook, Lesson 4.15.

LESSON SUMMARY

Estimate sign and size first, then check.

UNIT 4 · LESSON 4.16 · 7.NS.1, 7.NS.3, 7.EE.3

Lesson 4.16: Rational Number Problem Solving

I can solve problems with signed fractions and decimals.

LET'S EXPLORE

A science class recorded temperature changes every hour: −1.5°, −2.25°, +0.75°, +3.5°. Starting at −4°C, what is the final temperature? What was the lowest?

WORKED EXAMPLE · TRACK

Start −4

1 −4 − 1.5 = −5.5.

2 −5.5 − 2.25 = −7.75 (lowest).

3 −7.75 + 0.75 = −7.

4 −7 + 3.5 = −3.5.

Answer: final −3.5°C; lowest −7.75°C

KEY IDEA

Organize with a table; check the reasonableness of every step.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-\frac{3}{8} + \frac{5}{6}$
2. −7.05 − (−2.6)
3. Distance between $-\frac{3}{4}$ and 2.5

More practice: Workbook, Lesson 4.16.

LESSON SUMMARY

Organize, convert, compute with the sign rules, and check.

UNIT 4 REVIEW

Unit 4 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
4.1	Rational numbers include signed fractions and decimals. For negatives, farther from 0 is less.
4.2	Find a common denominator, then follow the integer sign rules.
4.3	Subtract fractions by adding the opposite.
4.4	Change signed mixed numbers to improper fractions, then use the sign rules. Remember $-3\,{}^{1}\!/_{2} = -3 - {}^{1}\!/_{2}$.
4.5	Divide numerator by denominator; the decimal terminates or repeats.
4.6	Convert to decimals and compare on a number line.
4.7	Line up decimals and apply the sign rules.
4.8	Subtract decimals by adding the opposite.
4.9	Use signed decimals for deposits and purchases and keep a running balance.
4.10	Use signed fractions and decimals for real measurements.
4.11	Regroup opposites and friendly numbers to compute mentally.
4.12	Convert to one form and combine step by step.
4.13	Distance between rational numbers is $\lvert a - b \rvert$.
4.14	Substitute in parentheses and follow the sign rules.
4.15	Estimate sign and size first, then check.
4.16	Organize, convert, compute with the sign rules, and check.

UNIT 5

Multiplying and Dividing Rational Numbers

Sign rules and why they work, signed fractions and decimals, order of operations, rates, and averages.

You can add and subtract any rational number. Now you will multiply and divide them, discover why the sign rules work, and use them to model repeated change, rates, and averages.

In this unit you will…

- Multiply and divide integers and explain the sign rules.
- Multiply and divide signed fractions and decimals.
- Use order of operations and the distributive property with negatives.
- Model repeated change and rates with signed numbers.
- Solve multi-step problems with all four operations.

MATH IN REAL LIFE

Weather forecasters, divers, pilots, and investors all use rates of change that can be negative.

UNIT VOCABULARY

product	Result of multiplication.
quotient	Result of division.
reciprocal	Number whose product with another is 1.
mean	Sum ÷ count.

Lessons in this unit

Lesson	Title
5.1	Multiplying Integers: Patterns
5.2	Multiplying Integers: Rules and Reasons
5.3	Dividing Integers

Lesson	Title
5.4	Negative Fractions and Signs of Quotients
5.5	Multiplying Signed Fractions
5.6	Dividing Signed Fractions
5.7	Multiplying and Dividing Signed Decimals
5.8	Rational Number Operations Practice
5.9	Order of Operations with Rational Numbers
5.10	The Distributive Property with Rational Numbers
5.11	Repeated Change: Rate × Time
5.12	Choosing the Operation
5.13	Average Change
5.14	Rate Problems with Negatives
5.15	Multi-Step Rational Number Problems
5.16	Rational Number Operations Problem Solving

UNIT 5 · LESSON 5.1 · 7.NS.2a

Lesson 5.1: Multiplying Integers: Patterns

I can use patterns to discover how to multiply positive and negative integers.

WORDS TO KNOW

product	The result of multiplication.

LET'S EXPLORE

A pool drains 3 inches of water each hour. What is the change in water level after 4 hours?

Draining means the level changes by −3 each hour. Four hours: $4 \times (-3) = -3 + (-3) + (-3) + (-3) = -12$. The level drops 12 inches.

Pattern	Product
$3 \times (-2)$	−6
$2 \times (-2)$	−4
$1 \times (-2)$	−2
$0 \times (-2)$	0
$-1 \times (-2)$	2
$-2 \times (-2)$	4

Look at the pattern: each time the first factor goes down by 1, the product goes up by 2. Continue past 0 and the products become positive. So a negative times a negative is positive.

KEY IDEA

Positive × negative = negative. Negative × negative = positive.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $5 \times (-6)$
2. $(-7) \times (-3)$
3. $(-9) \times 4$

More practice: Workbook, Lesson 5.1.

LESSON SUMMARY

Same signs → positive product. Different signs → negative product.

UNIT 5 · LESSON 5.2 · 7.NS.2a, 7.NS.2c

Lesson 5.2: Multiplying Integers: Rules and Reasons

I can multiply integers, including more than two factors, and explain the sign rules.

LET'S EXPLORE

What is (−2) × (−3) × (−5)? How can you tell the sign before multiplying?

Multiply the absolute values, then decide the sign. Each pair of negatives makes a positive. So an even number of negative factors gives a positive product, and an odd number gives a negative product.

WORKED EXAMPLE · THREE FACTORS

(−2)(−3)(−5)

1 Three negative factors: odd → negative.

2 2 × 3 × 5 = 30.

Answer: −30

WORKED EXAMPLE · USING THE DISTRIBUTIVE PROPERTY

Why is (−1)(−1) = 1?

1 (−1)(1 + (−1)) = (−1)(0) = 0.

2 So (−1)(1) + (−1)(−1) = 0, which means −1 + (−1)(−1) = 0.

3 (−1)(−1) must be 1.

Answer: The rules keep the properties true.

KEY IDEA

Count the negative factors: even → positive, odd → negative.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. (−4)(5)(−2)

2. $(-1)(-1)(-1)(-1)(-1)$

3. $(-3)^2$

More practice: Workbook, Lesson 5.2.

LESSON SUMMARY

Even number of negative factors → positive; odd → negative.

UNIT 5 · LESSON 5.3 · 7.NS.2b

Lesson 5.3: Dividing Integers

I can divide integers using the related multiplication fact.

WORDS TO KNOW

quotient The result of division.

LET'S EXPLORE

A diver descends 60 feet in 5 minutes at a steady rate. What is the change in depth per minute?

$-60 \div 5 = ?$ asks: $5 \times ? = -60$. Since $5 \times (-12) = -60$, the answer is -12 feet per minute.

Division	Because	Quotient
$-60 \div 5$	$5 \times (-12) = -60$	-12
$60 \div (-5)$	$-5 \times (-12) = 60$	-12
$-60 \div (-5)$	$-5 \times 12 = -60$	12

KEY IDEA

Division sign rules match multiplication: same signs → positive; different signs → negative. You can never divide by 0.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-42 \div 6$

2. $56 \div (-8)$

3. $-81 \div (-9)$

More practice: Workbook, Lesson 5.3.

LESSON SUMMARY

Divide absolute values; same signs → positive, different signs → negative.

UNIT 5 · LESSON 5.4 · 7.NS.2b

Lesson 5.4: Negative Fractions and Signs of Quotients

I can explain why $-(\frac{p}{q}) = \frac{-p}{q} = \frac{p}{-q}$.

LET'S EXPLORE

Are $-\frac{3}{4}$, $\frac{-3}{4}$, and $\frac{3}{-4}$ the same number?

Yes. A fraction is a division: $\frac{-3}{4} = -3 \div 4 = -0.75$ and $\frac{3}{-4} = 3 \div (-4) = -0.75$. Both equal $-\frac{3}{4}$.

WORKED EXAMPLE · WHERE THE SIGN GOES

$\frac{-12}{-3}$

1 $-12 \div (-3) = 4$.

Answer: 4 (two negatives make a positive)

KEY IDEA

$-(\frac{p}{q}) = \frac{-p}{q} = \frac{p}{-q}$. A fraction with one negative sign is negative.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Simplify $\frac{-18}{6}$
2. Simplify $\frac{20}{-5}$
3. Simplify $\frac{-7}{-14}$

More practice: Workbook, Lesson 5.4.

LESSON SUMMARY

One negative sign anywhere makes the fraction negative; two make it positive.

UNIT 5 · LESSON 5.5 · 7.NS.2a, 7.NS.2c

Lesson 5.5: Multiplying Signed Fractions

I can multiply positive and negative fractions and mixed numbers.

LET'S EXPLORE

A recipe needs $\frac{3}{4}$ cup of sugar, but you're making $\frac{2}{3}$ of the recipe. A water tank loses $\frac{1}{2}$ gallon per minute for $\frac{3}{4}$ minute. What are the products?

WORKED EXAMPLE · FRACTIONS

$\frac{2}{3} \times \frac{3}{4}$

1 Multiply numerators and denominators: $\frac{6}{12} = \frac{1}{2}$.

Answer: $\frac{1}{2}$ cup

WORKED EXAMPLE · WITH A NEGATIVE

$(-\frac{1}{2}) \times \frac{3}{4}$

1 Different signs → negative.

2 $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$.

Answer: $-\frac{3}{8}$ gallon

WORKED EXAMPLE · MIXED NUMBERS

$(-1\frac{1}{2}) \times (-2\frac{2}{3})$

1 $= \frac{3}{2} \times \frac{8}{3}$ (two negatives → positive).

2 $= \frac{24}{6} = 4$.

Answer: 4

KEY IDEA

Decide the sign, multiply the absolute values, and simplify. Change mixed numbers to improper fractions.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $(-\frac{2}{5}) \times \frac{3}{4}$
2. $(-\frac{5}{6}) \times (-\frac{3}{10})$

3. $(-2\,{}^{1}\!/\!_{4}) \times {}^{2}\!/\!_{3}$

More practice: Workbook, Lesson 5.5.

LESSON SUMMARY

Sign first; then multiply numerators and denominators.

UNIT 5 · LESSON 5.6 · 7.NS.2b, 7.NS.2c

Lesson 5.6: Dividing Signed Fractions

I can divide positive and negative fractions.

WORDS TO KNOW

reciprocal — Two numbers whose product is 1, like ${}^{2}\!/\!_{3}$ and ${}^{3}\!/\!_{2}$.

LET'S EXPLORE

A tank's level fell ${}^{3}\!/\!_{4}$ ft over ${}^{3}\!/\!_{8}$ hour. What is the change per hour?

WORKED EXAMPLE · DIVIDE

$(-{}^{3}\!/\!_{4}) \div {}^{3}\!/\!_{8}$

1 Different signs → negative.

2 ${}^{3}\!/\!_{4} \times {}^{8}\!/\!_{3} = {}^{24}\!/\!_{12} = 2$.

Answer: −2 ft per hour

WORKED EXAMPLE · TWO NEGATIVES

$(-{}^{5}\!/\!_{6}) \div (-{}^{5}\!/\!_{12})$

1 Positive. ${}^{5}\!/\!_{6} \times {}^{12}\!/\!_{5} = 2$.

Answer: 2

KEY IDEA

To divide, multiply by the reciprocal of the divisor. Sign rules are the same.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $(-{}^{2}\!/\!_{3}) \div {}^{4}\!/\!_{9}$
2. ${}^{7}\!/\!_{8} \div (-{}^{7}\!/\!_{16})$

3. $(-2\frac{1}{2}) \div (-\frac{3}{4})$

More practice: Workbook, Lesson 5.6.

LESSON SUMMARY

Multiply by the reciprocal; use the same sign rules.

UNIT 5 · LESSON 5.7 · 7.NS.2

Lesson 5.7: Multiplying and Dividing Signed Decimals

I can multiply and divide positive and negative decimals.

LET'S EXPLORE

A stock fell $1.25 per share. Ellie owns 8 shares. What is the total change? Over 4 days, what was the average daily change?

WORKED EXAMPLE · MULTIPLY

$8 \times (-1.25)$

1 $8 \times 1.25 = 10$. Different signs → negative.

Answer: −$10

WORKED EXAMPLE · DIVIDE

$-10 \div 4$

1 $10 \div 4 = 2.5$. Negative.

Answer: −$2.50 per day

KEY IDEA

Multiply or divide the decimals, then apply the sign rules.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $(-0.6) \times 4.5$
2. $(-7.2) \div (-0.9)$
3. $(-2.5) \times (-0.4)$

More practice: Workbook, Lesson 5.7.

LESSON SUMMARY

Compute with decimals, then apply sign rules.

UNIT 5 · LESSON 5.8 · 7.NS.2

Lesson 5.8: Rational Number Operations Practice

I can multiply and divide any rational numbers fluently.

LET'S EXPLORE

Which is greater: $(-\frac{3}{4})(-0.8)$ or $(-1.2) \div (-\frac{3}{2})$?

WORKED EXAMPLE · COMPARE

Both products/quotients

1 $(-\frac{3}{4})(-0.8) = 0.75 \times 0.8 = 0.6$.

2 $(-1.2) \div (-1.5) = 0.8$.

Answer: the second ($0.8 > 0.6$)

KEY IDEA

Convert to one form (all fractions or all decimals) when operations mix.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $(-0.5) \times \frac{4}{5}$
2. $\frac{9}{10} \div (-0.3)$
3. $(-1\frac{1}{4})(-0.8)$

More practice: Workbook, Lesson 5.8.

LESSON SUMMARY

Convert to one form, decide the sign, then compute.

UNIT 5 · LESSON 5.9 · 7.NS.3, 7.EE.3

Lesson 5.9: Order of Operations with Rational Numbers

I can use the order of operations with positive and negative numbers.

WORDS TO KNOW

order of operations	Parentheses, exponents, multiplication and division (left to right), addition and subtraction (left to right).

LET'S EXPLORE

Evaluate $-3 + 4 \times (-2)^2 \div 8$.

WORKED EXAMPLE · ORDER

$-3 + 4 \times (-2)^2 \div 8$

1. Exponent: $(-2)^2 = 4$.
2. Multiply/divide left to right: $4 \times 4 = 16$; $16 \div 8 = 2$.
3. Add: $-3 + 2 = -1$.

Answer: -1

WORKED EXAMPLE · PARENTHESES WITH NEGATIVES

$(-6 - 4) \div (-5) + 3$

1. $(-10) \div (-5) = 2$.
2. $2 + 3 = 5$.

Answer: 5

KEY IDEA

Same order of operations as always; keep track of every sign.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-8 + 3 \times (-2)$
2. $(-4)^2 - 10$
3. $12 \div (-3) \times (-2)$

More practice: Workbook, Lesson 5.9.

LESSON SUMMARY

Follow the order of operations and track signs at each step.

UNIT 5 · LESSON 5.10 · 7.NS.2a, 7.EE.1

Lesson 5.10: The Distributive Property with Rational Numbers

I can use the distributive property with negative numbers.

WORDS TO KNOW

distributive property $a(b + c) = ab + ac$

LET'S EXPLORE

How could you compute -4×2.75 in your head?

WORKED EXAMPLE · MENTAL MATH

-4×2.75

1 $2.75 = 3 - 0.25$.

2 $-4(3 - 0.25) = -12 + 1 = -11$.

Answer: -11

WORKED EXAMPLE · DISTRIBUTE A NEGATIVE

$-3(x - 5)$

1 $-3 \cdot x + (-3)(-5) = -3x + 15$.

Answer: $-3x + 15$

KEY IDEA

Distribute the sign too: $-a(b - c) = -ab + ac$.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $-5(2 + 0.4)$
2. $-2(y - 7)$
3. $-\frac{1}{2}(8 - 6)$

More practice: Workbook, Lesson 5.10.

LESSON SUMMARY

Distribute the factor (with its sign) to every term.

UNIT 5 · LESSON 5.11 · 7.NS.2a, 7.NS.3

Lesson 5.11: Repeated Change: Rate × Time

I can model repeated change with multiplication of signed numbers.

LET'S EXPLORE

A balloon descends 4.5 meters per minute. Where was it 3 minutes ago, compared to now? Where will it be 3 minutes from now?

Use (rate) × (time). Descending is a negative rate: −4.5 m/min. Future time is positive; past time is negative.

WORKED EXAMPLE · FUTURE

(−4.5)(3)

1 Different signs → negative.

Answer: −13.5 m (13.5 m lower)

WORKED EXAMPLE · PAST

(−4.5)(−3)

1 Same signs → positive.

Answer: +13.5 m (it was 13.5 m higher)

KEY IDEA

Negative × negative = positive makes sense: going back in time while descending means you were higher.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rate −2°F/hour; 5 hours later
2. Rate −2°F/hour; 5 hours ago
3. Rate +3 cm/day; 4 days ago

More practice: Workbook, Lesson 5.11.

LESSON SUMMARY

Rate × time; negative times negative means going back in time while decreasing.

UNIT 5 · LESSON 5.12 · 7.NS.3

Lesson 5.12: Choosing the Operation

I can decide which operation fits a situation with rational numbers.

LET'S EXPLORE

A hiker's elevation changed by −1,240 ft over 4 hours. Is the average change per hour found by multiplying or dividing?

Situation	Operation
combine changes	add
difference / change between	subtract
repeated equal change	multiply
equal share / average rate	divide

WORKED EXAMPLE · AVERAGE CHANGE

−1,240 ft over 4 hours

1 Equal share per hour → divide: −1,240 ÷ 4.

Answer: −310 ft per hour

KEY IDEA

Think about what is happening, not just key words.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Lost 3.5 lb a week for 6 weeks: total change?
2. From −8°F to 5°F: change?
3. −$84 over 7 days: average daily change?

More practice: Workbook, Lesson 5.12.

LESSON SUMMARY

Combine → add; compare → subtract; repeat → multiply; share or average → divide.

UNIT 5 · LESSON 5.13 · 7.NS.3

Lesson 5.13: Average Change

I can find the mean of positive and negative numbers.

WORDS TO KNOW

mean (average)	The sum of the values divided by how many values there are.

LET'S EXPLORE

Daily temperature changes: −4°, +2°, −7°, +1°, −2°. What was the average daily change?

WORKED EXAMPLE · MEAN

−4, 2, −7, 1, −2

1 Sum: −4 + 2 − 7 + 1 − 2 = −10.

2 Divide by 5: −10 ÷ 5 = −2.

Answer: −2° per day

KEY IDEA

Mean = sum ÷ count. A negative mean means a typical decrease.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Mean of −3, 5, −8, 2
2. Mean of −1.5, −2.5, 1
3. Golf scores −2, 0, −3, +1: mean

More practice: Workbook, Lesson 5.13.

LESSON SUMMARY

Mean = sum ÷ count; signs matter in the sum.

UNIT 5 · LESSON 5.14 · 7.NS.3, 7.RP.1

Lesson 5.14: Rate Problems with Negatives

I can solve rate problems that involve decreases.

LET'S EXPLORE

A cold front drops the temperature from 18°F to −6°F in 8 hours. What is the rate of change per hour?

WORKED EXAMPLE · RATE OF CHANGE

18 → −6 in 8 hours

1 Change: −6 − 18 = −24.

2 Rate: −24 ÷ 8 = −3.

Answer: −3°F per hour

WORKED EXAMPLE · PREDICT

At −3°F per hour, when will it reach −15°F from −6°F?

1 Change needed: −15 − (−6) = −9.

2 −9 ÷ (−3) = 3 hours.

Answer: 3 hours later

KEY IDEA

Rate of change = change ÷ time. Change = end − start.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A pool level drops from 60 in. to 48 in. in 4 days. Rate?

2. A balance goes from $50 to −$25 in 5 weeks. Rate?

3. Elevation 2,400 ft to 600 ft in 3 hours. Rate?

More practice: Workbook, Lesson 5.14.

LESSON SUMMARY

Rate of change = (end − start) ÷ time.

UNIT 5 · LESSON 5.15 · 7.EE.3, 7.NS.3

Lesson 5.15: Multi-Step Rational Number Problems

I can solve multi-step problems using all four operations with rational numbers.

LET'S EXPLORE

A business lost $1,250 per month for 3 months, then earned $2,400 per month for 2 months. What is the average monthly result?

WORKED EXAMPLE · ALL OPERATIONS

3 months at −$1,250, 2 months at +$2,400

1 Total: 3(−1,250) + 2(2,400) = −3,750 + 4,800 = 1,050.

2 Average: 1,050 ÷ 5 = 210.

Answer: +$210 per month

KEY IDEA

Organize the steps; estimate to check the sign and size.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4(−2.5) + 3(1.5)
2. $(-\frac{3}{4} + \frac{1}{2}) \times 8$
3. (−12 − 6) ÷ (−3) + 2

More practice: Workbook, Lesson 5.15.

LESSON SUMMARY

Break the problem into steps, use the right operation for each, and check.

UNIT 5 · LESSON 5.16 · 7.NS.1–3

Lesson 5.16: Rational Number Operations Problem Solving

I can use all four operations with rational numbers to solve problems.

LET'S EXPLORE

Design a 'temperature challenge': start at 5°F, apply four changes so that the final temperature is −7°F and the average change is −3°F.

WORKED EXAMPLE · WORK BACKWARD

Average −3 over 4 changes

1 Total change = 4 × (−3) = −12.

2 5 + (−12) = −7 ✓. Any 4 changes summing to −12 work, e.g., −5, +2, −6, −3.

Answer: −5, +2, −6, −3

KEY IDEA

Use multiplication and division to plan totals and averages.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. (−2.5)(−4) ÷ (−5)
2. $-\frac{2}{3} \div \frac{4}{9}$
3. Mean of −4.2, 1.8, −3

More practice: Workbook, Lesson 5.16.

LESSON SUMMARY

Plan totals with multiplication, find averages with division, and track signs.

UNIT 5 REVIEW

Unit 5 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
5.1	Same signs → positive product. Different signs → negative product.
5.2	Even number of negative factors → positive; odd → negative.
5.3	Divide absolute values; same signs → positive, different signs → negative.
5.4	One negative sign anywhere makes the fraction negative; two make it positive.
5.5	Sign first; then multiply numerators and denominators.
5.6	Multiply by the reciprocal; use the same sign rules.
5.7	Compute with decimals, then apply sign rules.
5.8	Convert to one form, decide the sign, then compute.
5.9	Follow the order of operations and track signs at each step.
5.10	Distribute the factor (with its sign) to every term.
5.11	Rate × time; negative times negative means going back in time while decreasing.
5.12	Combine → add; compare → subtract; repeat → multiply; share or average → divide.
5.13	Mean = sum ÷ count; signs matter in the sum.
5.14	Rate of change = (end − start) ÷ time.
5.15	Break the problem into steps, use the right operation for each, and check.
5.16	Plan totals with multiplication, find averages with division, and track signs.

UNIT 6

Expressions, Equations, and Inequalities

Writing and simplifying expressions, solving two-step equations, and solving and graphing inequalities.

Algebra lets you describe real situations with symbols and solve for unknowns. You will write and simplify expressions, solve equations like $3x + 5 = 20$ and $2(x - 4) = 10$, and solve and graph inequalities.

In this unit you will…

- Write and evaluate expressions with rational numbers.
- Combine like terms, expand, and factor.
- Solve equations of the forms $px + q = r$ and $p(x + q) = r$.
- Write, solve, and graph inequalities.
- Interpret solutions in real situations.

MATH IN REAL LIFE

Phone plans, taxi fares, savings goals, and grade averages can all be written as equations and inequalities.

UNIT VOCABULARY

variable	A letter for a number.
expression	No equals sign.
equation	Two expressions are equal.
inequality	Comparison with $<$, $>$, $\leq$, $\geq$.
like terms	Same variable part.

Lessons in this unit

Lesson	Title
6.1	Writing Expressions

Lesson	Title
6.2	Evaluating Expressions
6.3	Combining Like Terms
6.4	Expressions in Context
6.5	The Distributive Property: Expanding
6.6	Factoring Expressions
6.7	Simplifying with Rational Coefficients
6.8	Equivalent Expressions
6.9	One-Step Equations
6.10	Two-Step Equations
6.11	Equations with Parentheses
6.12	Equation Word Problems
6.13	Writing and Graphing Inequalities
6.14	Solving One-Step Inequalities
6.15	Two-Step Inequalities
6.16	Inequality Word Problems

UNIT 6 · LESSON 6.1 · 7.EE.2, 7.EE.4

Lesson 6.1: Writing Expressions

I can write algebraic expressions from words and real situations.

WORDS TO KNOW

variable	A letter that stands for a number.
expression	Numbers, variables, and operations without an equals sign, like 3x + 5.
coefficient	The number multiplied by a variable. In 3x, the coefficient is 3.
term	A part of an expression separated by + or −.

LET'S EXPLORE

A movie ticket costs $9 and popcorn costs $6. Write an expression for the cost of t tickets and one popcorn.

An expression uses a variable for an unknown or changing number. The cost is 9 for each ticket, so 9t, plus 6: 9t + 6.

Words	Expression
5 more than a number n	n + 5
3 less than twice a number	2n − 3
the product of −4 and x	−4x
a number divided by 6	n ÷ 6
half of the sum of a number and 8	$\frac{1}{2}(n + 8)$

WATCH OUT

"3 less than x" is x − 3, not 3 − x. The order matters for subtraction and division.

KEY IDEA

Read the words carefully. Use a variable for the unknown and write operations in the correct order.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 7 less than a number m
2. the quotient of a number and −3
3. A $25 bill split among k friends

More practice: Workbook, Lesson 6.1.

LESSON SUMMARY

Use a variable for the unknown and translate words in the correct order.

UNIT 6 · LESSON 6.2 · 7.EE.3

Lesson 6.2: Evaluating Expressions

I can evaluate expressions for given values, including negative and fractional values.

WORDS TO KNOW

evaluate	Substitute values for variables and compute.

LET'S EXPLORE

The expression 1.8C + 32 converts Celsius to Fahrenheit. What is −10°C in Fahrenheit?

WORKED EXAMPLE · SUBSTITUTE

1.8C + 32 when C = −10

1 1.8(−10) + 32 = −18 + 32.

Answer: 14°F

WORKED EXAMPLE · TWO VARIABLES

3a − 2b when a = −4, b = 1.5

1 3(−4) − 2(1.5) = −12 − 3.

Answer: −15

KEY IDEA

Substitute with parentheses, then follow the order of operations.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $4x - 7$ when $x = -2$
2. $\frac{1}{2}n + 3$ when $n = -8$
3. $a^2 - b$ when $a = -3$, $b = 4$

More practice: Workbook, Lesson 6.2.

LESSON SUMMARY

Substitute in parentheses and follow the order of operations.

UNIT 6 · LESSON 6.3 · 7.EE.1

Lesson 6.3: Combining Like Terms

I can simplify expressions by combining like terms, including rational coefficients.

WORDS TO KNOW

like terms	Terms with the same variable part, like 3x and −5x.
constant	A term without a variable.

LET'S EXPLORE

Simplify 4x + 7 − 6x + 2.

Like terms have the same variable part. Add their coefficients. Constants combine with constants.

WORKED EXAMPLE · COMBINE

4x + 7 − 6x + 2

1 x-terms: 4x − 6x = −2x.

2 Constants: 7 + 2 = 9.

Answer: −2x + 9

WORKED EXAMPLE · RATIONAL COEFFICIENTS

$\frac{1}{2}a - 0.75a + 3$

1 $\frac{1}{2} - 0.75 = 0.5 - 0.75 = -0.25$.

Answer: −0.25a + 3

KEY IDEA

Only like terms combine. 3x + 2 is already simplified.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 5y − 8y + 3

2. −2m + 6 + 7m − 10

3. 1.5x − 2 + 0.5x

More practice: Workbook, Lesson 6.3.

LESSON SUMMARY

Add the coefficients of like terms; combine constants.

UNIT 6 · LESSON 6.4 · 7.EE.2

Lesson 6.4: Expressions in Context

I can rewrite an expression to show what it means in a situation.

LET'S EXPLORE

A store raises a price p by 5%. Two students write p + 0.05p and 1.05p. Are both right?

Both are right! p + 0.05p shows "the price plus 5% of the price." 1.05p shows "105% of the price." Rewriting an expression can reveal a new way of seeing the problem.

WORKED EXAMPLE · DISCOUNT

A 20% discount on price p

1 p − 0.2p = 0.8p.

2 0.8p means you pay 80% of the price.

Answer: 0.8p

WORKED EXAMPLE · PERIMETER

A rectangle has length L and width 3. Perimeter?

1 L + 3 + L + 3 = 2L + 6 = 2(L + 3).

Answer: 2(L + 3): twice the sum of length and width

KEY IDEA

Equivalent expressions can show different meanings of the same quantity.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A 15% tip on bill b, two ways

2. A 30% markdown on price m

3. Equal sides: s + s + s

More practice: Workbook, Lesson 6.4.

LESSON SUMMARY

Rewriting expressions reveals different meanings of the same quantity.

UNIT 6 · LESSON 6.5 · 7.EE.1

Lesson 6.5: The Distributive Property: Expanding

I can expand expressions using the distributive property.

WORDS TO KNOW

expand	Rewrite a product as a sum: $3(x + 4) = 3x + 12$.

LET'S EXPLORE

Expand $-2(3x - 5)$.

WORKED EXAMPLE · EXPAND

$-2(3x - 5)$

1 $-2 \cdot 3x = -6x$.

2 $-2 \cdot (-5) = 10$.

Answer: $-6x + 10$

WORKED EXAMPLE · FRACTIONS

$\frac{3}{4}(8y + 12)$

1 $\frac{3}{4} \cdot 8y = 6y$; $\frac{3}{4} \cdot 12 = 9$.

Answer: $6y + 9$

WORKED EXAMPLE · EXPAND AND COMBINE

$4(x + 2) - 3(x - 1)$

1 $4x + 8 - 3x + 3$.

Answer: $x + 11$

KEY IDEA

Multiply the outside factor (with its sign) by every term inside.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $5(2a - 3)$
2. $-(4m + 1)$
3. $-\frac{1}{2}(6n - 10)$

More practice: Workbook, Lesson 6.5.

LESSON SUMMARY

Multiply the factor by every term inside, keeping signs.

UNIT 6 · LESSON 6.6 · 7.EE.1

Lesson 6.6: Factoring Expressions

I can factor an expression by finding a common factor.

WORDS TO KNOW

factor (an expression)	Rewrite a sum as a product: $6x + 9 = 3(2x + 3)$.
greatest common factor (GCF)	The largest number that divides every term.

LET'S EXPLORE

A rectangle has area $6x + 9$. One side is 3. What is the other side?

WORKED EXAMPLE · FACTOR OUT THE GCF

$6x + 9$

1 GCF of 6 and 9 is 3.

2 $6x = 3 \cdot 2x$; $9 = 3 \cdot 3$.

Answer: $3(2x + 3)$

WORKED EXAMPLE · NEGATIVE FACTOR

$-4y + 12$

1 Factor -4: $-4(y - 3)$. Check: $-4y + 12$ ✓.

Answer: $-4(y - 3)$

KEY IDEA

Factoring undoes the distributive property. Always check by expanding.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $10a + 15$

2. $8m - 12$

3. $-7n - 21$

More practice: Workbook, Lesson 6.6.

LESSON SUMMARY

Find the GCF, write it outside, and divide each term by it.

UNIT 6 · LESSON 6.7 · 7.EE.1

Lesson 6.7: Simplifying with Rational Coefficients

I can simplify expressions with fraction and decimal coefficients.

LET'S EXPLORE

Simplify $\frac{1}{2}(4x - 6) - \frac{1}{3}(3x + 9)$.

WORKED EXAMPLE · EXPAND THEN COMBINE

$\frac{1}{2}(4x - 6) - \frac{1}{3}(3x + 9)$

1 $2x - 3 - x - 3$.

Answer: $x - 6$

WORKED EXAMPLE · DECIMALS

$0.4(5n + 10) - 1.5n$

1 $2n + 4 - 1.5n$.

Answer: $0.5n + 4$

KEY IDEA

Distribute carefully, then combine like terms.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{1}{4}(8a + 4) + a$

2. $0.2(10y - 5) - y$

3. $\frac{2}{5}(10 - 5m) + 3m$

More practice: Workbook, Lesson 6.7.

LESSON SUMMARY

Distribute rational factors, then combine like terms.

UNIT 6 · LESSON 6.8 · 7.EE.1, 7.EE.2

Lesson 6.8: Equivalent Expressions

I can tell whether two expressions are equivalent and justify my answer.

WORDS TO KNOW

equivalent expressions	Expressions that are equal for every value of the variable.

LET'S EXPLORE

Are 2(x + 3) + x and 3x + 6 equivalent? What about 3(x + 2) and 3x + 2?

WORKED EXAMPLE · SIMPLIFY BOTH

2(x + 3) + x vs 3x + 6

1 2x + 6 + x = 3x + 6.

2 Same.

Answer: Equivalent

WORKED EXAMPLE · TEST A VALUE

3(x + 2) vs 3x + 2

1 x = 1: 9 vs 5.

2 Different for x = 1.

Answer: Not equivalent

KEY IDEA

Simplify both to prove equivalence. One counterexample proves they are not equivalent.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4(x − 1) and 4x − 4

2. 5 − 2x and 2x − 5

3. $x + x + x + 2$ and $3(x + 1) - 1$

More practice: Workbook, Lesson 6.8.

LESSON SUMMARY

Simplify both to show equivalence; a counterexample shows they're not.

UNIT 6 · LESSON 6.9 · 7.EE.4a

Lesson 6.9: One-Step Equations

I can solve one-step equations with rational numbers.

WORDS TO KNOW

equation	A statement that two expressions are equal.
solution	A value that makes an equation true.
inverse operations	Operations that undo each other.

LET'S EXPLORE

$x - 4.5 = -2$. What number is x?

Use inverse operations to get the variable alone. Do the same thing to both sides to keep the equation balanced.

WORKED EXAMPLE · ADD

$x - 4.5 = -2$

1 Add 4.5 to both sides: $x = -2 + 4.5$.

Answer: $x = 2.5$

WORKED EXAMPLE · DIVIDE

$-3y = 18$

1 Divide both sides by -3.

Answer: $y = -6$

WORKED EXAMPLE · MULTIPLY

$\frac{n}{4} = -5$

1 Multiply both sides by 4.

Answer: $n = -20$

KEY IDEA

Undo the operation on the variable. Check by substituting.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $x + 7 = -3$
2. $-5m = -35$
3. $\frac{2}{3}k = 8$

More practice: Workbook, Lesson 6.9.

LESSON SUMMARY

Use inverse operations on both sides; check by substituting.

UNIT 6 · LESSON 6.10 · 7.EE.4a

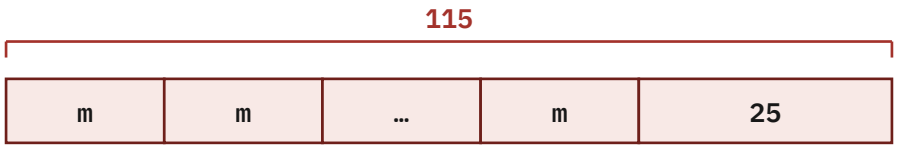
Lesson 6.10: Two-Step Equations

I can solve equations of the form $px + q = r$.

LET'S EXPLORE

A gym charges a $25 sign-up fee plus $15 per month. Ellie paid $115. How many months?

$15m + 25 = 115$. Undo the operations in reverse order: subtract 25 first, then divide by 15.



WORKED EXAMPLE · SOLVE

$15m + 25 = 115$

1 Subtract 25: $15m = 90$.

2 Divide by 15: $m = 6$.

3 Check: $15(6) + 25 = 115$ ✓.

Answer: 6 months

WORKED EXAMPLE · NEGATIVE COEFFICIENT

$-4x + 7 = 31$

1 $-4x = 24$.

2 $x = -6$.

Answer: $x = -6$

KEY IDEA

Undo addition/subtraction first, then multiplication/division.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $3x + 5 = 20$
2. $-2x + 9 = -7$
3. $5x - 12 = 13$

More practice: Workbook, Lesson 6.10.

LESSON SUMMARY

Undo addition or subtraction first, then multiplication or division.

UNIT 6 · LESSON 6.11 · 7.EE.4a

Lesson 6.11: Equations with Parentheses

I can solve equations of the form $p(x + q) = r$.

LET'S EXPLORE

Three friends each buy a $6 ticket plus the same snack. They spend $33 total. How much was each snack?

$3(s + 6) = 33$. You can divide first or distribute first.

WORKED EXAMPLE · DIVIDE FIRST

$3(s + 6) = 33$

1 Divide by 3: $s + 6 = 11$.

2 Subtract 6: $s = 5$.

Answer: $5

WORKED EXAMPLE · DISTRIBUTE FIRST

$3(s + 6) = 33$

1 $3s + 18 = 33$.

2 $3s = 15$; $s = 5$.

Answer: $5

KEY IDEA

For $p(x + q) = r$: divide by p first, or distribute and solve. Both work.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $2(x - 4) = 10$
2. $-5(y + 1) = 20$
3. $\frac{1}{2}(n + 6) = -3$

More practice: Workbook, Lesson 6.11.

LESSON SUMMARY

Divide first or distribute first; both give the same solution.

UNIT 6 · LESSON 6.12 · 7.EE.4a

Lesson 6.12: Equation Word Problems

I can write and solve equations for word problems and compare to an arithmetic solution.

LET'S EXPLORE

The perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?

WORKED EXAMPLE · ALGEBRAIC

$2(6) + 2w = 54$

1 $12 + 2w = 54$.

2 $2w = 42$; $w = 21$.

Answer: 21 cm

WORKED EXAMPLE · ARITHMETIC

Same problem without a variable

1 $54 - 12 = 42$ for both widths.

2 $42 \div 2 = 21$.

Answer: 21 cm (same steps!)

KEY IDEA

Equation steps match arithmetic steps. Writing the equation helps organize harder problems.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A plumber charges $60 plus $45 per hour. The bill is $240. Hours?
2. Lillie has $18 and saves $7.50 a week. When will she have $63?
3. The sum of three consecutive integers is 57. The smallest?

More practice: Workbook, Lesson 6.12.

LESSON SUMMARY

Define the variable, write the equation, solve, and check in the story.

UNIT 6 · LESSON 6.13 · 7.EE.4b

Lesson 6.13: Writing and Graphing Inequalities

I can write inequalities for situations and graph them on a number line.

WORDS TO KNOW

inequality	A statement that compares with <, >, ≤, or ≥.
solution set	All values that make an inequality true.

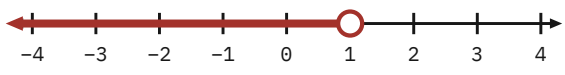
LET'S EXPLORE

You must be at least 48 inches tall to ride a roller coaster. Write and graph an inequality.

"At least 48" means 48 or more: h ≥ 48. A closed circle means the number is included; an open circle means it is not.



h ≥ 48 (closed circle)



x < 1 (open circle)

Words	Symbol
at least, no less than	≥
at most, no more than	≤
more than, greater than	>
fewer than, less than	<

KEY IDEA

≤ and ≥ use closed circles; < and > use open circles. The arrow points toward the solutions.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. No more than 30 students: write it
2. Temperature below −5
3. Graph $x \geq -2$: open or closed circle?

More practice: Workbook, Lesson 6.13.

LESSON SUMMARY

Translate words to <, >, ≤, ≥; graph with open or closed circles.

UNIT 6 · LESSON 6.14 · 7.EE.4b

Lesson 6.14: Solving One-Step Inequalities

I can solve one-step inequalities, including multiplying or dividing by a negative.

LET'S EXPLORE

Solve $-3x < 12$. Is $x = -5$ a solution? Is $x = 0$?

Solve inequalities like equations, with one big difference: when you multiply or divide both sides by a negative number, reverse the inequality symbol.

WORKED EXAMPLE · WHY REVERSE?

$2 < 5$. Multiply both sides by -1.

1 -2 and -5: now $-2 > -5$.

2 Multiplying by a negative flips the order on the number line.

Answer: the symbol must flip

WORKED EXAMPLE · SOLVE

$-3x < 12$

1 Divide by -3 and flip: $x > -4$.

2 Check $x = 0$: $-3(0) = 0 < 12$ ✓. Check $x = -5$: $15 < 12$ ✗.

Answer: $x > -4$

KEY IDEA

Flip the inequality symbol when multiplying or dividing by a negative number.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $x + 6 \geq 2$
2. $-5y \leq 30$
3. $\frac{n}{-2} > 3$

More practice: Workbook, Lesson 6.14.

LESSON SUMMARY

Solve like equations, but flip the symbol when multiplying or dividing by a negative.

UNIT 6 · LESSON 6.15 · 7.EE.4b

Lesson 6.15: Two-Step Inequalities

I can solve inequalities of the form $px + q > r$ and graph the solutions.

LET'S EXPLORE

Lillie has $20 and earns $8 per hour. How many hours must she work to have at least $100?

WORKED EXAMPLE · SOLVE

$8h + 20 \geq 100$

1 Subtract 20: $8h \geq 80$.

2 Divide by 8: $h \geq 10$.

Answer: at least 10 hours

WORKED EXAMPLE · NEGATIVE COEFFICIENT

$-2x + 5 > 11$

1 $-2x > 6$.

2 Divide by -2 and flip: $x < -3$.

Answer: $x < -3$

KEY IDEA

Use the same steps as two-step equations; flip when dividing by a negative.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $3x - 4 \leq 11$
2. $-4y + 2 > 14$
3. $\frac{1}{2}n + 3 \geq -1$

More practice: Workbook, Lesson 6.15.

LESSON SUMMARY

Two-step inequalities: undo +/− then ×/÷; flip for negatives.

UNIT 6 · LESSON 6.16 · 7.EE.4b

Lesson 6.16: Inequality Word Problems

I can write, solve, and interpret inequalities for real situations.

LET'S EXPLORE

A school club has $150 for a pizza party. Each pizza costs $12.50 and delivery is $10. How many pizzas can they buy?

WORKED EXAMPLE · WRITE, SOLVE, INTERPRET

$12.5p + 10 \leq 150$

1 $12.5p \leq 140$.

2 $p \leq 11.2$.

Answer: at most 11 pizzas (you can't buy part of a pizza)

KEY IDEA

Interpret the solution in context: whole numbers only? Must be positive?

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A phone plan costs $30 + $5 per GB. Budget $60. Max GB?
2. Ellie needs at least 85 points. She has 62. Each task is worth 4 points. How many tasks?
3. An elevator holds at most 2,000 lb. A 180-lb operator rides with boxes of 65 lb. Max boxes?

More practice: Workbook, Lesson 6.16.

LESSON SUMMARY

Write, solve, then interpret the solution in context.

UNIT 6 REVIEW

Unit 6 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
6.1	Use a variable for the unknown and translate words in the correct order.
6.2	Substitute in parentheses and follow the order of operations.
6.3	Add the coefficients of like terms; combine constants.
6.4	Rewriting expressions reveals different meanings of the same quantity.
6.5	Multiply the factor by every term inside, keeping signs.
6.6	Find the GCF, write it outside, and divide each term by it.
6.7	Distribute rational factors, then combine like terms.
6.8	Simplify both to show equivalence; a counterexample shows they're not.
6.9	Use inverse operations on both sides; check by substituting.
6.10	Undo addition or subtraction first, then multiplication or division.
6.11	Divide first or distribute first; both give the same solution.
6.12	Define the variable, write the equation, solve, and check in the story.
6.13	Translate words to $<$, $>$, $\leq$, $\geq$; graph with open or closed circles.
6.14	Solve like equations, but flip the symbol when multiplying or dividing by a negative.
6.15	Two-step inequalities: undo +/− then ×/÷; flip for negatives.
6.16	Write, solve, then interpret the solution in context.

UNIT 7

Statistics and Probability

Sampling, comparing populations, probability models, and compound events.

How can a survey of 100 people tell us about a whole city? How likely is it to roll doubles? In this unit you will learn to sample fairly, compare groups with data, and measure chance with probability.

In this unit you will…

- Use random samples to make inferences about populations.
- Compare two data sets using center and variability.
- Find theoretical and experimental probabilities.
- Build probability models.
- Find probabilities of compound events with lists, tables, trees, and simulations.

MATH IN REAL LIFE

Election polls, weather forecasts, quality control in factories, and sports statistics all use sampling and probability.

UNIT VOCABULARY

population	The whole group.
sample	Part of the population.
MAD	Average distance from the mean.
probability	A number 0–1 for likelihood.
sample space	All possible outcomes.

Lessons in this unit

Lesson	Title
7.1	Populations and Samples

Lesson	Title
7.2	Random Samples and Bias
7.3	Making Inferences from Samples
7.4	Variability in Samples
7.5	Measures of Center and Variability
7.6	Comparing Dot Plots
7.7	Comparing Box Plots
7.8	Comparing Populations
7.9	Probability and Likelihood
7.10	Theoretical Probability
7.11	Experimental Probability
7.12	Probability Models
7.13	Sample Spaces for Compound Events
7.14	Tree Diagrams
7.15	Probability of Compound Events
7.16	Simulations

UNIT 7 · LESSON 7.1 · 7.SP.1

Lesson 7.1: Populations and Samples

I can tell the difference between a population and a sample.

WORDS TO KNOW

population	The entire group you want to know about.
sample	A part of the population that is actually studied.
census	Data collected from every member of a population.

LET'S EXPLORE

You want to know the favorite lunch of every 7th grader in your state. Could you ask all of them? What could you do instead?

A population is the whole group you are curious about. Studying everyone is usually too slow or expensive, so statisticians study a sample, a smaller part of the population, and use it to learn about the whole.

WORKED EXAMPLE · IDENTIFY

A library asks 80 of its 2,000 members how often they visit.

1. Population: all 2,000 members.
2. Sample: the 80 members surveyed.

Answer: Population: 2,000 members; sample: 80 members

KEY IDEA

A sample is useful only when it represents the population well.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A chef tastes one spoonful of soup. Population? Sample?
2. A factory tests 50 of 10,000 light bulbs. Population? Sample?
3. When is a census a better choice than a sample?

More practice: Workbook, Lesson 7.1.

LESSON SUMMARY

Study a sample to learn about a population.

UNIT 7 · LESSON 7.2 · 7.SP.1

Lesson 7.2: Random Samples and Bias

I can tell whether a sample is random and representative, or biased.

WORDS TO KNOW

random sample	A sample in which every member of the population has an equal chance of being chosen.
biased sample	A sample that over- or under-represents part of the population.

LET'S EXPLORE

To find the town's favorite sport, Ellie surveys people leaving a soccer game. Is that fair?

Ellie's sample is biased: people at a soccer game probably like soccer more than the town as a whole. A random sample, such as drawing names from the whole town's list, gives everyone the same chance and tends to represent the population.

Sampling method	Random?	Why
Draw 50 names from a hat with every student's name	Yes	Everyone has an equal chance
Ask the first 50 students who arrive	No	Early arrivers may differ
Ask volunteers to answer online	No	Only people who care respond
Number students; use a random-number generator	Yes	Equal chance

WATCH OUT

A large sample is not automatically fair. 1,000 biased answers are still biased.

KEY IDEA

Random samples tend to be representative; samples that leave out groups are biased.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Survey the basketball team about favorite sports.

2. Roll a die to pick one student from each table group of 6.

3. Ask your friends whom they will vote for.

More practice: Workbook, Lesson 7.2.

LESSON SUMMARY

Random samples give everyone an equal chance; biased samples leave groups out.

UNIT 7 · LESSON 7.3 · 7.SP.2

Lesson 7.3: Making Inferences from Samples

I can use data from a random sample to estimate something about a population.

WORDS TO KNOW

inference	A conclusion about a population based on sample data.

LET'S EXPLORE

In a random sample of 50 students, 18 prefer pizza for lunch. The school has 600 students. About how many prefer pizza?

WORKED EXAMPLE · USE A PROPORTION

18 of 50 prefer pizza; 600 students in all

1 Sample rate: $^{18}/_{50}$ = 36%.

2 36% of 600 = 0.36 × 600 = 216.

Answer: about 216 students

WORKED EXAMPLE · DEFECTS

4 of 200 sampled phones are defective. Estimate for 15,000 phones.

1 $^{4}/_{200}$ = 2%.

2 0.02 × 15,000 = 300.

Answer: about 300 phones

KEY IDEA

Sample rate × population size = estimate. Estimates are approximate, not exact.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 12 of 40 sampled students walk to school. School: 500. Estimate walkers.
2. 3 of 60 sampled apples are bruised. Shipment: 2,400. Estimate bruised.
3. 45 of 75 sampled voters favor a plan. Percent?

More practice: Workbook, Lesson 7.3.

LESSON SUMMARY

Multiply the sample rate by the population size.

UNIT 7 · LESSON 7.4 · 7.SP.2

Lesson 7.4: Variability in Samples

I can use several samples to judge how much an estimate might vary.

WORDS TO KNOW

sampling variability	The natural differences between results from different samples.

LET'S EXPLORE

Five students each scoop 20 candies from the same jar and count the red ones. Will they all get the same number?

No! Different random samples give different results. That is sampling variability. Looking at several samples helps you see how much an estimate might change.

Sample	1	2	3	4	5
Red (of 20)	6	8	5	7	9
Percent red	30%	40%	25%	35%	45%

WORKED EXAMPLE · COMBINE SAMPLES

Estimate the percent of red candies in the jar.

1 Mean red per sample: (6 + 8 + 5 + 7 + 9) ÷ 5 = 7.

2 7 ÷ 20 = 35%.

Answer: about 35%

WORKED EXAMPLE · USE IT

The jar holds 500 candies. About how many are red?

1 35% of 500.

Answer: about 175

KEY IDEA

Estimates from samples vary. Larger samples and more samples give more reliable estimates.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. What is the range of the five percents above?
2. Which gives a more reliable estimate: one sample of 20 or five samples of 20?
3. A sixth sample has 7 red. Does it change the estimate much?

More practice: Workbook, Lesson 7.4.

LESSON SUMMARY

Samples vary; combine several or use larger samples for better estimates.

UNIT 7 · LESSON 7.5 · 7.SP.4

Lesson 7.5: Measures of Center and Variability

I can find the mean, median, mean absolute deviation, range, and interquartile range of a data set.

WORDS TO KNOW

mean	The sum of the values divided by the number of values.
median	The middle value when data are in order.
mean absolute deviation (MAD)	The average distance of the values from the mean.
interquartile range (IQR)	Q3 – Q1, the spread of the middle half of the data.

LET'S EXPLORE

Two quizzes have scores 2, 4, 6, 8, 10 and 12, 15, 15, 18, 20. Which set is more spread out?

Center (mean or median) describes a typical value. Variability (MAD, range, or IQR) describes how spread out the values are.

WORKED EXAMPLE · MEAN AND MAD

2, 4, 6, 8, 10

1 Mean: 30 ÷ 5 = 6.

2 Distances from 6: 4, 2, 0, 2, 4.

3 MAD: 12 ÷ 5 = 2.4.

Answer: mean 6, MAD 2.4

WORKED EXAMPLE · MEDIAN AND IQR

52, 54, 55, 56, 58, 59, 60, 61, 62, 64, 66

1 Median (6th value): 59.

2 Q1 = median of lower half = 55; Q3 = 62.

3 IQR = 62 − 55 = 7.

Answer: median 59, IQR 7

KEY IDEA

Use mean and MAD for symmetric data; median and IQR when there are outliers or skew.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Mean and MAD of 12, 15, 15, 18, 20

2. Median of 7, 3, 9, 4, 10, 6

3. IQR of 1, 3, 4, 6, 7, 9, 12, 15

More practice: Workbook, Lesson 7.5.

LESSON SUMMARY

Center: mean or median. Variability: MAD, range, or IQR.

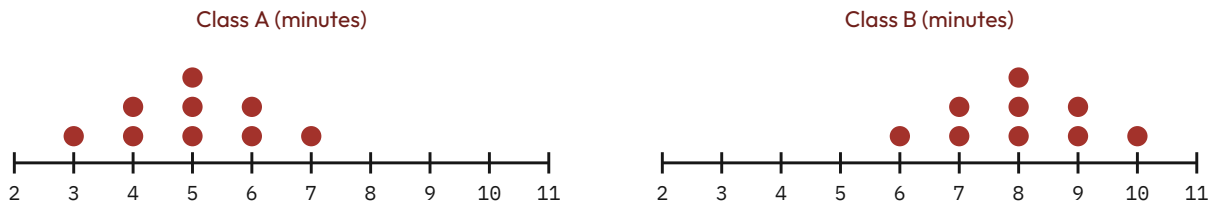
UNIT 7 · LESSON 7.6 · 7.SP.3

Lesson 7.6: Comparing Dot Plots

I can compare two data sets using their centers and variability.

LET'S EXPLORE

Two classes record minutes of homework. Class B seems to do more. How much more, compared to the spread?



WORKED EXAMPLE · COMPARE

Class A vs Class B

1 Means: A = 5, B = 8; difference 3.

2 MAD of each set: ≈ 0.89.

3 $3 \div {}^{8}/_{9} \approx 3.38$.

Answer: The centers differ by about 3.4 MADs: a big difference

KEY IDEA

Divide the difference in centers by the variability (MAD). A result above about 2 means the groups are clearly different.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Do the dot plots overlap?
2. Which class has more variability?
3. Is a typical B student doing more homework?

More practice: Workbook, Lesson 7.6.

LESSON SUMMARY

Compare centers relative to the spread.

UNIT 7 · LESSON 7.7 · 7.SP.3, 7.SP.4

Lesson 7.7: Comparing Box Plots

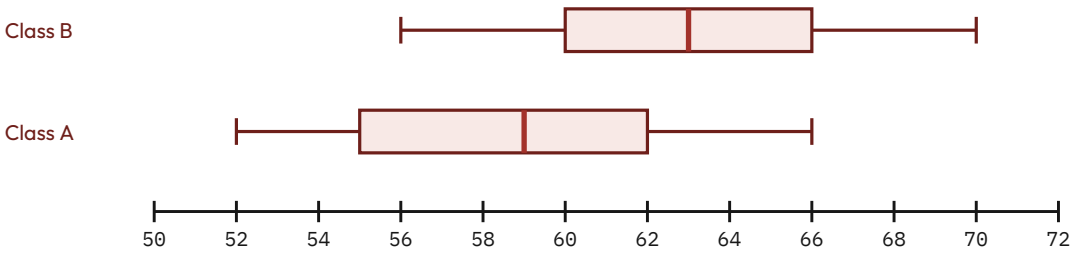
I can compare two data sets displayed in box plots.

WORDS TO KNOW

box plot — A display of the five-number summary: minimum, Q1, median, Q3, maximum.

LET'S EXPLORE

Two 7th-grade classes measured heights in inches. Which class tends to be taller?



WORKED EXAMPLE · READ THE PLOTS

Class A vs Class B

1 Medians: A = 59, B = 63.

2 IQRs: A = 7, B = 6.

Answer: Class B's median is 4 in. greater; the spreads are similar

KEY IDEA

Compare medians for center and IQRs for spread.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Range of Class A
2. Range of Class B
3. About what percent of Class B is taller than Class A's median?

More practice: Workbook, Lesson 7.7.

LESSON SUMMARY

Compare medians for center and IQRs for spread.

UNIT 7 · LESSON 7.8 · 7.SP.3, 7.SP.4

Lesson 7.8: Comparing Populations

I can use random samples to compare two populations.

LET'S EXPLORE

Random samples of 7th and 4th graders report weekly reading hours. Do 7th graders read more?

Group	Sample data (hours)	Mean	MAD
Grade 7	4, 5, 6, 6, 7, 8	6	1
Grade 4	2, 3, 3, 4, 4, 5	3.5	≈ 0.83

WORKED EXAMPLE · COMPARE

Difference in means vs MAD

1 Difference: 6 − 3.5 = 2.5.

2 Divide by the larger MAD: 2.5 ÷ 1 = 2.5.

Answer: about 2.5 MADs: 7th graders likely read more

KEY IDEA

If the samples are random and the difference is more than about 2 MADs, the populations probably differ.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. If the difference were 0.5 MAD, what would you conclude?
2. Why do the samples need to be random?
3. Name one other way to compare these groups.

More practice: Workbook, Lesson 7.8.

LESSON SUMMARY

Random samples + difference in means ÷ MAD → is there a real difference?

UNIT 7 · LESSON 7.9 · 7.SP.5

Lesson 7.9: Probability and Likelihood

I can describe the likelihood of an event with a number from 0 to 1.

WORDS TO KNOW

probability	A number from 0 to 1 that tells how likely an event is.
event	A set of outcomes.
outcome	One possible result of an experiment.

LET'S EXPLORE

Is it likely, unlikely, certain, or impossible that the sun rises tomorrow? That you roll a 7 on a number cube?

Probability measures likelihood on a scale from 0 (impossible) to 1 (certain). A probability of $\frac{1}{2}$ means equally likely to happen or not.

0	near 0	$\frac{1}{2}$	near 1	1
impossible	unlikely	equally likely	likely	certain

KEY IDEA

Probabilities near 0 are unlikely; near 1 are likely.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rolling a 7 on a standard number cube
2. A coin landing heads
3. Drawing a red marble from a bag of 9 red, 1 blue

More practice: Workbook, Lesson 7.9.

LESSON SUMMARY

Probability runs from 0 (impossible) to 1 (certain).

UNIT 7 · LESSON 7.10 · 7.SP.7a

Lesson 7.10: Theoretical Probability

I can find the theoretical probability of an event.

WORDS TO KNOW

theoretical probability	Number of favorable outcomes ÷ number of equally likely outcomes.
sample space	The set of all possible outcomes.

LET'S EXPLORE

A bag holds 3 red, 5 blue, and 2 green marbles. What is the probability of drawing blue?

When all outcomes are equally likely, P(event) = favorable outcomes ÷ total outcomes.

WORKED EXAMPLE · MARBLES

3 red, 5 blue, 2 green

1 Total: 10. Blue: 5.

Answer: P(blue) = $^{1}/_{2}$

WORKED EXAMPLE · NUMBER CUBE

P(number greater than 4)

1 Favorable: 5, 6. Total: 6.

Answer: P = $^{1}/_{3}$

WORKED EXAMPLE · COMPLEMENT

P(not green)

1 P(green) = $^{1}/_{5}$, so P(not green) = 1 − $^{1}/_{5}$.

Answer: $^{4}/_{5}$

KEY IDEA

P(not A) = 1 − P(A).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. P(red)
2. P(even) on a number cube

3. P(not 6) on a number cube

More practice: Workbook, Lesson 7.10.

LESSON SUMMARY

P(event) = favorable ÷ total equally likely outcomes.

UNIT 7 · LESSON 7.11 · 7.SP.6, 7.SP.7b

Lesson 7.11: Experimental Probability

I can find experimental probability and compare it to theoretical probability.

WORDS TO KNOW

experimental probability	Number of times an event happens ÷ number of trials.
trial	One repetition of an experiment.

LET'S EXPLORE

Ellie rolled a number cube 60 times. She got a 6 fifteen times. Is the cube fair?

Result	1	2	3	4	5	6
Frequency	7	9	11	8	10	15

WORKED EXAMPLE · EXPERIMENTAL

P(6) from the data

1 6 occurred 15 times in 60 rolls.

Answer: $\frac{1}{4}$ = 25%

WORKED EXAMPLE · COMPARE

Theoretical P(6)

1 $\frac{1}{6} \approx 17\%$.

2 25% is higher, but 60 rolls is a small number of trials.

Answer: Roll many more times before deciding the cube is unfair

KEY IDEA

As the number of trials grows, experimental probability tends to get closer to theoretical probability.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Experimental P(3)
2. Experimental P(odd)
3. Predict how many 6s in 600 rolls of a fair cube

More practice: Workbook, Lesson 7.11.

LESSON SUMMARY

Experimental: occurrences ÷ trials. It approaches theoretical with more trials.

UNIT 7 · LESSON 7.12 · 7.SP.7a, 7.SP.7b

Lesson 7.12: Probability Models

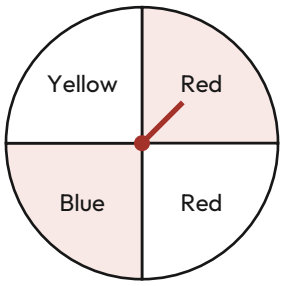
I can build uniform and non-uniform probability models.

WORDS TO KNOW

probability model	A list of outcomes and their probabilities, which add to 1.
uniform model	All outcomes are equally likely.

LET'S EXPLORE

A spinner has 4 equal sections: red, red, blue, yellow. Are all colors equally likely?



WORKED EXAMPLE · NON-UNIFORM MODEL

Spinner above

1 P(red) = $^{2}/_{4}$ = $^{1}/_{2}$, P(blue) = $^{1}/_{4}$, P(yellow) = $^{1}/_{4}$.

2 Sum: $^{1}/_{2} + {}^{1}/_{4} + {}^{1}/_{4} = 1$ ✓.

Answer: Not uniform: red is twice as likely

WORKED EXAMPLE · MODEL FROM DATA

A survey of 40: 18 dogs, 14 cats, 8 fish (favorite pet)

1 P(dog) = $^{9}/_{20}$, P(cat) = $^{7}/_{20}$, P(fish) = $^{1}/_{5}$.

Answer: probabilities add to 1

KEY IDEA

In every probability model, the probabilities add up to 1.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Uniform model for a coin

2. Spinner P(not red)

3. A model lists 0.3, 0.5, and ? Find ?

More practice: Workbook, Lesson 7.12.

LESSON SUMMARY

A probability model lists every outcome; the probabilities sum to 1.

UNIT 7 · LESSON 7.13 · 7.SP.8b

Lesson 7.13: Sample Spaces for Compound Events

I can list the sample space of a compound event using lists and tables.

WORDS TO KNOW

compound event — An event made of two or more simple events.

LET'S EXPLORE

You roll two number cubes. How many outcomes are possible?

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

WORKED EXAMPLE · COUNT OUTCOMES

Two number cubes

1 $6 \times 6 = 36$ outcomes.

Answer: 36

WORKED EXAMPLE · USE THE TABLE

P(sum is 7)

1 Count 7s in the table: 6.

Answer: $\frac{1}{6}$

KEY IDEA

Organized lists and tables make sure you count every outcome once.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. P(sum is 2)
2. P(sum is 10)
3. P(doubles)

More practice: Workbook, Lesson 7.13.

LESSON SUMMARY

Lists and tables organize every outcome of a compound event.

UNIT 7 · LESSON 7.14 · 7.SP.8b

Lesson 7.14: Tree Diagrams

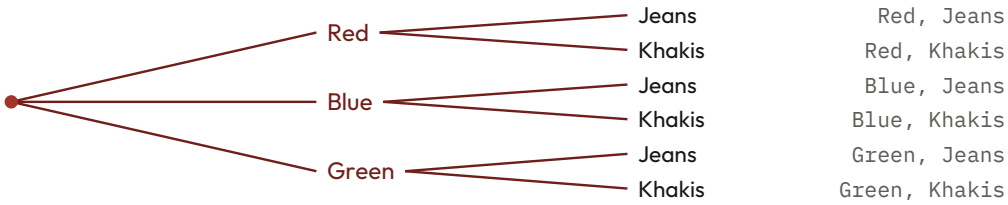
I can use a tree diagram to find the sample space of a compound event.

WORDS TO KNOW

tree diagram	A branching diagram that shows every outcome of a multi-step experiment.

LET'S EXPLORE

Lillie has a red, blue, or green shirt and jeans or khakis. How many outfits can she make?



WORKED EXAMPLE · COUNT

3 shirts, 2 pants

1 Each of 3 shirts branches to 2 pants.

Answer: 6 outfits

WORKED EXAMPLE · PROBABILITY

Outfit chosen at random: P(blue shirt with jeans)

1 1 favorable outcome of 6.

Answer: $\frac{1}{6}$

KEY IDEA

Multiply the number of choices at each stage to count outcomes (the counting principle).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. P(outfit with khakis)
2. Add a yellow shirt. How many outfits?
3. Coin flipped 3 times: how many outcomes?

More practice: Workbook, Lesson 7.14.

LESSON SUMMARY

Tree diagrams and the counting principle show every outcome.

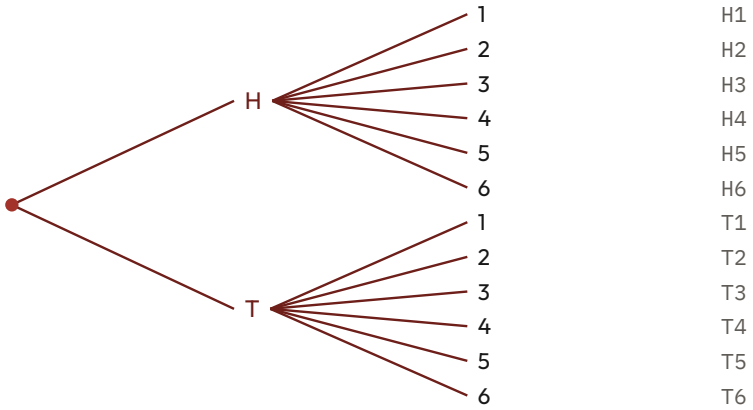
UNIT 7 · LESSON 7.15 · 7.SP.8a, 7.SP.8c

Lesson 7.15: Probability of Compound Events

I can find the probability of a compound event using the sample space.

LET'S EXPLORE

Flip a coin and roll a number cube. What is P(heads and an even number)?



WORKED EXAMPLE · COUNT

P(heads and even)

1 Sample space: 2 × 6 = 12.

2 Favorable: H2, H4, H6 = 3.

Answer: $\frac{1}{4}$

WORKED EXAMPLE · MULTIPLY (INDEPENDENT EVENTS)

Same event

1 P(H) × P(even) = $\frac{1}{2} \times \frac{1}{2}$.

Answer: $\frac{1}{4}$

KEY IDEA

Probability of a compound event = favorable outcomes ÷ sample space size. For independent events you can also multiply.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. P(tails and 5)
2. P(heads and greater than 2)
3. P(tails or 6)

More practice: Workbook, Lesson 7.15.

LESSON SUMMARY

Count favorable outcomes in the sample space, or multiply for independent events.

UNIT 7 · LESSON 7.16 · 7.SP.8c

Lesson 7.16: Simulations

I can design and use a simulation to estimate the probability of a compound event.

WORDS TO KNOW

simulation	An experiment that models a real situation using random tools.

LET'S EXPLORE

A cereal box holds one of 4 equally likely prizes. About how many boxes must you buy to get all 4?

Instead of buying cereal, simulate: roll a 4-sided die (or spin a 4-part spinner) where each number stands for a prize. Count rolls until all four appear. Repeat many times.

Trial	Rolls	Boxes needed
1	2, 1, 2, 4, 1, 3	6
2	3, 3, 1, 4, 2	5
3	1, 4, 4, 2, 4, 1, 3	7
4	4, 2, 2, 1, 3	5
5	2, 2, 3, 2, 1, 1, 4	7

WORKED EXAMPLE · ESTIMATE

Mean of the five trials

1 $(6 + 5 + 7 + 5 + 7) \div 5 = 6$.

Answer: about 6 boxes (more trials improve the estimate)

KEY IDEA

A simulation matches each real outcome to a random-tool outcome with the same probability.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Experimental P(needing more than 6 boxes) from the table
2. Which tool simulates a 50% event?

3. Which tool simulates a 1-in-6 event?

More practice: Workbook, Lesson 7.16.

LESSON SUMMARY

Simulations use random tools to estimate real probabilities.

UNIT 7 REVIEW

Unit 7 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
7.1	Study a sample to learn about a population.
7.2	Random samples give everyone an equal chance; biased samples leave groups out.
7.3	Multiply the sample rate by the population size.
7.4	Samples vary; combine several or use larger samples for better estimates.
7.5	Center: mean or median. Variability: MAD, range, or IQR.
7.6	Compare centers relative to the spread.
7.7	Compare medians for center and IQRs for spread.
7.8	Random samples + difference in means ÷ MAD → is there a real difference?
7.9	Probability runs from 0 (impossible) to 1 (certain).
7.10	P(event) = favorable ÷ total equally likely outcomes.
7.11	Experimental: occurrences ÷ trials. It approaches theoretical with more trials.
7.12	A probability model lists every outcome; the probabilities sum to 1.
7.13	Lists and tables organize every outcome of a compound event.
7.14	Tree diagrams and the counting principle show every outcome.
7.15	Count favorable outcomes in the sample space, or multiply for independent events.
7.16	Simulations use random tools to estimate real probabilities.

UNIT 8

Scale Drawings and Geometric Figures

Scaled copies, scale drawings and maps, drawing triangles from conditions, and cross sections.

Architects, mapmakers, and game designers all work with scale. In this unit you will make and use scale drawings and maps, discover which measurements determine a triangle, and imagine slicing solids to see what shapes appear.

In this unit you will...

- Find and use scale factors, including their effect on area.
- Use scales to find actual lengths, areas, and distances.
- Redraw scale drawings at a new scale.
- Decide when conditions give one, many, or no triangles.
- Describe cross sections of prisms, pyramids, cylinders, and cones.

MATH IN REAL LIFE

Blueprints, road maps, model trains, and CT scans (which are cross sections of the body!) all use the ideas in this unit.

UNIT VOCABULARY

scale factor	Multiplier for every length.
scale	Drawing length : actual length.
triangle inequality	Two sides sum to more than the third.
cross section	The shape of a slice.

Lessons in this unit

Lesson	Title
8.1	Scaled Copies
8.2	Scale Factors

Lesson	Title
8.3	Scale Factors and Area
8.4	Scale Drawings
8.5	Finding Drawing Lengths
8.6	Area in Scale Drawings
8.7	Maps and Scale
8.8	Changing the Scale
8.9	Triangle Side Lengths
8.10	Triangle Angles
8.11	Drawing Triangles from Conditions
8.12	Drawing Quadrilaterals with Conditions
8.13	Cross Sections of Prisms
8.14	Cross Sections of Pyramids
8.15	Cross Sections of Cylinders and Cones
8.16	Scale Drawing Project

UNIT 8 · LESSON 8.1 · 7.G.1

Lesson 8.1: Scaled Copies

I can tell whether one figure is a scaled copy of another.

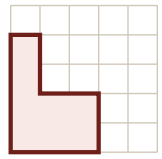
WORDS TO KNOW

scaled copy	A copy of a figure in which every length is multiplied by the same number.
corresponding parts	Matching sides or angles in a figure and its copy.

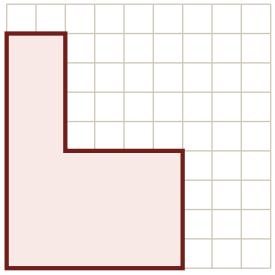
LET'S EXPLORE

Ellie draws a 3-by-4 rectangle, then a 6-by-8 rectangle and a 5-by-6 rectangle. Which one is a scaled copy?

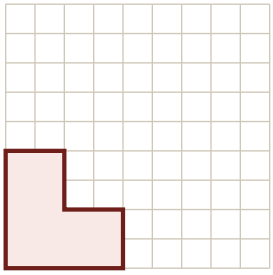
In a scaled copy, every length is multiplied by the same number and every angle stays the same. Adding the same amount to each side does not make a scaled copy.



Original



Copy A: every length × 2



Copy B: not scaled

WORKED EXAMPLE · CHECK THE RECTANGLES

3×4 vs 6×8 and 5×6

1 $6 \div 3 = 2$ and $8 \div 4 = 2$: same factor.

2 $5 \div 3 = {}^{5}/_{3}$ but $6 \div 4 = {}^{3}/_{2}$: different.

Answer: 6×8 is a scaled copy; 5×6 is not

KEY IDEA

Scaled copy: all lengths multiplied by the same factor, all angles unchanged.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Is a 4×10 rectangle a scaled copy of 2×5?
2. Is a 5×7 rectangle a scaled copy of 3×5?
3. Is any square a scaled copy of any other square?

More practice: Workbook, Lesson 8.1.

LESSON SUMMARY

Scaled copies multiply every length by one factor; angles stay the same.

UNIT 8 · LESSON 8.2 · 7.G.1

Lesson 8.2: Scale Factors

I can find and use a scale factor.

WORDS TO KNOW

scale factor	The number every length is multiplied by to make a scaled copy.

LET'S EXPLORE

A photo 4 in. by 6 in. is enlarged so the short side is 10 in. How long is the long side?

WORKED EXAMPLE · FIND THE FACTOR

4 in. → 10 in.

1 Scale factor = 10 ÷ 4 = 2.5.

Answer: 2.5

WORKED EXAMPLE · USE THE FACTOR

Long side: 6 × 2.5

1 6 × 2.5 = 15.

Answer: 15 in.

A scale factor greater than 1 makes an enlargement. A scale factor between 0 and 1 makes a reduction. A scale factor of 1 makes a same-size copy.

KEY IDEA

Scale factor = copy length ÷ original length.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Original 5 cm, copy 15 cm. Scale factor?
2. Original 12 ft, copy 3 ft. Scale factor?
3. Side 8 m, scale factor 0.75. Copy length?

More practice: Workbook, Lesson 8.2.

LESSON SUMMARY

Scale factor = copy ÷ original; multiply each length by it.

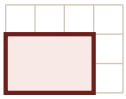
UNIT 8 · LESSON 8.3 · 7.G.1

Lesson 8.3: Scale Factors and Area

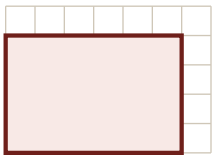
I can find how area changes when a figure is scaled.

LET'S EXPLORE

A 2-by-3 rectangle is scaled by a factor of 2. Does its area double?



Area 6



Area 24

No! The original area is 6 square units; the copy is 4 × 6 = 24 square units, which is 4 times as much. Length and width both doubled, so area is multiplied by 2 × 2.

WORKED EXAMPLE · FACTOR 3

Area 10 cm², scale factor 3

1 Area factor: $3^2 = 9$.

2 $10 \times 9 = 90$.

Answer: 90 cm²

WORKED EXAMPLE · REDUCTION

Area 48 in², scale factor $\frac{1}{2}$

1 Area factor: $(\frac{1}{2})^2 = \frac{1}{4}$.

2 $48 \times \frac{1}{4} = 12$.

Answer: 12 in²

KEY IDEA

If lengths are multiplied by k, area is multiplied by k^2.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Area 5, scale factor 4. New area?
2. Area 36, scale factor $\frac{1}{3}$
3. Area changed from 7 to 63. Scale factor?

More practice: Workbook, Lesson 8.3.

LESSON SUMMARY

Lengths × k means area × k^2.

UNIT 8 · LESSON 8.4 · 7.G.1

Lesson 8.4: Scale Drawings

I can use a scale to find actual lengths from a scale drawing.

WORDS TO KNOW

scale drawing	A drawing that represents a real object with lengths in proportion.
scale	A ratio that compares drawing length to actual length, such as 1 in. : 8 ft.

LET'S EXPLORE

A floor plan uses the scale 1 in. : 8 ft. A bedroom measures 1.5 in. by 1.75 in. on the plan. How big is the real room?

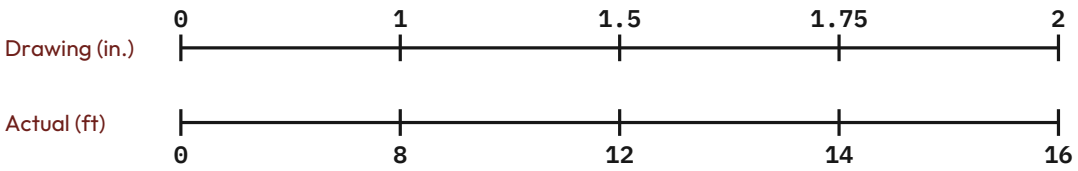
WORKED EXAMPLE · USE THE SCALE

1.5 in. by 1.75 in.; 1 in. : 8 ft

1 1.5 × 8 = 12 ft.

2 1.75 × 8 = 14 ft.

Answer: 12 ft by 14 ft



KEY IDEA

Multiply drawing lengths by the scale's actual-per-unit value.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Scale 1 cm : 5 m. Drawing 7 cm. Actual?
2. Scale 1 in. : 20 mi. Drawing 3.5 in. Actual?
3. Scale 2 cm : 1 m. Drawing 9 cm. Actual?

More practice: Workbook, Lesson 8.4.

LESSON SUMMARY

Drawing length × actual-per-unit = actual length.

UNIT 8 · LESSON 8.5 · 7.G.1

Lesson 8.5: Finding Drawing Lengths

I can find the length on a scale drawing for a given actual length.

LET'S EXPLORE

A soccer field is 100 m long. Using 1 cm : 5 m, how long is it on the drawing?

WORKED EXAMPLE · DIVIDE BY THE SCALE

100 m with 1 cm : 5 m

1 100 ÷ 5 = 20.

Answer: 20 cm

WORKED EXAMPLE · SCALE WITH A FRACTION

A 30-ft wall with $\frac{1}{2}$ in. : 3 ft

1 30 ÷ 3 = 10 groups of 3 ft.

2 10 × $\frac{1}{2}$ in. = 5 in.

Answer: 5 in.

KEY IDEA

To go from actual to drawing, divide by how much 1 drawing unit represents.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Scale 1 in. : 4 ft. Actual 22 ft. Drawing?
2. Scale 1 cm : 20 km. Actual 130 km. Drawing?
3. Scale 2 cm : 3 m. Actual 12 m. Drawing?

More practice: Workbook, Lesson 8.5.

LESSON SUMMARY

Actual ÷ actual-per-unit = drawing length.

UNIT 8 · LESSON 8.6 · 7.G.1

Lesson 8.6: Area in Scale Drawings

I can find actual areas from scale drawings.

LET'S EXPLORE

A park drawing is 4 cm by 6 cm with 1 cm : 50 m. What is the park's actual area?

WORKED EXAMPLE · CONVERT LENGTHS FIRST

4 cm by 6 cm; 1 cm : 50 m

1 Actual: 4 × 50 = 200 m and 6 × 50 = 300 m.

2 Area: 200 × 300 = 60,000 m².

Answer: 60,000 m²

WORKED EXAMPLE · AREA FACTOR

Same park

1 Drawing area: 24 cm².

2 Each cm² represents 50 × 50 = 2,500 m².

3 24 × 2,500 = 60,000.

Answer: 60,000 m²

WATCH OUT

Do not multiply the drawing area by 50. Each square centimeter stands for 50 m × 50 m.

KEY IDEA

Convert each length first, then compute the area. Or multiply drawing area by (scale)².

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Drawing 3 in. × 5 in.; 1 in. : 4 ft. Actual area?
2. Drawing 2 cm × 2 cm; 1 cm : 10 m. Actual area?
3. Drawing area 6 in²; 1 in. : 3 ft. Actual area?

More practice: Workbook, Lesson 8.6.

LESSON SUMMARY

Convert lengths first, or multiply the drawing area by the scale squared.

UNIT 8 · LESSON 8.7 · 7.G.1, 7.RP.2

Lesson 8.7: Maps and Scale

I can use a map scale to find real distances and travel times.

LET'S EXPLORE

On a map with scale 1 cm : 25 km, two cities are 7.2 cm apart. How far apart are they? How long would a 90 km/h drive take?

WORKED EXAMPLE · DISTANCE

7.2 cm; 1 cm : 25 km

1 7.2 × 25 = 180.

Answer: 180 km

WORKED EXAMPLE · TIME

180 km at 90 km/h

1 180 ÷ 90 = 2.

Answer: 2 hours

WORKED EXAMPLE · SCALE WITHOUT UNITS

Scale 1 : 50,000. Map distance 6 cm.

1 Actual = 6 × 50,000 = 300,000 cm.

2 300,000 cm = 3,000 m = 3 km.

Answer: 3 km

KEY IDEA

A scale like 1 : 50,000 has no units: 1 of any unit on the map is 50,000 of the same unit in real life.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Map 1 in. : 30 mi. Distance 4.5 in.?

2. Scale 1 : 100,000. Map 4 cm. Actual km?

3. 150 mi at 60 mph: hours?

More practice: Workbook, Lesson 8.7.

LESSON SUMMARY

Multiply map distance by the scale; convert units if needed.

UNIT 8 · LESSON 8.8 · 7.G.1

Lesson 8.8: Changing the Scale

I can reproduce a scale drawing at a different scale.

LET'S EXPLORE

A garden plan uses 1 cm : 2 m and is 10 cm by 6 cm. Redraw it at 1 cm : 4 m. What size is the new drawing?

WORKED EXAMPLE · FIND ACTUAL, THEN REDRAW

10 cm × 6 cm at 1 cm : 2 m

1 Actual: 20 m by 12 m.

2 New drawing at 1 cm : 4 m: 20 ÷ 4 = 5 cm; 12 ÷ 4 = 3 cm.

Answer: 5 cm by 3 cm

WORKED EXAMPLE · COMPARE SCALES

1 cm : 2 m vs 1 cm : 4 m

1 Each cm now stands for twice as much.

2 So the new drawing is $\frac{1}{2}$ the size.

Answer: drawing lengths × $\frac{1}{2}$

KEY IDEA

Find actual lengths first, then convert to the new scale.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Plan 8 in. at 1 in. : 5 ft; redraw at 1 in. : 10 ft

2. Plan 3 cm at 1 cm : 6 m; redraw at 1 cm : 2 m

3. Plan 12 cm at 1 cm : 1 m; redraw at 1 cm : 3 m

More practice: Workbook, Lesson 8.8.

LESSON SUMMARY

Find actual lengths, then convert to the new scale.

UNIT 8 · LESSON 8.9 · 7.G.2

Lesson 8.9: Triangle Side Lengths

I can decide whether three side lengths can form a triangle.

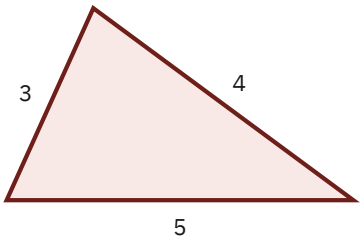
WORDS TO KNOW

Triangle Inequality	The sum of any two side lengths of a triangle is greater than the third side.

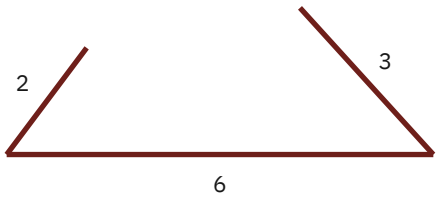
LET'S EXPLORE

Can you make a triangle with straws 2 in., 3 in., and 6 in. long?

No! The two short straws together reach only 5 in., which is shorter than 6 in. They can't meet.



3, 4, 5: forms a triangle



2, 3, 6: the sides can't meet

WORKED EXAMPLE · TEST

4, 7, 9

1 Shortest two: 4 + 7 = 11.

2 11 > 9 ✓.

Answer: Yes, a triangle

WORKED EXAMPLE · TEST

5, 5, 10

1 5 + 5 = 10, not greater than 10.

Answer: No triangle (it lies flat)

KEY IDEA

Check only the two shortest sides: their sum must be greater than the longest side.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3, 3, 5

2. 1, 2, 4

3. 8, 8, 8

More practice: Workbook, Lesson 8.9.

LESSON SUMMARY

The two shorter sides must add to more than the longest side.

UNIT 8 · LESSON 8.10 · 7.G.2

Lesson 8.10: Triangle Angles

I can use the angle sum of a triangle to decide whether angles can form a triangle.

WORDS TO KNOW

angle sum	The angles in any triangle add up to 180°.

LET'S EXPLORE

Can a triangle have angles of 90°, 60°, and 40°?

Tear the three corners off any paper triangle and line them up: they always form a straight angle, 180°.

WORKED EXAMPLE · CHECK

90°, 60°, 40°

1 $90 + 60 + 40 = 190 \neq 180$.

Answer: No triangle

WORKED EXAMPLE · MISSING ANGLE

Two angles are 48° and 75°

1 $180 - 48 - 75 = 57$.

Answer: 57°

WORKED EXAMPLE · HOW MANY TRIANGLES?

Angles 50°, 60°, 70°

1 They sum to 180°.

2 Triangles of many sizes have these angles (scaled copies).

Answer: Many triangles

KEY IDEA

Angles alone can't fix a triangle's size: three angles that sum to 180° give infinitely many triangles.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 30°, 60°, 90°: triangle?

2. Two angles 110° and 80°: triangle?

3. Missing angle: 35°, 35°, ?

More practice: Workbook, Lesson 8.10.

LESSON SUMMARY

Triangle angles sum to 180°; angles alone don't fix the size.

UNIT 8 · LESSON 8.11 · 7.G.2

Lesson 8.11: Drawing Triangles from Conditions

I can draw triangles with a ruler and protractor and tell whether the conditions give one, many, or no triangles.

WORDS TO KNOW

unique	Exactly one; any triangle drawn with those conditions is the same shape and size.

LET'S EXPLORE

Draw a triangle with sides 5 cm and 7 cm and a 40° angle between them. Will your triangle match a classmate's?

Yes. When two sides and the angle between them are given, there is only one possible third side, so the triangle is unique.

Given	Example	How many triangles?
3 sides (that pass the inequality)	5, 6, 7 cm	one
2 sides + angle between them	5 cm, 40°, 7 cm	one
2 angles + side between them	50°, 6 cm, 60°	one
3 angles that sum to 180°	50°, 60°, 70°	many
Conditions that break a rule	2, 3, 6 cm; or 100°, 90°	none

WORKED EXAMPLE · STEPS TO DRAW SAS

5 cm, 40°, 7 cm

1. Draw a 7 cm segment.
2. At one end, measure 40° with a protractor and draw a ray.
3. Mark 5 cm on the ray.
4. Connect the endpoints.

Answer: one triangle

KEY IDEA

SSS, SAS, and ASA give one triangle. AAA gives many. Broken rules give none.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Sides 4, 4, 4 cm
2. Angles 45°, 45°, 90°
3. Angles 100°, 100°, ?

More practice: Workbook, Lesson 8.11.

LESSON SUMMARY

SSS, SAS, ASA → one triangle; AAA → many; broken rules → none.

UNIT 8 · LESSON 8.12 · 7.G.2

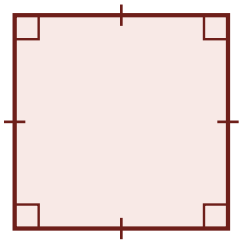
Lesson 8.12: Drawing Quadrilaterals with Conditions

I can draw quadrilaterals with given conditions and decide whether they are unique.

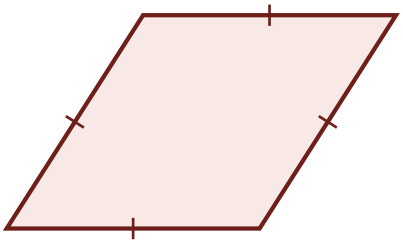
LET'S EXPLORE

Draw a quadrilateral with all sides 4 cm. Is there only one?

No: a square works, but so does any rhombus with 4 cm sides. Push a square's corner and it leans into a rhombus. Quadrilaterals need more conditions than triangles to be unique.



square



rhombus

WORKED EXAMPLE · UNIQUE?

A rectangle 5 cm by 3 cm

1 Right angles fix the shape; lengths fix the size.

Answer: one rectangle

KEY IDEA

Side lengths alone don't fix a quadrilateral. Angles help.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rectangle with sides 6 and 2: how many?
2. Parallelogram with sides 6 and 2: how many?
3. Square with sides 3 cm: how many?

More practice: Workbook, Lesson 8.12.

LESSON SUMMARY

Quadrilaterals need angle conditions to be unique.

UNIT 8 · LESSON 8.13 · 7.G.3

Lesson 8.13: Cross Sections of Prisms

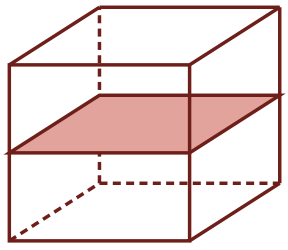
I can describe the cross sections made by slicing a rectangular prism.

WORDS TO KNOW

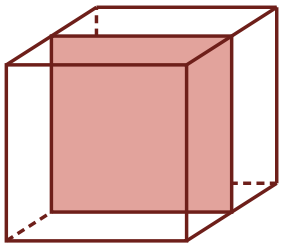
cross section	The two-dimensional shape made when a plane slices through a solid.
plane	A flat surface that extends forever.

LET'S EXPLORE

Slice a block of cheese straight across, parallel to the bottom. What shape is the cut face?



Parallel to the base: a rectangle the same size as the base



Perpendicular to the base: a rectangle

WORKED EXAMPLE · SLICE A CUBE

Cube, slice parallel to a face

1 The cut matches the face.

Answer: a square

WORKED EXAMPLE · SLANTED SLICE

Cube, slice through one corner

1 Cutting off one corner makes three cut edges.

Answer: a triangle

KEY IDEA

Slices of a rectangular prism parallel or perpendicular to the base are rectangles. Slanted slices can make triangles and other polygons.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Slice of a 4 × 6 × 10 box parallel to the 4 × 6 base
2. Vertical slice of a cube
3. Slice cutting off one corner of a cube

More practice: Workbook, Lesson 8.13.

LESSON SUMMARY

Prism slices parallel or perpendicular to the base are rectangles (or the base shape).

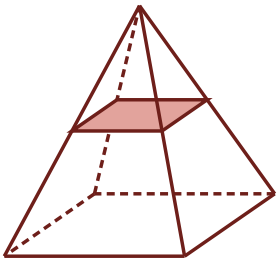
UNIT 8 · LESSON 8.14 · 7.G.3

Lesson 8.14: Cross Sections of Pyramids

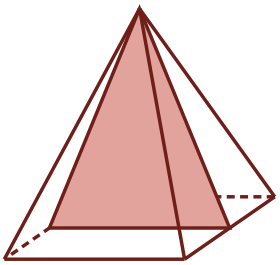
I can describe the cross sections made by slicing a pyramid.

LET'S EXPLORE

Slice a square pyramid parallel to its base. What shape is the cut? What if you slice straight down through the top?



Parallel to the base: a smaller square



Perpendicular through the top: a triangle

WORKED EXAMPLE · SIZE CHANGES

Slice parallel to the base, halfway up

1 Each side of the cut is half the base side.

Answer: a square half as wide

KEY IDEA

Parallel slices of a pyramid have the base's shape but get smaller toward the top. Vertical slices through the top are triangles.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Square pyramid, horizontal slice
2. Triangular pyramid, slice parallel to base
3. Square pyramid, vertical slice not through the top

More practice: Workbook, Lesson 8.14.

LESSON SUMMARY

Horizontal slices: base shape, smaller; vertical slices through the apex: triangles.

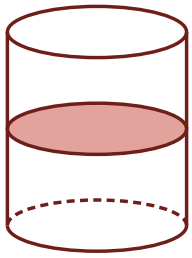
UNIT 8 · LESSON 8.15 · 7.G.3

Lesson 8.15: Cross Sections of Cylinders and Cones

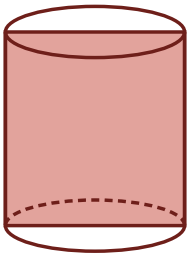
I can describe the cross sections of cylinders and cones.

LET'S EXPLORE

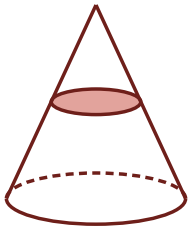
What shape do you see when you cut a can straight across? Straight down the middle?



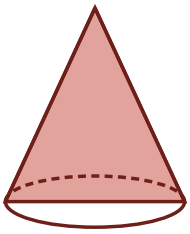
circle



rectangle



smaller circle



triangle

KEY IDEA

Horizontal slices of cylinders and cones are circles. Vertical slices through the center are rectangles (cylinder) or triangles (cone).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Slice of a cylinder parallel to the base
2. Vertical slice of a cone through its tip
3. Vertical slice of a cylinder through its center

More practice: Workbook, Lesson 8.15.

LESSON SUMMARY

Round solids: horizontal slices are circles; vertical slices are rectangles or triangles.

UNIT 8 · LESSON 8.16 · 7.G.1

Lesson 8.16: Scale Drawing Project

I can create a scale drawing of a real space and use it to solve problems.

LET'S EXPLORE

Design a scale floor plan of your bedroom or kitchen.

Measure the room and its furniture. Choose a scale that fits your paper (for example, 1 in. : 2 ft or 1 cm : 50 cm). Draw the room and furniture to scale, and label each length.

WORKED EXAMPLE · SAMPLE

A room 14 ft by 12 ft on 8.5 × 11 in. paper

1 1 in. : 2 ft gives 7 in. by 6 in.: fits!

2 A 6 ft bed becomes 3 in.

Answer: Scale 1 in. : 2 ft

Object	Actual size	Drawing size (1 in. : 2 ft)
Room	14 ft × 12 ft	7 in. × 6 in.
Bed	6 ft × 3.5 ft	3 in. × 1.75 in.
Desk	4 ft × 2 ft	2 in. × 1 in.

KEY IDEA

Choose a scale so the biggest length fits your paper.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rug 5 ft × 8 ft at 1 in. : 2 ft
2. Window 3 ft wide at 1 in. : 2 ft
3. Area of the room above

More practice: Workbook, Lesson 8.16.

LESSON SUMMARY

Measure, pick a scale that fits, draw, and label.

UNIT 8 REVIEW

Unit 8 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
8.1	Scaled copies multiply every length by one factor; angles stay the same.
8.2	Scale factor = copy ÷ original; multiply each length by it.
8.3	Lengths × k means area × k^2.
8.4	Drawing length × actual-per-unit = actual length.
8.5	Actual ÷ actual-per-unit = drawing length.
8.6	Convert lengths first, or multiply the drawing area by the scale squared.
8.7	Multiply map distance by the scale; convert units if needed.
8.8	Find actual lengths, then convert to the new scale.
8.9	The two shorter sides must add to more than the longest side.
8.10	Triangle angles sum to 180°; angles alone don't fix the size.
8.11	SSS, SAS, ASA → one triangle; AAA → many; broken rules → none.
8.12	Quadrilaterals need angle conditions to be unique.
8.13	Prism slices parallel or perpendicular to the base are rectangles (or the base shape).
8.14	Horizontal slices: base shape, smaller; vertical slices through the apex: triangles.
8.15	Round solids: horizontal slices are circles; vertical slices are rectangles or triangles.
8.16	Measure, pick a scale that fits, draw, and label.

UNIT 9

Angles, Circles, Area, and Volume

Angle relationships, circles, composite area, surface area and volume of prisms, and a review of the whole year.

Geometry is everywhere: wheels, windows, pools, and boxes. In this final unit you will use angle facts to find unknown angles, discover π, find areas of composite figures, and measure the surface area and volume of prisms. The last week reviews the whole year.

In this unit you will…

- Use complementary, supplementary, vertical, and adjacent angles to find unknowns.
- Find the circumference and area of circles.
- Find areas of composite figures.
- Find surface area and volume of prisms.
- Review and connect every unit from the year.

MATH IN REAL LIFE

Builders, engineers, and designers use these formulas daily: pizza sizes, pool covers, paint, packaging, and concrete all depend on area and volume.

UNIT VOCABULARY

supplementary	Sum to 180°.
complementary	Sum to 90°.
circumference	$C = \pi d$.
area of a circle	$A = \pi r^2$.
volume of a prism	$V = Bh$.

Lessons in this unit

Lesson	Title
9.1	Angle Relationships

Lesson	Title
9.2	Solving for Unknown Angles
9.3	Circles and Circumference
9.4	Area of a Circle
9.5	Problems with Circles
9.6	Area of Composite Figures
9.7	Surface Area of Prisms
9.8	Volume of Prisms
9.9	Volume and Surface Area in Context
9.10	Composite Solids
9.11	Comparing Volume and Surface Area
9.12	Geometry Problem Solving
9.13	Year Review: Ratios, Proportions, and Percents
9.14	Year Review: Rational Numbers
9.15	Year Review: Expressions, Equations, and Inequalities
9.16	Year Review: Statistics, Probability, and Geometry

UNIT 9 · LESSON 9.1 · 7.G.5

Lesson 9.1: Angle Relationships

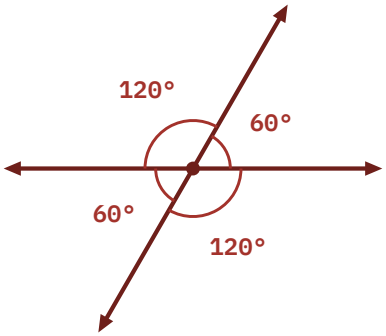
I can identify complementary, supplementary, vertical, and adjacent angles.

WORDS TO KNOW

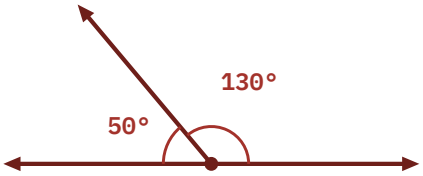
complementary angles	Two angles whose measures add to 90°.
supplementary angles	Two angles whose measures add to 180°.
vertical angles	Opposite angles formed by two intersecting lines. They are equal.
adjacent angles	Angles that share a vertex and a side but do not overlap.

LET'S EXPLORE

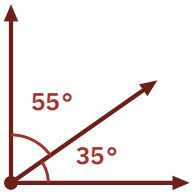
Two straight roads cross. One angle at the corner measures 60°. What are the other three angles?



Vertical angles are equal; neighbors are supplementary.



Supplementary: 130° + 50° = 180°



Complementary: 35° + 55° = 90°

WORKED EXAMPLE · CROSSING ROADS

One angle is 60°

1 The vertical angle is also 60°.

2 Each neighbor is 180° − 60° = 120°.

Answer: 60°, 120°, 60°, 120°

KEY IDEA

Complementary → 90°. Supplementary → 180°. Vertical angles are equal.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Complement of 28°
2. Supplement of 28°
3. Two lines cross; one angle is 75°. Its vertical angle?

More practice: Workbook, Lesson 9.1.

LESSON SUMMARY

Complementary sum to 90°, supplementary to 180°; vertical angles are equal.

UNIT 9 · LESSON 9.2 · 7.G.5, 7.EE.4a

Lesson 9.2: Solving for Unknown Angles

I can write and solve equations to find unknown angles.

LET'S EXPLORE

Two adjacent angles form a straight line. One is (2x + 10)° and the other is 70°. Find x.

WORKED EXAMPLE · SUPPLEMENTARY

(2x + 10) + 70 = 180

1 2x + 80 = 180.

2 2x = 100, so x = 50.

3 The angle is 2(50) + 10 = 110°.

Answer: x = 50

WORKED EXAMPLE · VERTICAL

Vertical angles 3x° and 105°

1 3x = 105.

Answer: x = 35

WORKED EXAMPLE · COMPLEMENTARY

Angles x° and 2x° are complementary

1 x + 2x = 90; 3x = 90.

Answer: x = 30; angles 30° and 60°

KEY IDEA

Write an equation from the relationship (sum to 90°, sum to 180°, or equal), then solve.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $x + 48 = 90$
2. Vertical angles $(4x - 20)°$ and $60°$
3. Supplementary: $5x$ and $4x$

More practice: Workbook, Lesson 9.2.

LESSON SUMMARY

Relationship → equation → solve → check the angle.

UNIT 9 · LESSON 9.3 · 7.G.4

Lesson 9.3: Circles and Circumference

I can find the circumference of a circle from its radius or diameter.

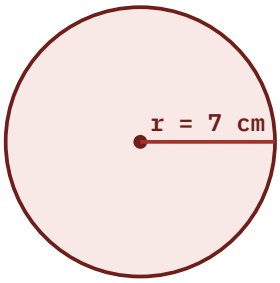
WORDS TO KNOW

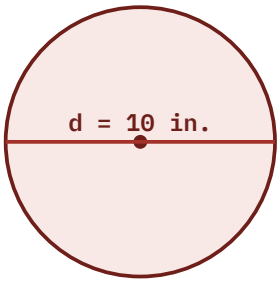
circumference	The distance around a circle.
diameter	A segment across a circle through its center; $d = 2r$.
radius	A segment from the center to the circle.
pi (π)	The ratio of any circle's circumference to its diameter, about 3.14.

LET'S EXPLORE

Measure around a can and across its top. Divide. Try another circle. What do you notice?

For every circle, circumference ÷ diameter is the same number, π (about 3.14). So $C = \pi d$, or $C = 2\pi r$. In this book use 3.14 for π unless told otherwise.





WORKED EXAMPLE · DIAMETER

d = 10 in.

1 C = 3.14 × 10 = 31.4.

Answer: 31.4 in.

WORKED EXAMPLE · RADIUS

r = 7 cm

1 d = 14; C = 3.14 × 14 = 43.96.

Answer: 43.96 cm

WORKED EXAMPLE · WORK BACKWARD

C = 62.8 m

1 d = 62.8 ÷ 3.14 = 20.

Answer: d = 20 m

KEY IDEA

C = πd = 2πr. Use π ≈ 3.14.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. C when d = 6 ft

2. C when r = 3 m

3. d when C = 31.4 cm

More practice: Workbook, Lesson 9.3.

LESSON SUMMARY

C = πd = 2πr, with π ≈ 3.14.

UNIT 9 · LESSON 9.4 · 7.G.4

Lesson 9.4: Area of a Circle

I can find the area of a circle.

LET'S EXPLORE

Cut a paper circle into 16 wedges and rearrange them. What shape do they almost make?

The wedges form a shape that is almost a parallelogram. Its base is half the circumference (πr) and its height is the radius (r). So its area is $\pi r \times r = \pi r^2$.

WORKED EXAMPLE · RADIUS

$r = 5$ cm

1 $A = 3.14 \times 5^2 = 3.14 \times 25 = 78.5$.

Answer: 78.5 cm²

WORKED EXAMPLE · DIAMETER

$d = 8$ in.

1 $r = 4$; $A = 3.14 \times 16 = 50.24$.

Answer: 50.24 in²

WATCH OUT

Square the radius before multiplying by π. And use the radius, not the diameter.

KEY IDEA

$A = \pi r^2$.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A when $r = 10$ m
2. A when $d = 6$ ft
3. A when $r = 1.5$ cm

More practice: Workbook, Lesson 9.4.

LESSON SUMMARY

$A = \pi r^2$: square the radius, then multiply by π.

UNIT 9 · LESSON 9.5 · 7.G.4

Lesson 9.5: Problems with Circles

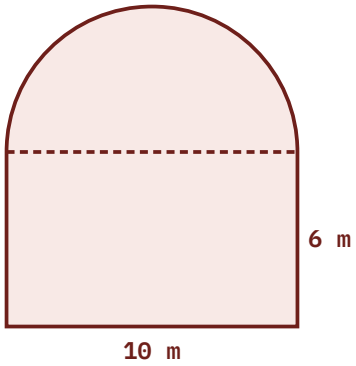
I can find areas and perimeters of figures made from parts of circles.

WORDS TO KNOW

semicircle	Half of a circle.

LET'S EXPLORE

A window is a 10 m by 6 m rectangle topped by a semicircle. How much glass does it need? How much trim goes around it?



WORKED EXAMPLE · AREA

Rectangle + half circle (r = 5)

1. Rectangle: 10 × 6 = 60.
2. Half circle: 3.14 × 25 ÷ 2 = 39.25.
3. Total: 60 + 39.25.

Answer: 99.25 m²

WORKED EXAMPLE · PERIMETER

Three sides of the rectangle + half the circumference

1. 10 + 6 + 6 = 22.
2. Half of 3.14 × 10 = 15.7.
3. 22 + 15.7.

Answer: 37.7 m

KEY IDEA

Break the figure into rectangles and circle parts. Include only the edges on the outside for perimeter.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Area of a semicircle, r = 4 cm
2. Area of a quarter circle, r = 10 ft
3. Perimeter of a semicircle, d = 10 in. (curve + diameter)

More practice: Workbook, Lesson 9.5.

LESSON SUMMARY

Split the figure into rectangles and circle parts; perimeter uses only outside edges.

UNIT 9 · LESSON 9.6 · 7.G.6

Lesson 9.6: Area of Composite Figures

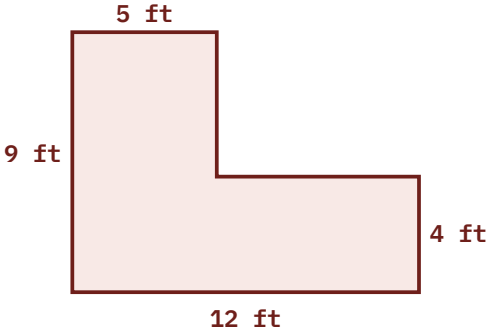
I can find the area of composite figures made of triangles, rectangles, and trapezoids.

WORDS TO KNOW

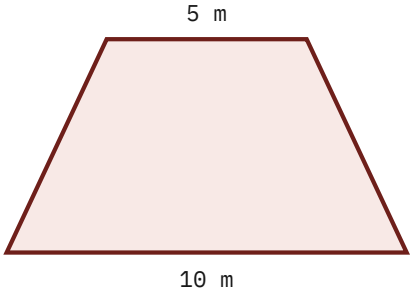
composite figure	A figure made of two or more simple shapes.

LET'S EXPLORE

A garden is shaped like an L. How can you find its area without a formula for L-shapes?



Split into two rectangles.



Trapezoid, height 4 m

WORKED EXAMPLE · L-SHAPE

Split into rectangles

1 Bottom: 12 × 4 = 48.

2 Top: 5 × 5 = 25.

Answer: 73 ft²

WORKED EXAMPLE · TRAPEZOID

Bases 10 m and 5 m, height 4 m

1 A = (10 + 5) ÷ 2 × 4 = 30.

Answer: 30 m²

Shape	Area formula
rectangle	$A = lw$
triangle	$A = \frac{1}{2}bh$
parallelogram	$A = bh$
trapezoid	$A = \frac{1}{2}(b_1 + b_2)h$

KEY IDEA

Decompose into shapes you know, find each area, and add (or subtract cut-out parts).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Triangle, b = 9, h = 6

2. Trapezoid, bases 7 and 11, h = 5

3. Rectangle 10 × 8 with a 3 × 3 square cut out

More practice: Workbook, Lesson 9.6.

LESSON SUMMARY

Split, find each area, then add or subtract.

UNIT 9 · LESSON 9.7 · 7.G.6

Lesson 9.7: Surface Area of Prisms

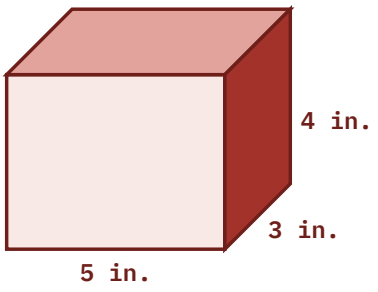
I can find the surface area of rectangular and triangular prisms.

WORDS TO KNOW

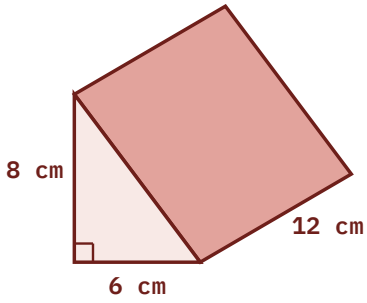
surface area	The total area of all the faces of a solid.
net	A flat pattern that folds into a solid.

LET'S EXPLORE

How much cardboard does a 5 in. by 3 in. by 4 in. box need?



Rectangular prism



Triangular prism: legs 6 and 8, hypotenuse 10, length 12

WORKED EXAMPLE · RECTANGULAR PRISM

$5 \times 3 \times 4$

1. Top and bottom: $2 \times (5 \times 3) = 30$.
2. Front and back: $2 \times (5 \times 4) = 40$.
3. Sides: $2 \times (3 \times 4) = 24$.

Answer: 94 in²

WORKED EXAMPLE · TRIANGULAR PRISM

Legs 6 and 8, hypotenuse 10, length 12

1. Two triangles: $2 \times \frac{1}{2} \times 6 \times 8 = 48$.
2. Three rectangles: $(6 + 8 + 10) \times 12 = 288$.

Answer: 336 cm²

KEY IDEA

Surface area = the sum of the areas of all faces. A net helps you see every face.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Cube with edge 3 cm
2. Box 10 × 2 × 4
3. Box 6 × 6 × 1

More practice: Workbook, Lesson 9.7.

LESSON SUMMARY

Add the areas of every face; nets help.

UNIT 9 · LESSON 9.8 · 7.G.6

Lesson 9.8: Volume of Prisms

I can find the volume of any right prism using V = Bh.

WORDS TO KNOW

volume	The amount of space inside a solid, in cubic units.
base area (B)	The area of the prism's base.

LET'S EXPLORE

How many 1-inch cubes fill a 5 in. by 3 in. by 4 in. box?

The base holds 5 × 3 = 15 cubes per layer, and there are 4 layers: 60 cubes. For any prism, V = Bh, where B is the area of the base and h is the prism's height (the distance between the bases).

WORKED EXAMPLE · RECTANGULAR PRISM

5 × 3 × 4

1 B = 15; h = 4.

Answer: 60 in³

WORKED EXAMPLE · TRIANGULAR PRISM

Base triangle legs 6 and 8; length 12

1 $B = \frac{1}{2} \times 6 \times 8 = 24$.

2 V = 24 × 12.

Answer: 288 cm³

WORKED EXAMPLE · TRAPEZOID BASE

Bases 4 and 6, height 3; prism length 10

1 B = 15.

2 V = 15 × 10.

Answer: 150 cubic units

KEY IDEA

V = Bh for every right prism. Volume is measured in cubic units.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Box 6 × 2 × 5

2. Cube edge 4

3. Prism with B = 18 cm², h = 7 cm

More practice: Workbook, Lesson 9.8.

LESSON SUMMARY

V = Bh: base area × height.

UNIT 9 · LESSON 9.9 · 7.G.6

Lesson 9.9: Volume and Surface Area in Context

I can decide whether a problem needs volume or surface area and solve it.

LET'S EXPLORE

A sandbox is 6 ft by 4 ft by 1 ft. You want to fill it with sand and paint its outside walls. Which needs volume? Which needs area?

WORKED EXAMPLE · FILL → VOLUME

$6 \times 4 \times 1$

1 $V = 24$.

Answer: 24 ft³ of sand

WORKED EXAMPLE · PAINT → AREA

Four outside walls (no top or bottom)

1 $2 \times (6 \times 1) + 2 \times (4 \times 1) = 12 + 8$.

Answer: 20 ft² of paint

Volume (cubic units)	Surface area (square units)
fill, hold, pour, pack inside	wrap, cover, paint, label outside

KEY IDEA

Filling uses volume; covering uses surface area. Decide which faces are included.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Water to fill a pool
2. Tiles to line the pool's walls and floor
3. Label around a soup can

More practice: Workbook, Lesson 9.9.

LESSON SUMMARY

Fill → volume; cover → surface area.

UNIT 9 · LESSON 9.10 · 7.G.6

Lesson 9.10: Composite Solids

I can find the volume and surface area of solids made from prisms.

LET'S EXPLORE

A set of steps is made of two boxes: a bottom box 6 ft × 3 ft × 1 ft and a top box 3 ft × 3 ft × 1 ft on one end. How much concrete is needed?

WORKED EXAMPLE · VOLUME

Add the boxes

1 Bottom: 18. Top: 9.

Answer: 27 ft³

WORKED EXAMPLE · ANOTHER SOLID

A 10 × 10 × 10 cube with a 2 × 2 × 10 hole drilled through

1 Cube: 1,000. Hole: 40.

Answer: 960 cubic units

KEY IDEA

Volume of a composite solid: add the parts, or subtract the parts removed.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Two cubes, edges 3 and 2, stacked
2. Box 8 × 5 × 4 with a 2 × 2 × 4 hole
3. Box 5 × 5 × 5 plus a 5 × 5 × 2 box on top

More practice: Workbook, Lesson 9.10.

LESSON SUMMARY

Add the parts, or subtract what is removed.

UNIT 9 · LESSON 9.11 · 7.G.6

Lesson 9.11: Comparing Volume and Surface Area

I can compare solids with the same volume but different surface areas.

LET'S EXPLORE

Boxes 8 × 1 × 1, 4 × 2 × 1, and 2 × 2 × 2 all hold 8 cubes. Which uses the least cardboard?

Box	Volume	Surface area
8 × 1 × 1	8	34
4 × 2 × 1	8	28
2 × 2 × 2	8	24

All three hold the same amount, but the cube uses the least material. The closer a box is to a cube, the less surface area it needs for its volume.

KEY IDEA

Same volume can have different surface areas. Cube-like shapes use the least surface area.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. SA of 12 × 1 × 1
2. SA of 3 × 2 × 2
3. Both have volume 12. Which uses less material?

More practice: Workbook, Lesson 9.11.

LESSON SUMMARY

Same volume can have different surface areas; cubes use the least.

UNIT 9 · LESSON 9.12 · 7.G.4, 7.G.5, 7.G.6

Lesson 9.12: Geometry Problem Solving

I can choose and use the right geometry formulas to solve multi-step problems.

LET'S EXPLORE

A circular pool (diameter 20 ft) is surrounded by a 3 ft wide concrete path. What is the area of the path?

WORKED EXAMPLE · RING AREA

Outer radius 13 ft, inner radius 10 ft

1 Outer: 3.14 × 169 = 530.66.

2 Inner: 3.14 × 100 = 314.

3 Subtract.

Answer: 216.66 ft²

Measure	Formula
Circumference	$C = \pi d = 2\pi r$
Circle area	$A = \pi r^2$
Triangle area	$A = {}^{1}\!/_{2}bh$
Trapezoid area	$A = {}^{1}\!/_{2}(b_1 + b_2)h$
Prism volume	$V = Bh$
Surface area	sum of all face areas

KEY IDEA

Plan: draw, label, choose the formula, compute, and check the units.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Circumference of the pool
2. Outer edge of the path
3. Volume of a 1 ft deep layer of water in the pool

More practice: Workbook, Lesson 9.12.

LESSON SUMMARY

Draw, label, choose a formula, compute, and check units.

UNIT 9 · LESSON 9.13 · 7.RP.1–3

Lesson 9.13: Year Review: Ratios, Proportions, and Percents

I can solve problems with unit rates, proportional relationships, and percents.

LET'S EXPLORE

Which is the better buy: 3 lb of apples for $4.47 or 5 lb for $7.25?

WORKED EXAMPLE · UNIT RATES

$4.47 ÷ 3 vs $7.25 ÷ 5

1 $1.49 per lb vs $1.45 per lb.

Answer: 5 lb for $7.25

WORKED EXAMPLE · PERCENT CHANGE

Price rises from $80 to $100

1 Change: 20.

2 20 ÷ 80 = 0.25.

Answer: 25% increase

KEY IDEA

Unit rates compare; proportions scale; percent = part ÷ whole × 100.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Unit rate: $\frac{3}{4}$ mi in $\frac{1}{4}$ hr

2. y is proportional to x; y = 10 when x = 4. Find k.

3. 20% off $65

More practice: Workbook, Lesson 9.13.

LESSON SUMMARY

Unit rates, proportions, and percents.

UNIT 9 · LESSON 9.14 · 7.NS.1–3

Lesson 9.14: Year Review: Rational Numbers

I can add, subtract, multiply, and divide rational numbers.

LET'S EXPLORE

The temperature was −6°F at dawn and rose 14°F by noon, then fell 9°F by night. What was the night temperature?

WORKED EXAMPLE · INTEGER STEPS

−6 + 14 − 9

1 −6 + 14 = 8.

2 8 − 9 = −1.

Answer: −1°F

WORKED EXAMPLE · FRACTIONS

$-\frac{3}{4} \div \frac{1}{2}$

1 $-\frac{3}{4} \times 2 = -\frac{6}{4}$.

Answer: $-1\frac{1}{2}$

KEY IDEA

Same signs multiply/divide to positive; different signs to negative. Subtracting means adding the opposite.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. −7 + 12
2. −3 − (−8)
3. −2.5 × 4

More practice: Workbook, Lesson 9.14.

LESSON SUMMARY

Sign rules for all four operations.

UNIT 9 · LESSON 9.15 · 7.EE.1–4

Lesson 9.15: Year Review: Expressions, Equations, and Inequalities

I can simplify expressions and solve equations and inequalities.

LET'S EXPLORE

A gym charges $20 to join and $15 per month. Lillie has $110. How many months can she afford?

WORKED EXAMPLE · INEQUALITY

$20 + 15m \le 110$

1 $15m \le 90$.

2 $m \le 6$.

Answer: at most 6 months

WORKED EXAMPLE · SIMPLIFY

$3(x - 4) + 2x$

1 $3x - 12 + 2x$.

Answer: $5x - 12$

KEY IDEA

Undo operations in reverse order; flip the inequality when multiplying or dividing by a negative.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Factor $18y + 12$

2. $4x - 7 = 21$

3. $-3x + 5 > 17$

More practice: Workbook, Lesson 9.15.

LESSON SUMMARY

Simplify, solve, and interpret.

UNIT 9 · LESSON 9.16 · 7.SP, 7.G

Lesson 9.16: Year Review: Statistics, Probability, and Geometry

I can solve problems using statistics, probability, and geometry.

LET'S EXPLORE

In a random sample of 40 students, 14 walk to school. About how many of the 600 students walk?

WORKED EXAMPLE · INFERENCE

14 of 40; 600 students

1 $^{14}/_{40}$ = 35%; 0.35 × 600.

Answer: about 210

WORKED EXAMPLE · PROBABILITY

Two number cubes: P(sum is 8)

1 (2,6), (3,5), (4,4), (5,3), (6,2): 5 outcomes of 36.

Answer: $^{5}/_{36}$

KEY IDEA

Use the right tool: proportions for samples, sample spaces for probability, formulas for geometry.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. P(heads, then heads)
2. Area of a circle, r = 6
3. Missing angle in a triangle: 50°, 75°

More practice: Workbook, Lesson 9.16.

LESSON SUMMARY

Pull together statistics, probability, and geometry.

UNIT 9 REVIEW

Unit 9 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
9.1	Complementary sum to 90°, supplementary to 180°; vertical angles are equal.
9.2	Relationship → equation → solve → check the angle.
9.3	$C = \pi d = 2\pi r$, with $\pi \approx 3.14$.
9.4	$A = \pi r^2$: square the radius, then multiply by π.
9.5	Split the figure into rectangles and circle parts; perimeter uses only outside edges.
9.6	Split, find each area, then add or subtract.
9.7	Add the areas of every face; nets help.
9.8	$V = Bh$: base area × height.
9.9	Fill → volume; cover → surface area.
9.10	Add the parts, or subtract what is removed.
9.11	Same volume can have different surface areas; cubes use the least.
9.12	Draw, label, choose a formula, compute, and check units.
9.13	Unit rates, proportions, and percents.
9.14	Sign rules for all four operations.
9.15	Simplify, solve, and interpret.
9.16	Pull together statistics, probability, and geometry.

Glossary

absolute value	Distance from 0.
additive inverse	A number's opposite; sums to 0.
adjacent angles	Angles that share a vertex and a side but do not overlap.
angle sum	The angles in any triangle add up to 180°.
area of a circle	$A = \pi r^2$.
associative property	Changing the grouping doesn't change the sum: $(a + b) + c = a + (b + c)$.
balance	The amount of money in an account; it can be negative if more is owed than saved.
base area (B)	The area of the prism's base.
biased sample	A sample that over- or under-represents part of the population.
box plot	A display of the five-number summary: minimum, Q1, median, Q3, maximum.
census	Data collected from every member of a population.
circumference	$C = \pi d$.
coefficient	The number multiplied by a variable. In $3x$, the coefficient is 3.
commission	Pay earned as a percent of sales.
commutative property	Changing the order of addends doesn't change the sum: $a + b = b + a$.
complementary	Sum to 90°.
complementary angles	Two angles whose measures add to 90°.
complex fraction	A fraction with a fraction in its numerator, denominator, or both.
composite figure	A figure made of two or more simple shapes.
compound event	An event made of two or more simple events.
constant	A term without a variable.
constant of proportionality	$k = y \div x$.
coordinate plane	A grid formed by a horizontal x-axis and a vertical y-axis.
corresponding parts	Matching sides or angles in a figure and its copy.
cross section	The shape of a slice.
diameter	A segment across a circle through its center; $d = 2r$.
discount	Amount off a price.

distance	How far apart two numbers are; always positive: $\|a - b\|$.
distributive property	$a(b + c) = ab + ac$
equation	A statement that two expressions are equal.
equivalent expressions	Expressions that are equal for every value of the variable.
equivalent ratios	Ratios that make the same comparison, like 3:2 and 6:4.
evaluate	To find the value of an expression by substituting numbers.
event	A set of outcomes.
expand	Rewrite a product as a sum: $3(x + 4) = 3x + 12$.
experimental probability	Number of times an event happens ÷ number of trials.
expression	No equals sign.
factor (an expression)	Rewrite a sum as a product: $6x + 9 = 3(2x + 3)$.
greatest common factor (GCF)	The largest number that divides every term.
inequality	Comparison with $<$, $>$, $\leq$, $\geq$.
inference	A conclusion about a population based on sample data.
integer	Whole numbers and their opposites.
interest	Money paid for using money: earned on savings or paid on loans.
interquartile range (IQR)	Q3 − Q1, the spread of the middle half of the data.
inverse operations	Operations that undo each other.
like terms	Same variable part.
MAD	Average distance from the mean.
markup	Amount added to cost.
mean	Sum ÷ count.
mean (average)	The sum of the values divided by how many values there are.
mean absolute deviation (MAD)	The average distance of the values from the mean.
median	The middle value when data are in order.
net	A flat pattern that folds into a solid.
opposites	Same distance from 0, opposite sides.
order of operations	Parentheses, exponents, multiplication and division (left to right), addition and subtraction (left to right).

origin	The point (0, 0) where the axes cross.
outcome	One possible result of an experiment.
percent	A rate per 100.
percent change	Change ÷ original.
percent equation	part = percent × whole
percent error	|estimate − actual| ÷ actual.
pi (π)	The ratio of any circle's circumference to its diameter, about 3.14.
plane	A flat surface that extends forever.
population	The whole group.
principal	The original amount saved or borrowed.
probability	A number 0–1 for likelihood.
probability model	A list of outcomes and their probabilities, which add to 1.
product	Result of multiplication.
proportion	An equation saying two ratios are equal, like $3/4 = 9/12$.
proportional relationship	y ÷ x is always the same.
quotient	Result of division.
radius	A segment from the center to the circle.
random sample	A sample in which every member of the population has an equal chance of being chosen.
rate	A ratio that compares two quantities with different units, like 120 miles per 2 hours.
ratio	A comparison of two quantities.
rational number	A number that can be written as a fraction of integers.
reciprocal	Number whose product with another is 1.
repeating decimal	A decimal with a repeating pattern.
sale price	The price after the discount.
sales tax	A percent of the price added at the register.
sample	Part of the population.
sample space	All possible outcomes.
sampling variability	The natural differences between results from different samples.

scale	Drawing length : actual length.
scale drawing	A drawing that represents a real object with lengths in proportion.
scale factor	Multiplier for every length.
scaled copy	A copy of a figure in which every length is multiplied by the same number.
sea level	Elevation 0; above is positive and below is negative.
semicircle	Half of a circle.
simple interest	$I = P \times r \times t$.
simulation	An experiment that models a real situation using random tools.
solution	A value that makes an equation true.
solution set	All values that make an inequality true.
supplementary	Sum to 180°.
supplementary angles	Two angles whose measures add to 180°.
surface area	The total area of all the faces of a solid.
term	A part of an expression separated by + or −.
terminating decimal	A decimal that ends.
theoretical probability	Number of favorable outcomes ÷ number of equally likely outcomes.
tip (gratuity)	Money paid for a service, often a percent of the bill.
tree diagram	A branching diagram that shows every outcome of a multi-step experiment.
trial	One repetition of an experiment.
triangle inequality	Two sides sum to more than the third.
Triangle Inequality	The sum of any two side lengths of a triangle is greater than the third side.
uniform model	All outcomes are equally likely.
unique	Exactly one; any triangle drawn with those conditions is the same shape and size.
unit price	The cost for one unit, such as dollars per ounce.
unit rate	Amount per one unit.
variable	A letter for a number.
vertical angles	Opposite angles formed by two intersecting lines. They are equal.
volume	The amount of space inside a solid, in cubic units.
volume of a prism	$V = Bh$.

zero pair	A positive and a negative chip together; they cancel to 0.

Grade 7 Mathematics

Student Textbook · 2026–2027

A full year of Grade 7 mathematics, taught one clear idea at a time. Every lesson opens with a question worth thinking about, walks through worked examples step by step, and builds from models and pictures to fluent, confident problem solving.

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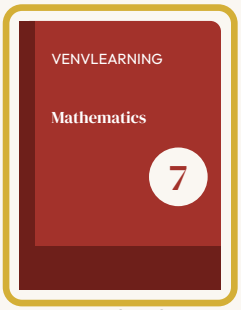
INSIDE THIS YEAR

01 Proportional Relationships
02 Percents
03 Adding and Subtracting Integers
04 Adding and Subtracting Rational Numbers
05 Multiplying and Dividing Rational Numbers
06 Expressions, Equations, and Inequalities
07 Statistics and Probability
08 Scale Drawings and Geometric Figures
09 Angles, Circles, Area, and Volume

IN THIS BOOK

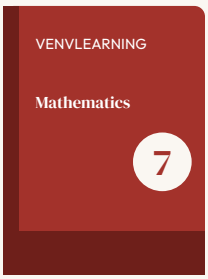
- Clear lesson goals: “I can…” statements
- Key vocabulary with student-friendly definitions
- Models, examples, and Key Idea summaries
- A study guide and glossary for every unit

THE COMPLETE SET

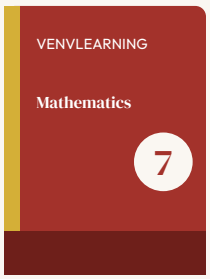


Textbook

You are holding this book



Workbook



Teacher Edition

Each lesson has the same number in all three books, and every day of the year is mapped to its lesson in the VenvLearning planner.